VELLORE INSTITUTE OF TECHNOLOGY, VELLORE



MINI PROJECT REPORT ON

ATLAS

Beyond the Ordinary

An Interactive Worldwide Photography Gallery

Course: **CSI3029 - Front End Design and Testing**

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1 Abstract and Project Overview

Abstract

ATLAS - Beyond the Ordinary is a static, interactive travel photography application built with HTML5, CSS3 and vanilla JavaScript. It presents 18 destinations across five continents through a responsive gallery, combined search and category filters, sorting, browser-local favorites, and a keyboard-accessible photograph viewer. The expanded version adds sourced destination guides, an interactive location map, two/three-place comparison, random discovery and an explicitly controlled slideshow. This report documents the implemented interface with 110 distinct screenshots captured from the running application and records 40 consolidated verification cases through TC-40. The project emphasizes frontend composition, meaningful mouse and keyboard events, state synchronization and evidence-based testing.

Problem statement

A passive photograph grid provides little help in finding a place, understanding its context, comparing interests or keeping a personal collection of favorites. Independent controls can produce inconsistent counts or lose user selections. A visual overlay can also become difficult to use when its keyboard focus, navigation boundaries and lifecycle are not managed explicitly.

Objectives

- Create a premium editorial photography interface with a consistent dark/blue/lime design system.
- Support meaningful mouse and keyboard exploration, accessible dialogs and responsive phone/tablet layouts.
- Connect discovery, guides, maps, comparison, favorites and playback through understandable browser state.

- Verify implementation with executable pure-logic tests and observed browser outcomes; report remaining coverage honestly.

Scope

The implementation is a build-free static frontend. Its single destination array supplies photographs, country/category metadata, editorial guides, source credits and representative map coordinates. Favorites persist in localStorage; comparison and filter state last for the current visit. Wikimedia Commons supplies remote photography and an on-demand OpenStreetMap embed supplies map interaction. Accounts, a backend database, uploads, bookings, personal travel journals and shareable destination links are outside this version.

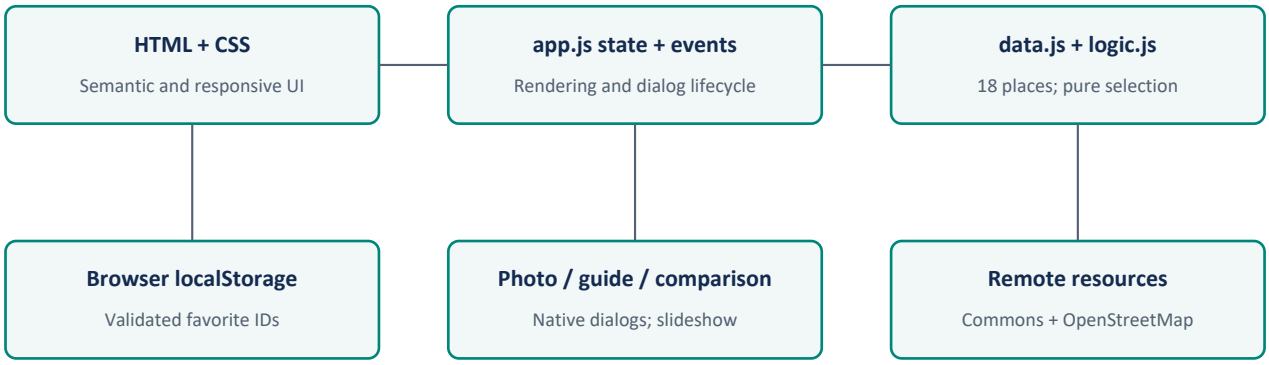
Measure (from the implementation)	Value
Destinations / continents	18 destinations / 5 continents
Categories	6: Mountains, Beaches, Cities, Nature, Architecture, Culture
Core technology	HTML5, CSS3 and 3 vanilla JavaScript files; no build framework
Guide and map coverage	18 detailed guides and 18 sourced coordinate pairs
Screenshot evidence	110 distinct browser screenshots in Section 4
Report verification	40 consolidated cases, TC-01 through TC-40
Automated suite	36 Node tests passed in the report preparation run

2 Project Links

Resource	Location / current status
Local running application	http://127.0.0.1:8000 (start the local server first)
Application entry point	index.html in the submitted project folder
Project documentation	README.md and docs/PROJECT_DOCUMENTATION.md
Source and image attribution	docs/DESTINATION_SOURCES.md and docs/IMAGE_CREDITS.md
Screenshots and test evidence	docs/report-evidence/manifest.json and node-results.tap
GitHub repository / public deployment	No public repository or hosted application URL has been supplied or verified.
Implementation video	Not supplied; no video link is claimed.

3 Technology Stack and System Architecture

Layer	Technology and implementation
Frontend structure	Semantic HTML landmarks, headings, native form controls, anchors and two dialog elements [1, 2].
Visual design	CSS Grid/Flexbox, custom-property tokens, clamp typography, image aspect ratios and responsive breakpoints.
Application logic	Vanilla JavaScript DOM construction, delegated clicks, individual mouse-enter/leave handlers and a centralized state object.
Data and persistence	data.js is the single source of destination/guide metadata. logic.js contains pure helpers; validated favorite IDs use browser localStorage [3].
External resources	Source-credited Commons thumbnails and an on-demand OpenStreetMap embed with visible contributor attribution [6, 7].
Verification and report	Node built-in test runner; actual browser actions; Python/ReportLab for this PDF. No browser package was added to the application.



Static browser application - no backend, authentication server or database

Figure 3.1: ATLAS system architecture

Browser data and application flow

The browser loads data.js, logic.js and app.js as ordered deferred scripts. User input updates search, category, sort, favorites or layout state. Pure selection logic returns a new eligible array; rendering updates cards, counters and pressed states without mutating the source data. Comparison choices remain independent of gallery filters. Opening the viewer snapshots the eligible order; a guide opened from comparison instead follows the selected comparison order. No application server or database is involved.

Mouse and keyboard event handling

Delegated card clicks distinguish photograph, favorite, guide and comparison controls. Native buttons provide Enter/Space activation. Mouse enter and leave apply restrained image feedback. One document key handler supports viewer arrows, F for saving, P for playback, modal Tab wrapping, Ctrl/Cmd+K for search and menu Escape. Editable fields and selects are excluded from shortcuts. Native dialog behavior makes background content inert; Close/Escape restore focus to an appropriate trigger.

Playback, maps and error handling

Playback is opt-in, requires two eligible photographs and uses a single interval with 3/5/8-second choices. Manual navigation, guide expansion, image failure, closing and tab backgrounding pause it. The map loads only on request, uses validated representative coordinates and preserves pan/zoom across unrelated state changes. Failed images use local labeled SVG artwork. Storage JSON is parsed defensively and invalid IDs are rejected; denied storage leaves current-visit favorites usable.

4 Feature-Wise Implementation and Screenshots

Every figure below is a screenshot captured from the running ATLAS application in the available Chromium-based in-app browser. The account label is [Visitor] because ATLAS does not implement authentication. Captures cover distinct destination or interaction states; responsive widths are emulated browser viewports, not claims of real-device testing. Photograph and map attributions appear in the application and are consolidated in Section 9. The evidence manifest retains all 110 image filenames and the associated verification record.

4.1 Landing Page, Collections and Footer

The wordmark and primary links lead to real sections. The cinematic hero introduces the edit, while lower collection shortcuts select meaningful categories. The footer links to documentation and genuine source credits.

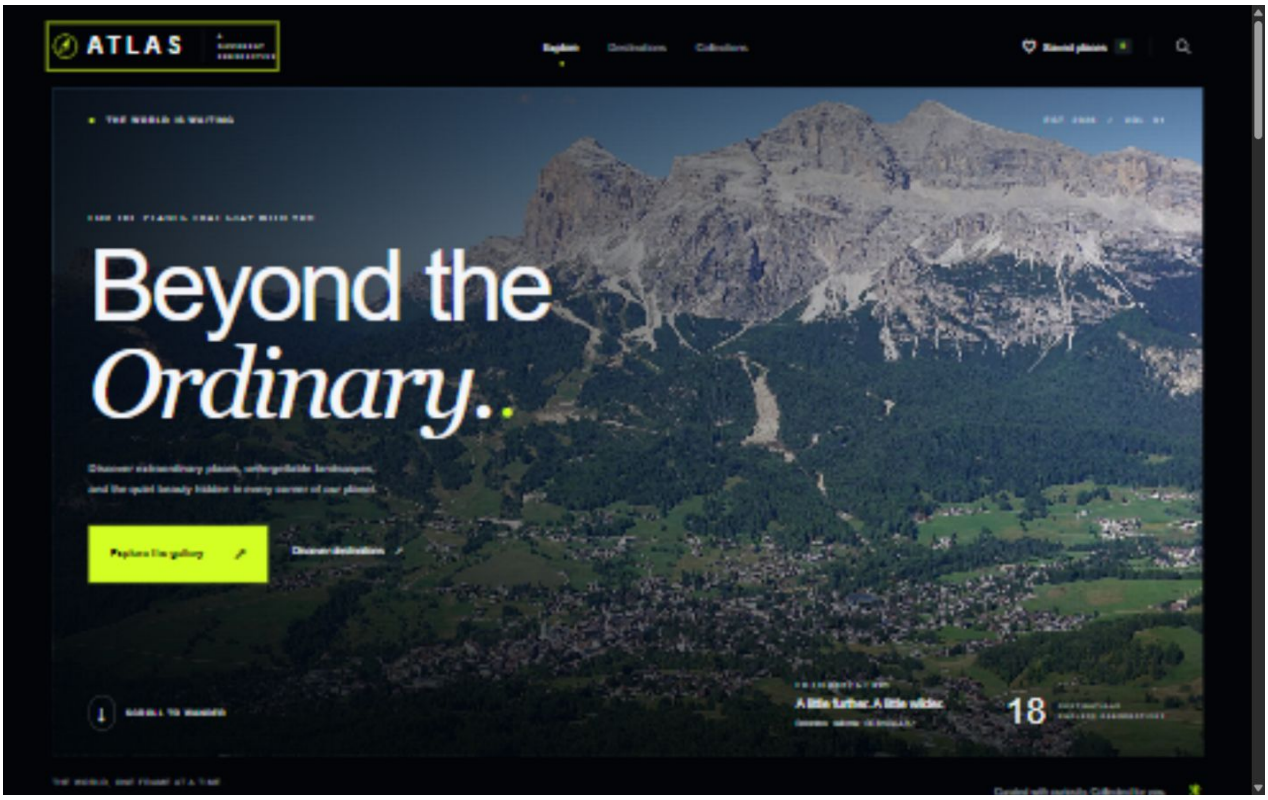


Figure 4.1: Desktop landing page at 1440 px [Visitor]

Editorial hero, branding and primary navigation.

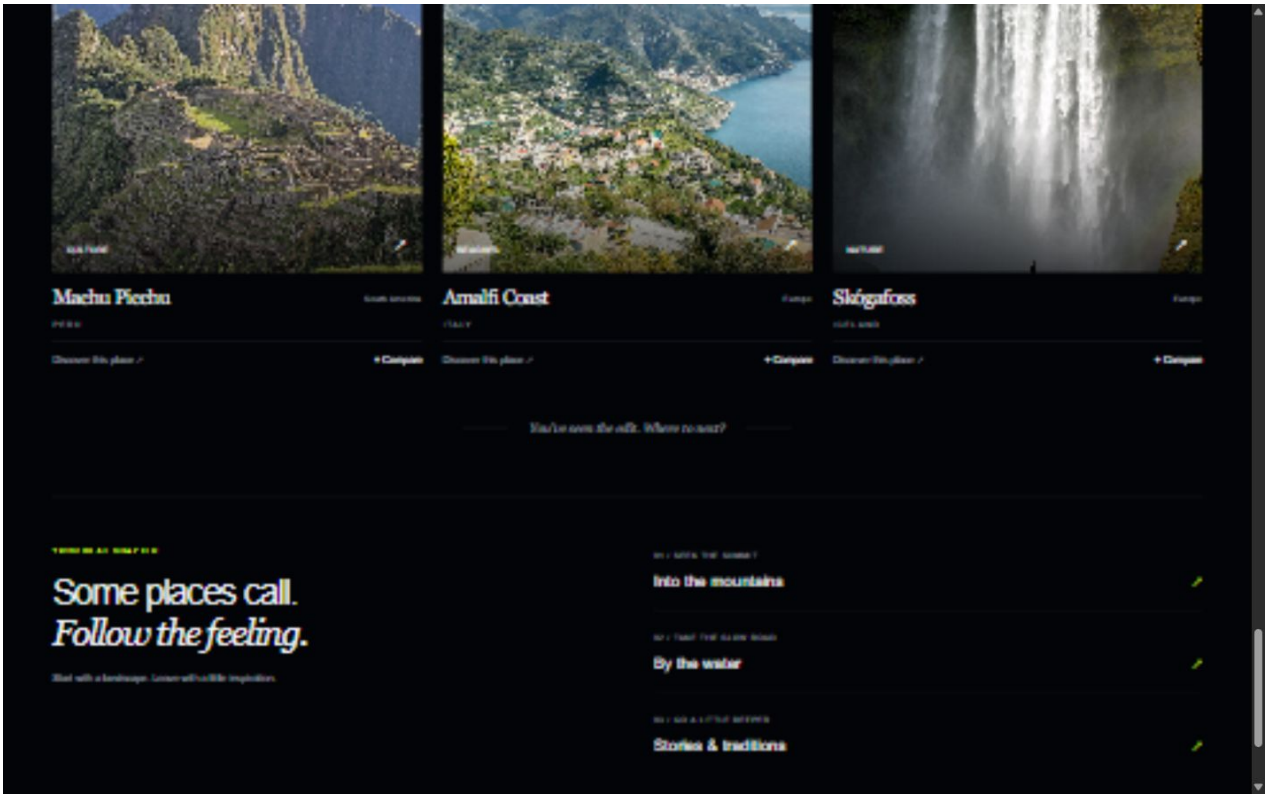


Figure 4.2: Editorial collections and footer [Visitor]

Category shortcuts, creator attribution and documentation links.

4.2 Dynamic Destination Gallery

All 18 destination records are rendered as native photo buttons with separate saving, guide and comparison controls. These searched-card captures show actual destination-specific results and retain the same card/data structure across continents.

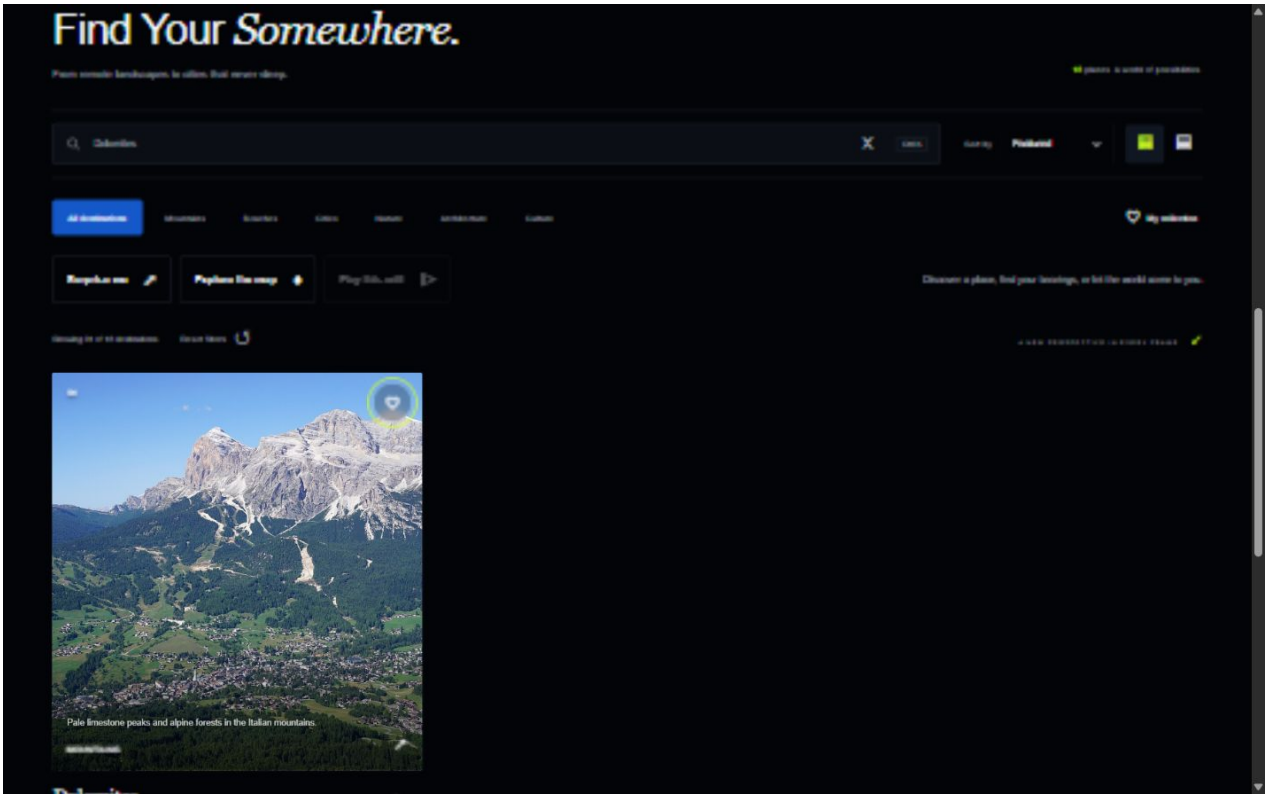


Figure 4.3: Dolomites: searched gallery card [Visitor]

Live search narrows the edit to the destination.

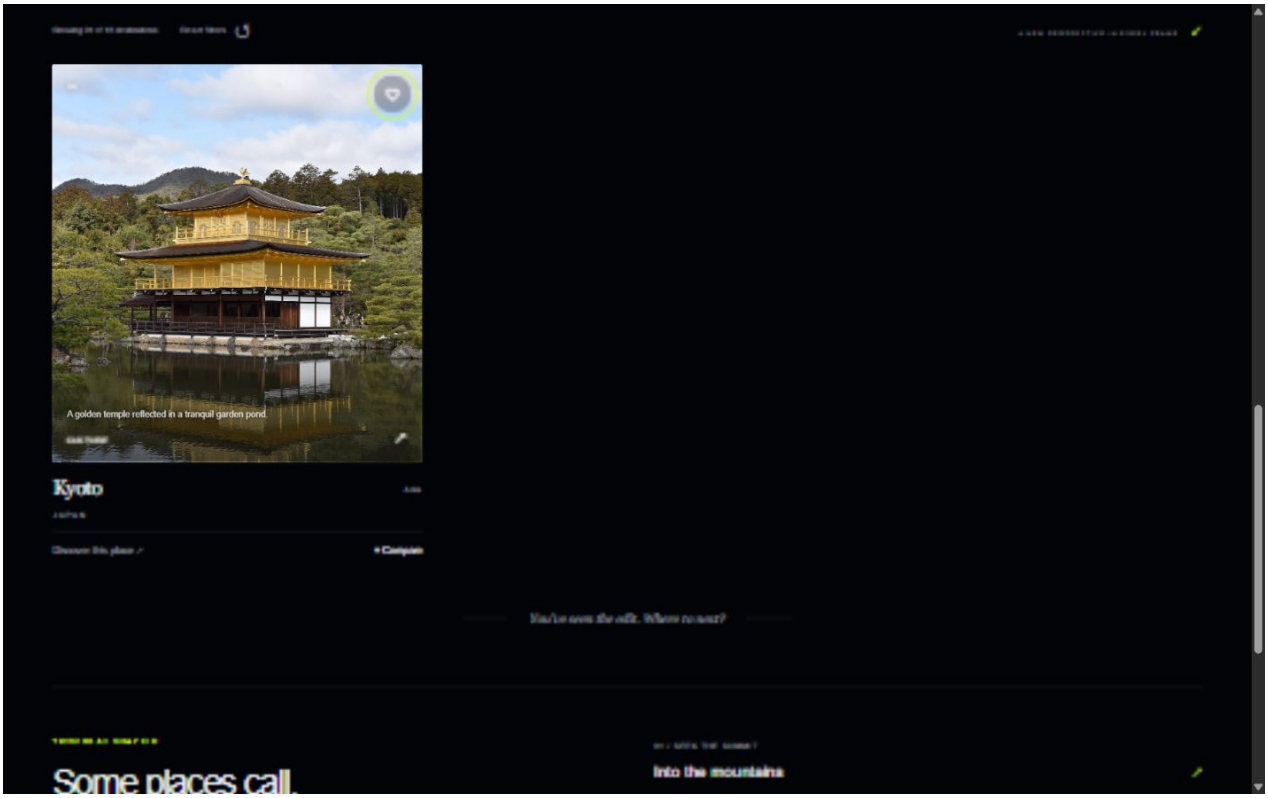


Figure 4.4: Kyoto: searched gallery card [Visitor]

Live search narrows the edit to the destination.

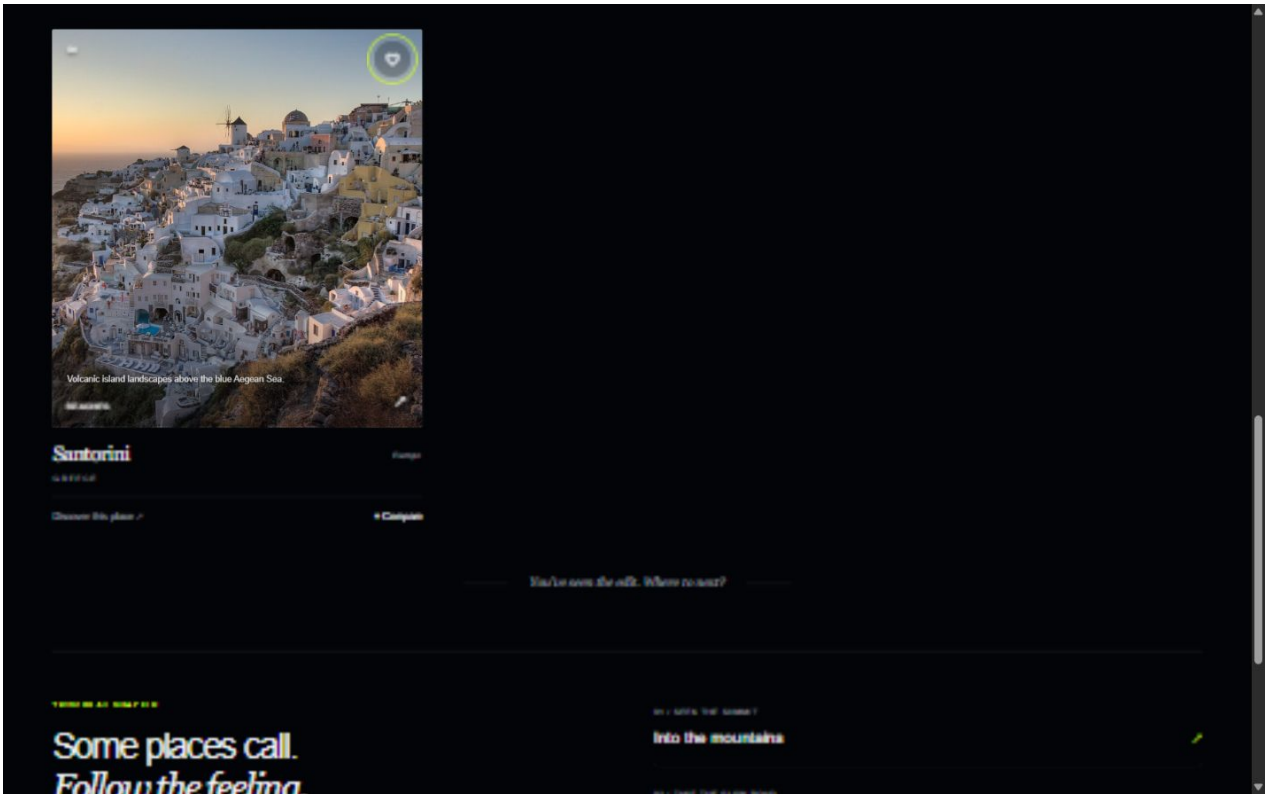


Figure 4.5: Santorini: searched gallery card [Visitor]

Live search narrows the edit to the destination.

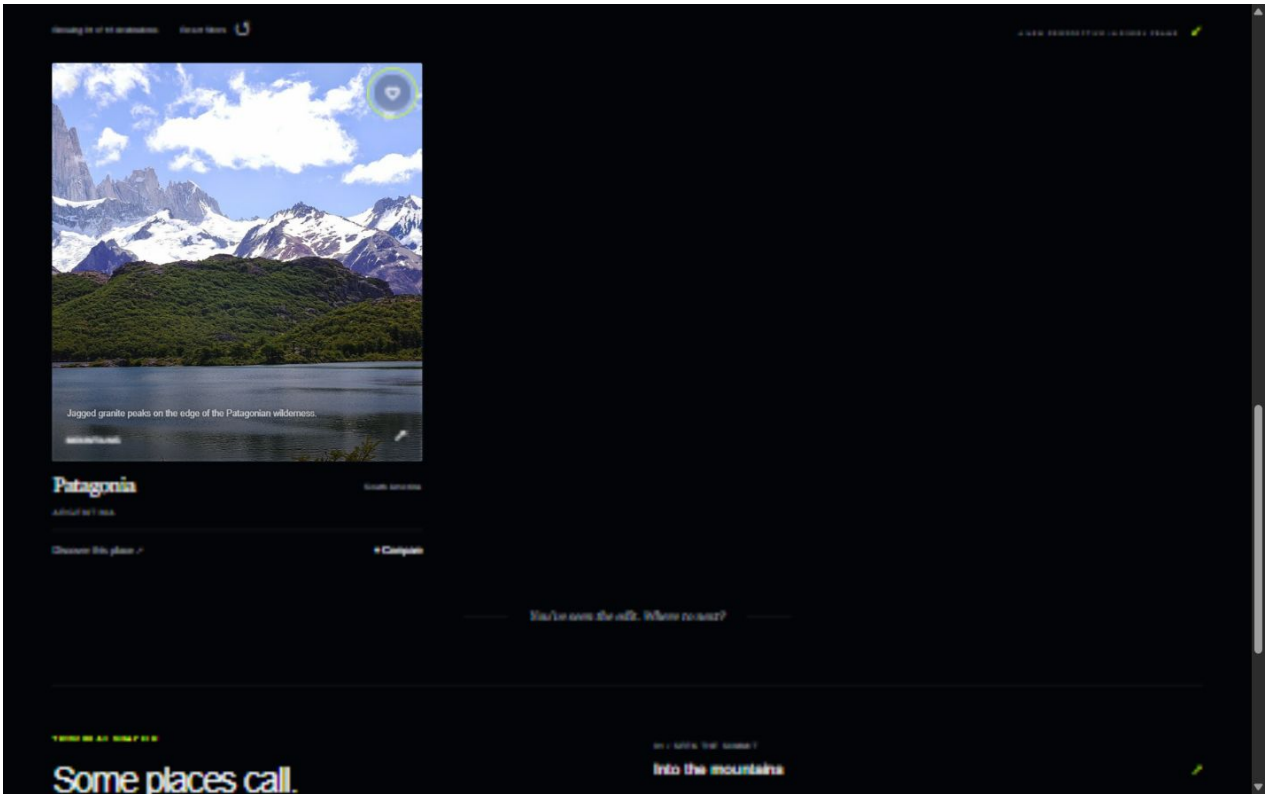


Figure 4.6: Patagonia: searched gallery card [Visitor]

Live search narrows the edit to the destination.

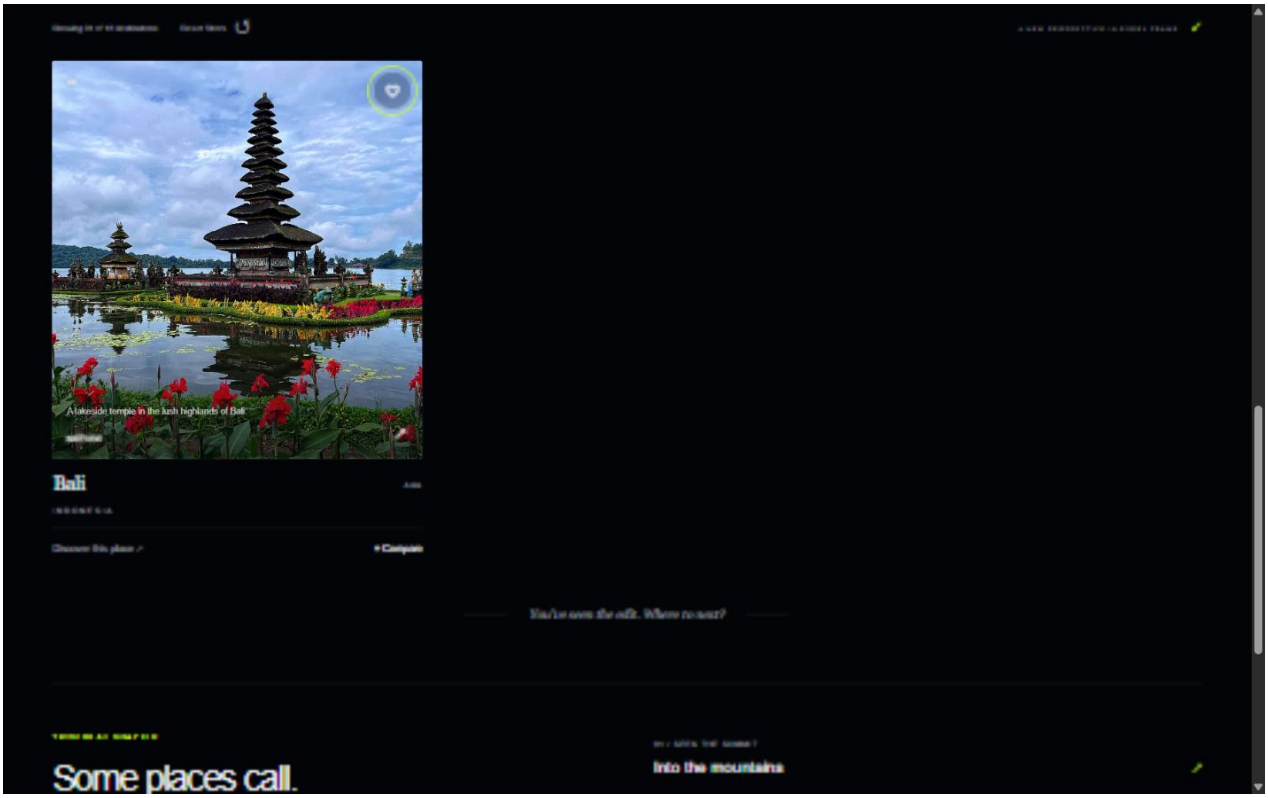


Figure 4.7: Bali: searched gallery card [Visitor]
Live search narrows the edit to the destination.

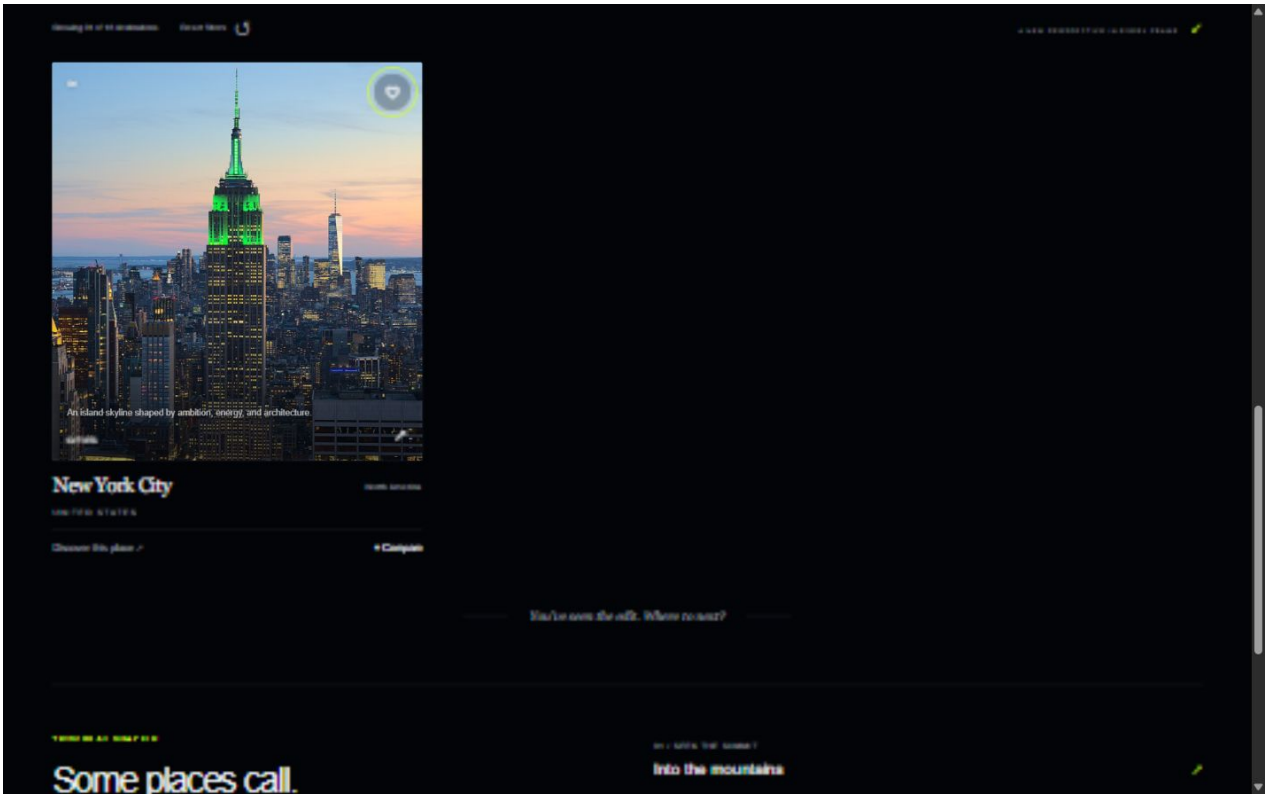


Figure 4.8: New York City: searched gallery card [Visitor]
Live search narrows the edit to the destination.

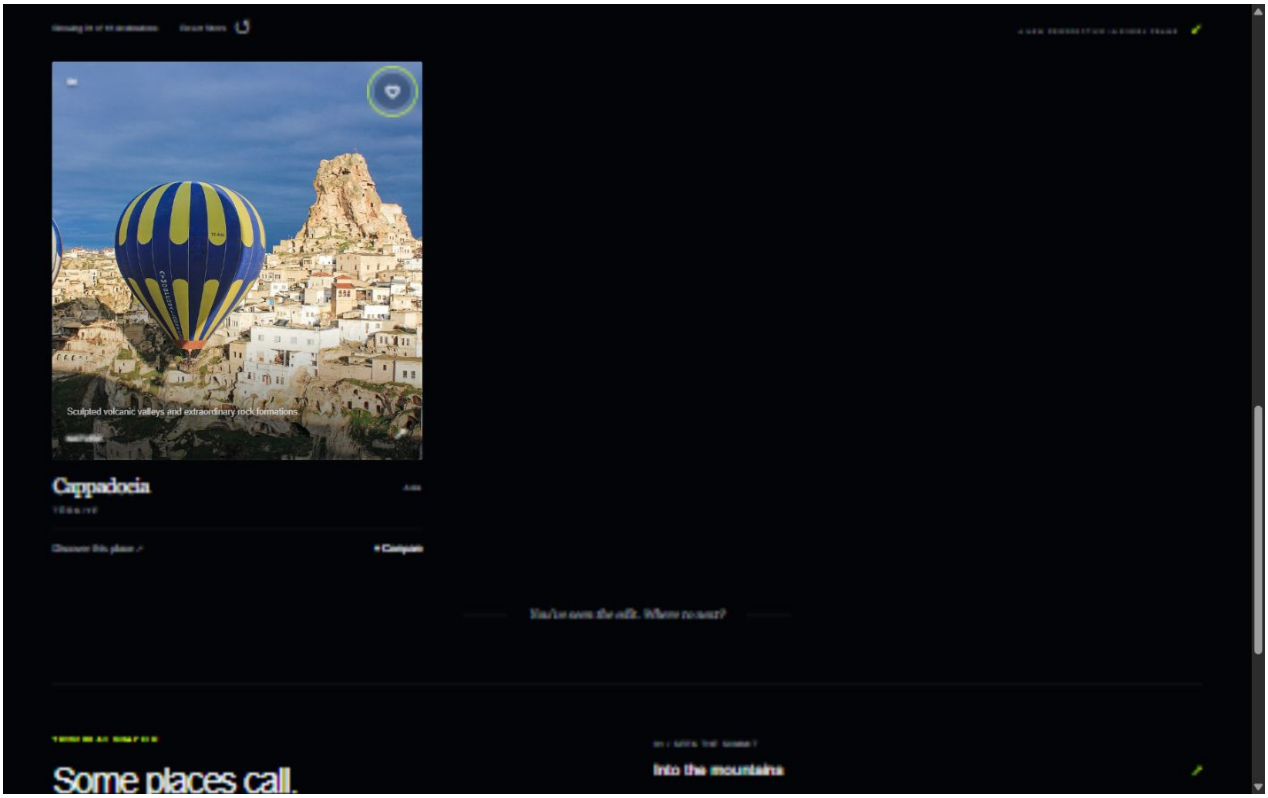


Figure 4.9: Cappadocia: searched gallery card [Visitor]

Case-insensitive live destination search.

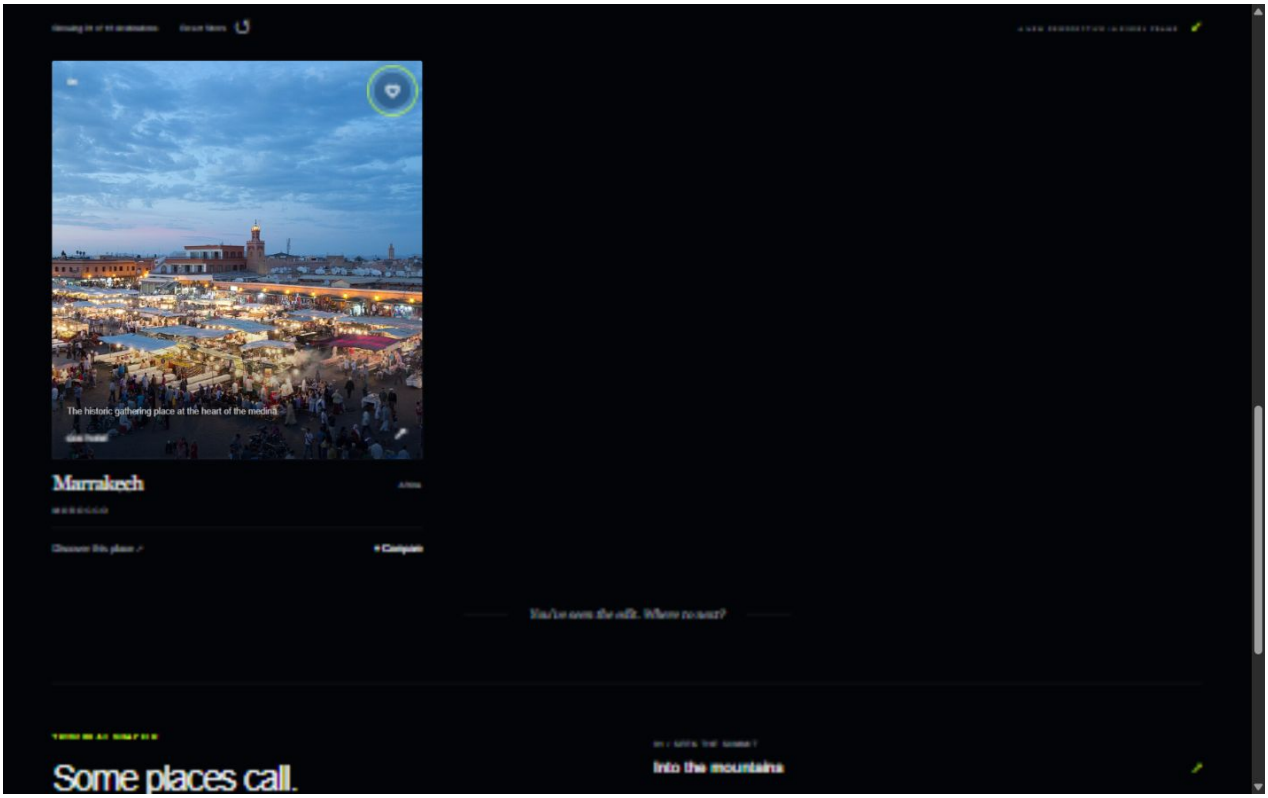


Figure 4.10: Marrakech: searched gallery card [Visitor]

Case-insensitive live destination search.

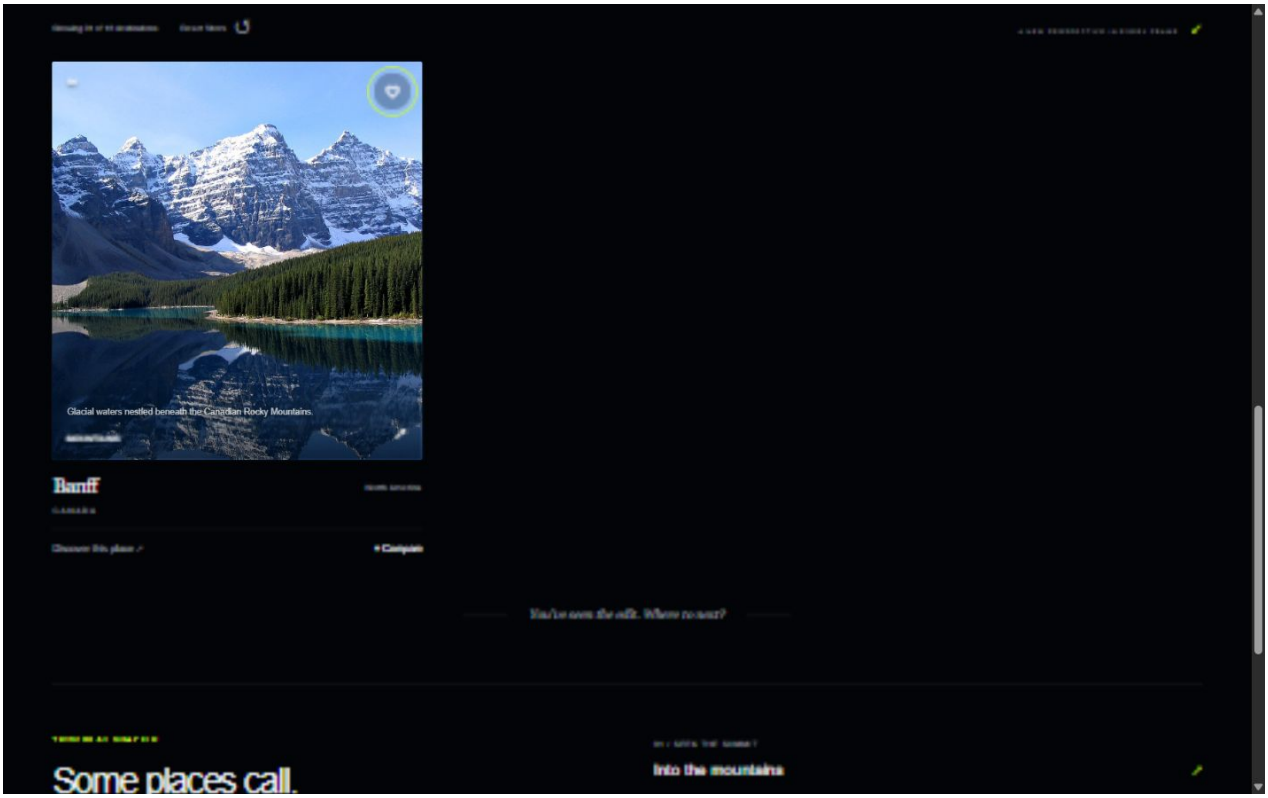


Figure 4.11: Banff: searched gallery card [Visitor]

Case-insensitive live destination search.

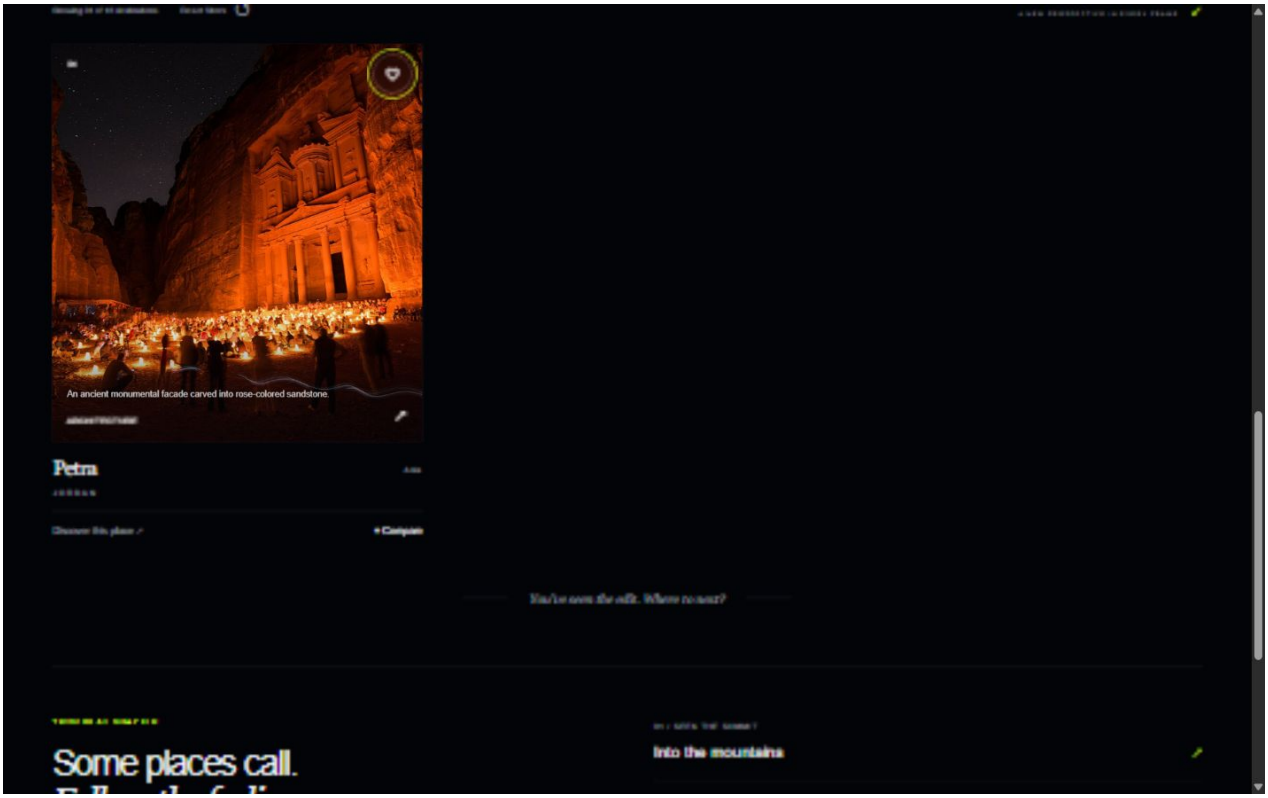


Figure 4.12: Petra: searched gallery card [Visitor]

Case-insensitive live destination search.

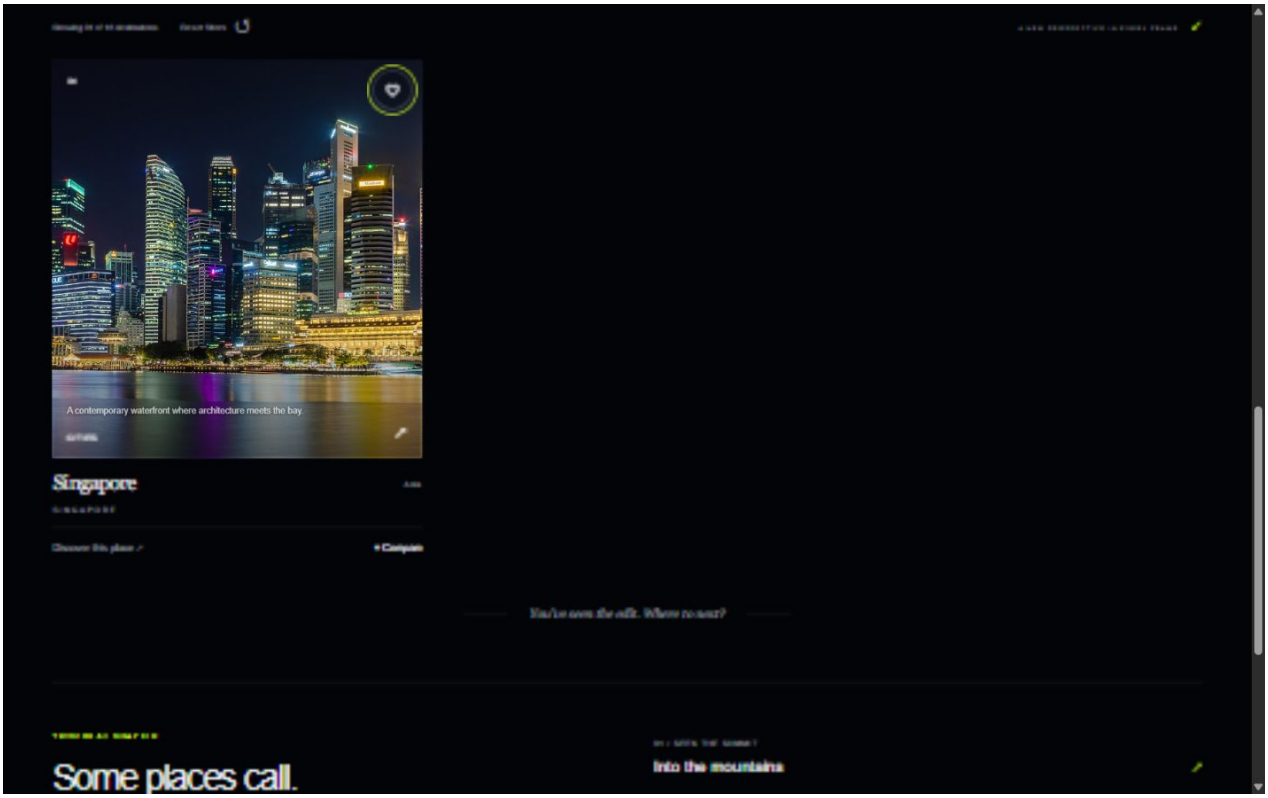


Figure 4.13: Singapore: searched gallery card [Visitor]

Case-insensitive live destination search.

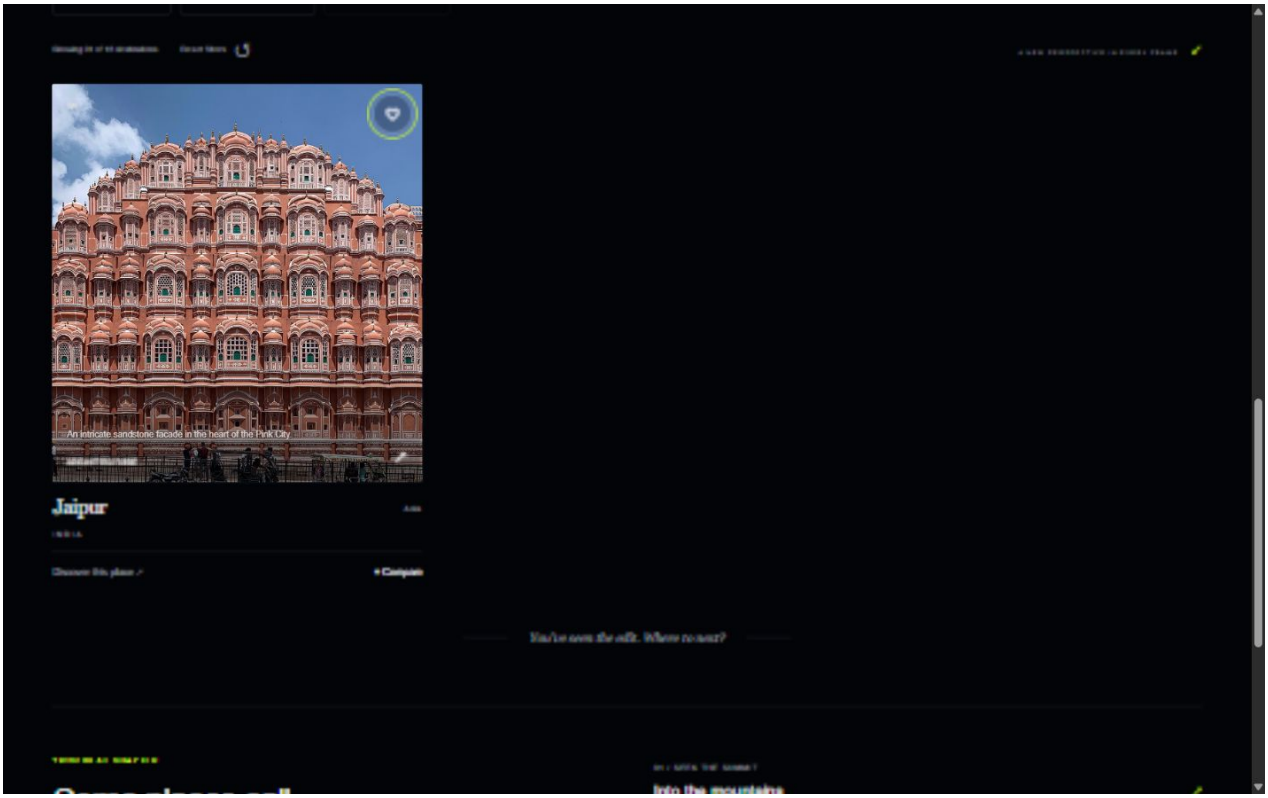


Figure 4.14: Jaipur: searched gallery card [Visitor]

Case-insensitive live destination search.

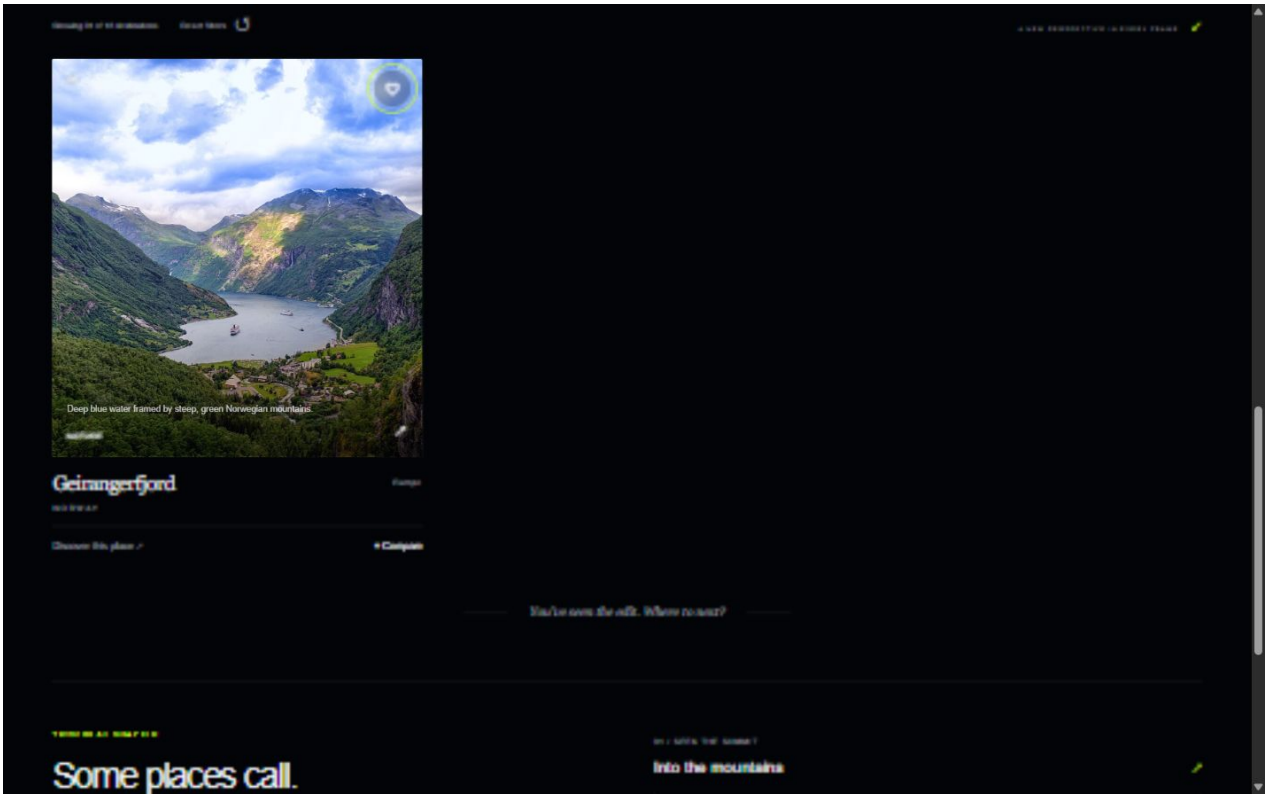


Figure 4.15: Geirangerfjord: searched gallery card [Visitor]

The eligible destination and current result count.

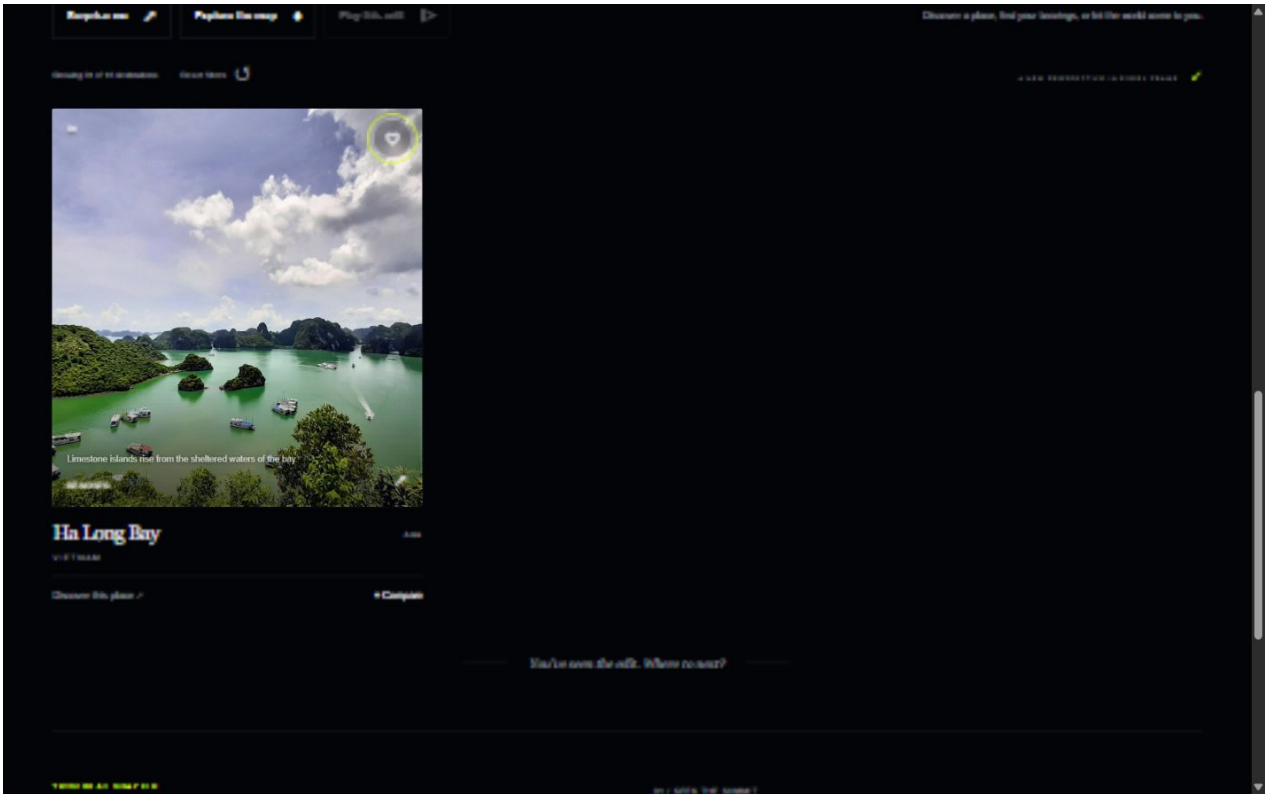


Figure 4.16: Ha Long Bay: searched gallery card [Visitor]

The eligible destination and current result count.

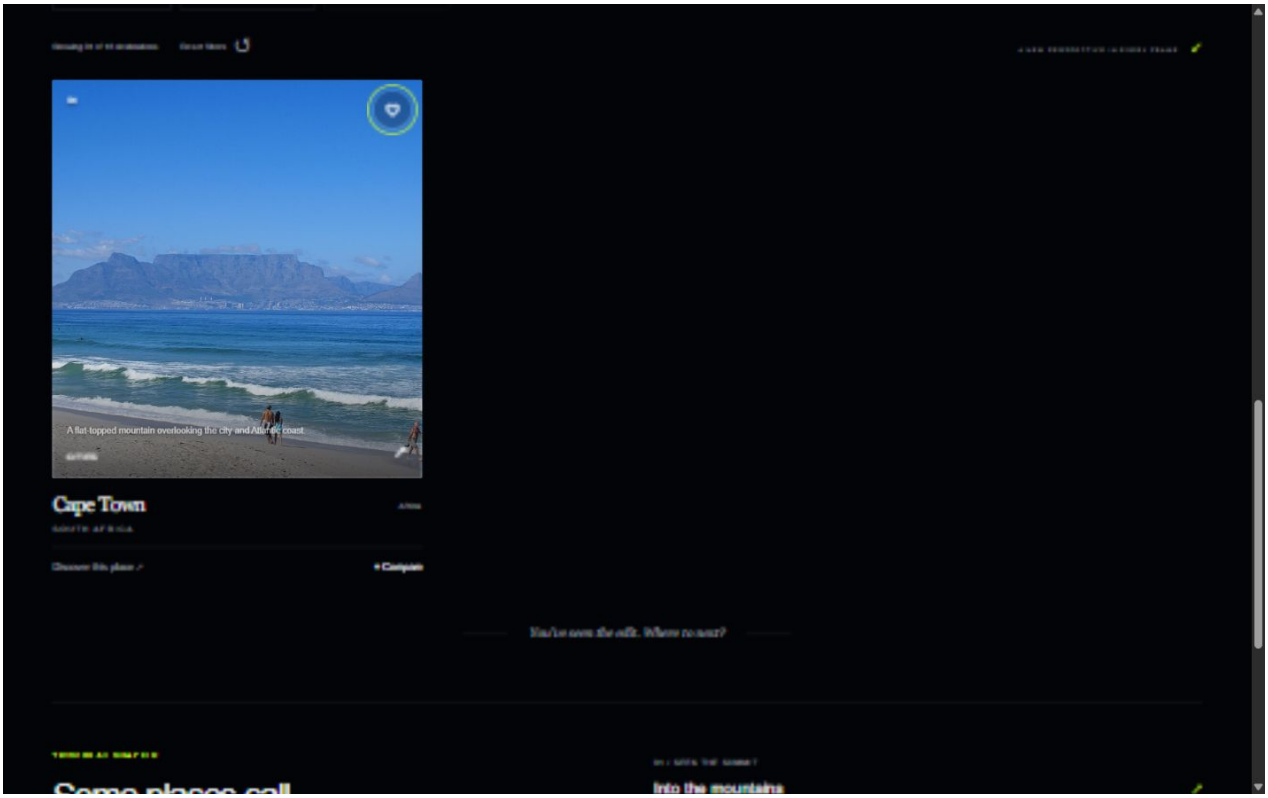


Figure 4.17: Cape Town: searched gallery card [Visitor]

The eligible destination and current result count.

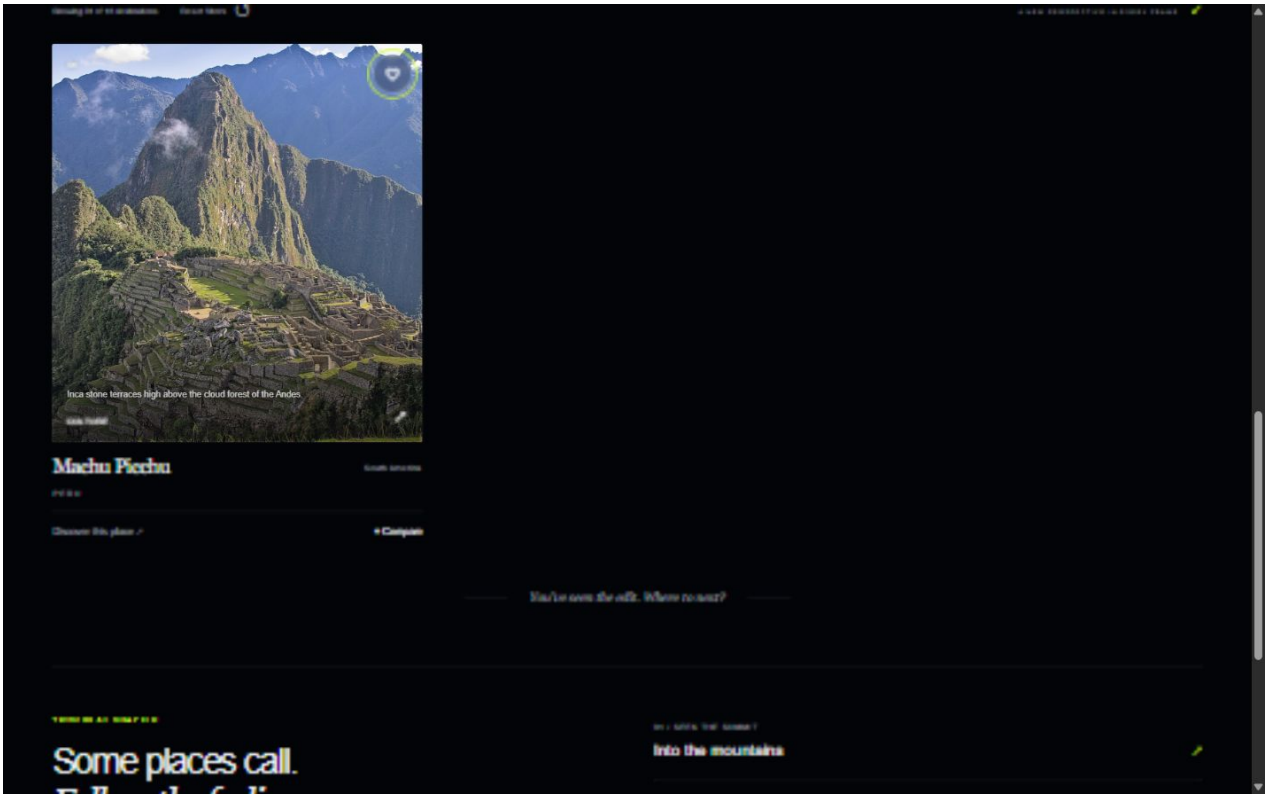


Figure 4.18: Machu Picchu: searched gallery card [Visitor]

The eligible destination and current result count.

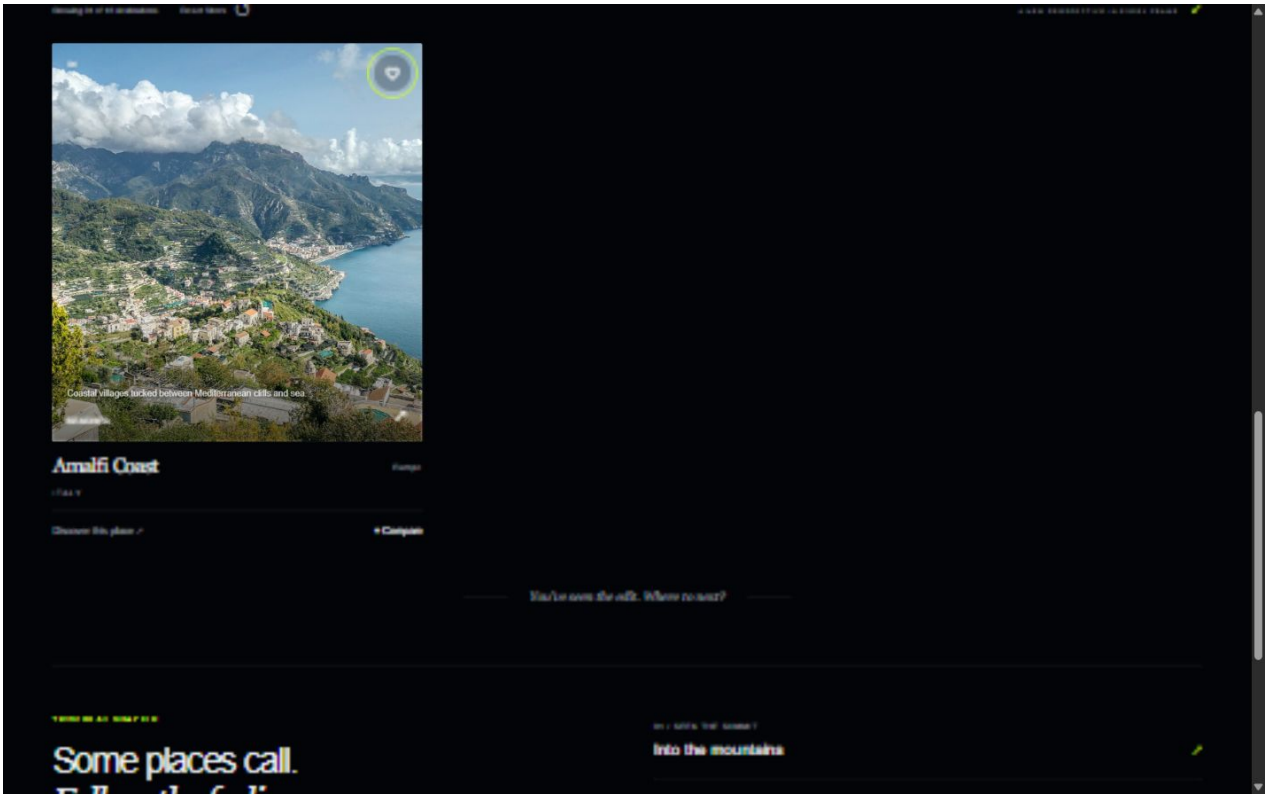


Figure 4.19: Amalfi Coast: searched gallery card [Visitor]

The eligible destination and current result count.

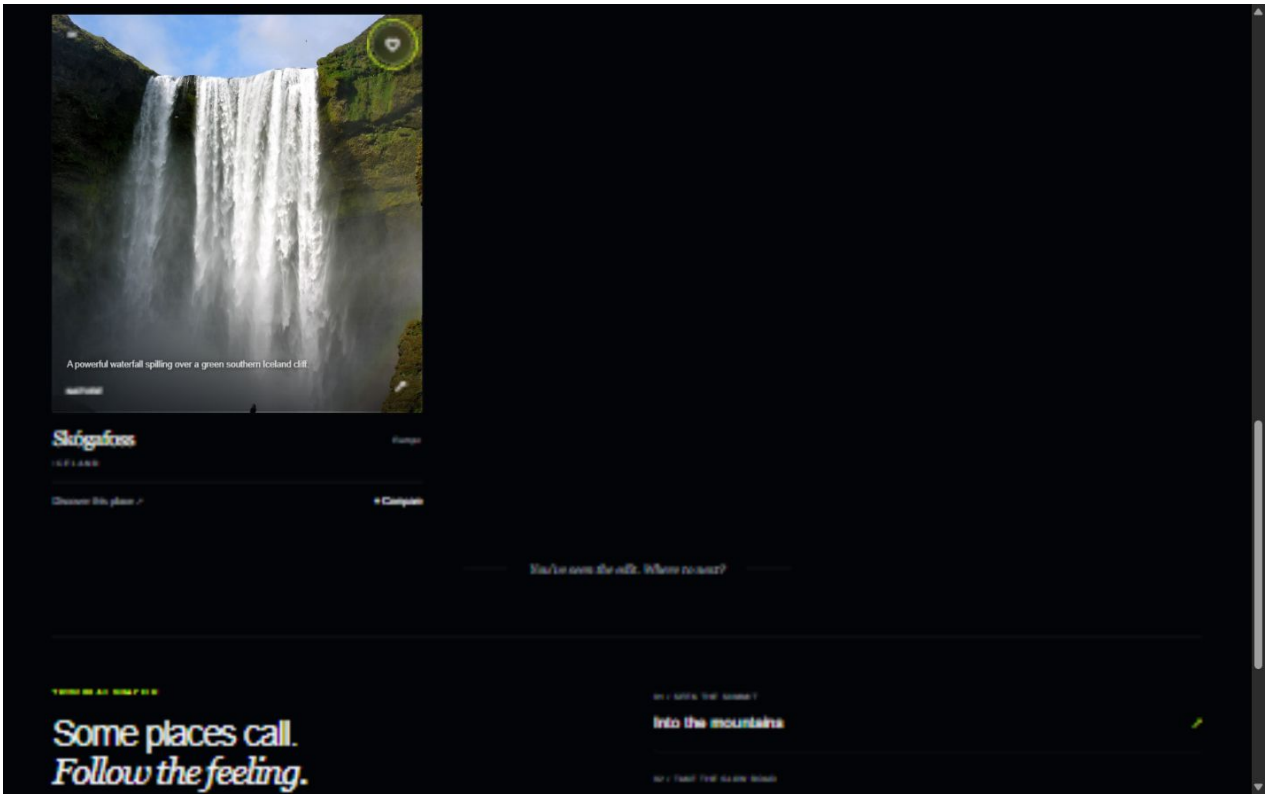


Figure 4.20: Skógafoss: searched gallery card [Visitor]

The eligible destination and current result count.

4.3 Search, Combined Selection and Empty States

Search matches destination, country, category and description without case sensitivity or surrounding-space sensitivity. It combines with the selected category, favorites-only mode and sort order. No-match states explain how to recover without disabling the rest of the application.

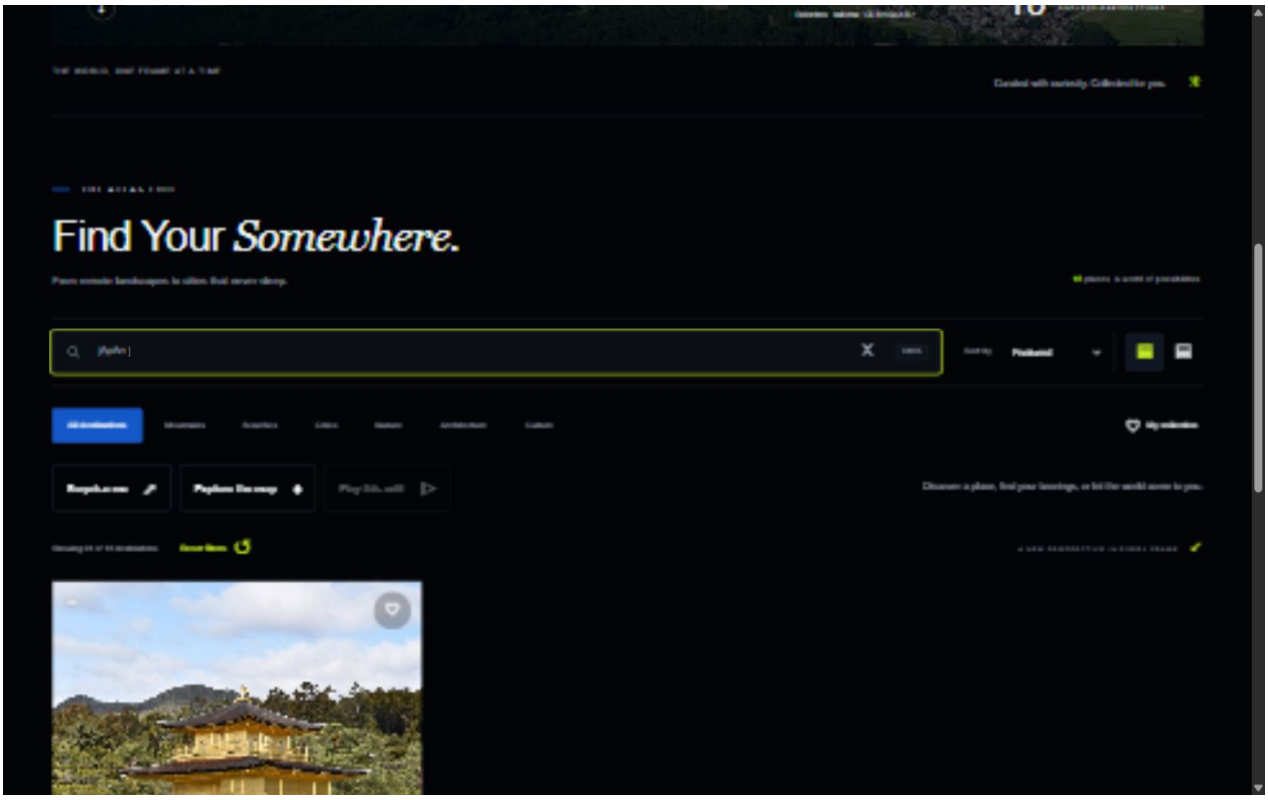


Figure 4.21: Country search with mixed case and surrounding spaces [Visitor]

The search result remains Kyoto, Japan.

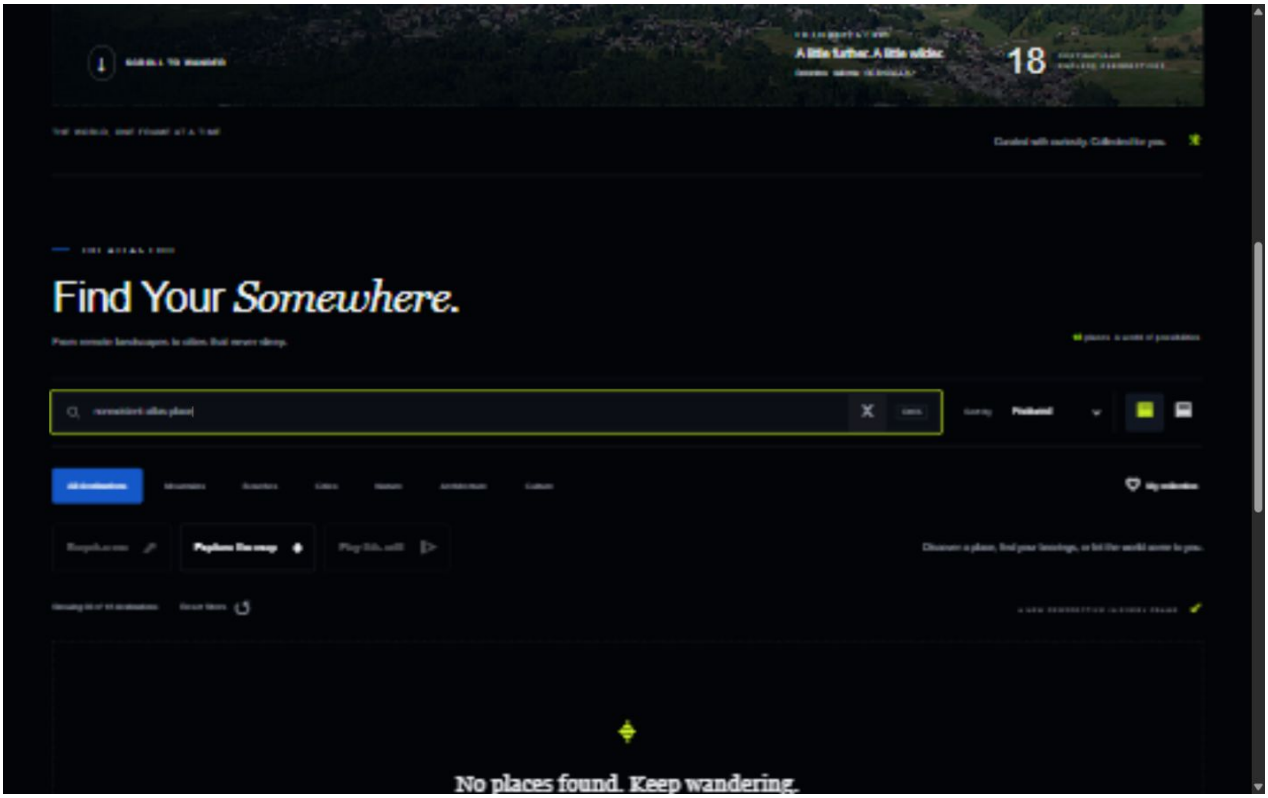


Figure 4.22: No results for an unmatched query [Visitor]

An explicit empty state provides a reset action.

4.4 Destination Category Filters

Six functional categories update aria-pressed state, result counts and eligible cards. All destinations removes only the category restriction, preserving other user selections.

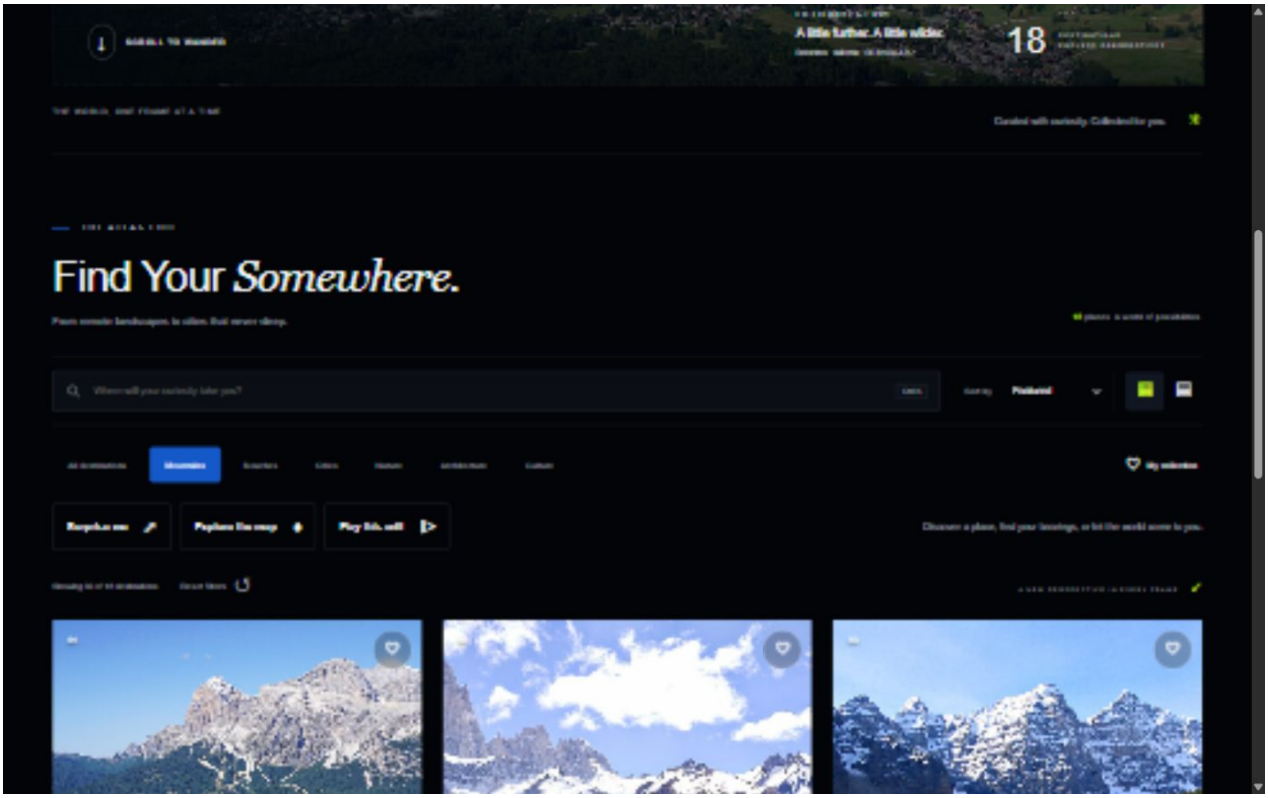


Figure 4.23: Mountains: active category and matching gallery [Visitor]

The selected category, live result count and eligible photographs.

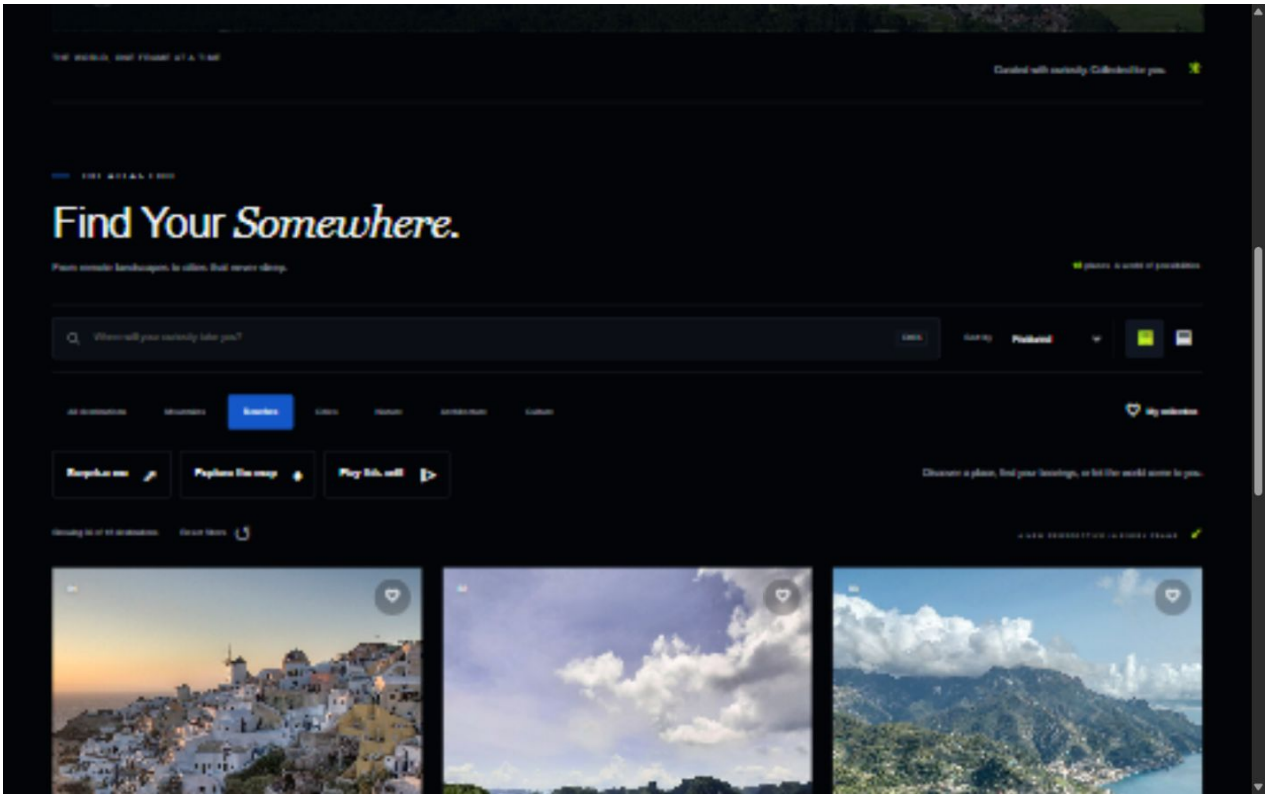


Figure 4.24: Beaches: active category and matching gallery [Visitor]

The selected category, live result count and eligible photographs.

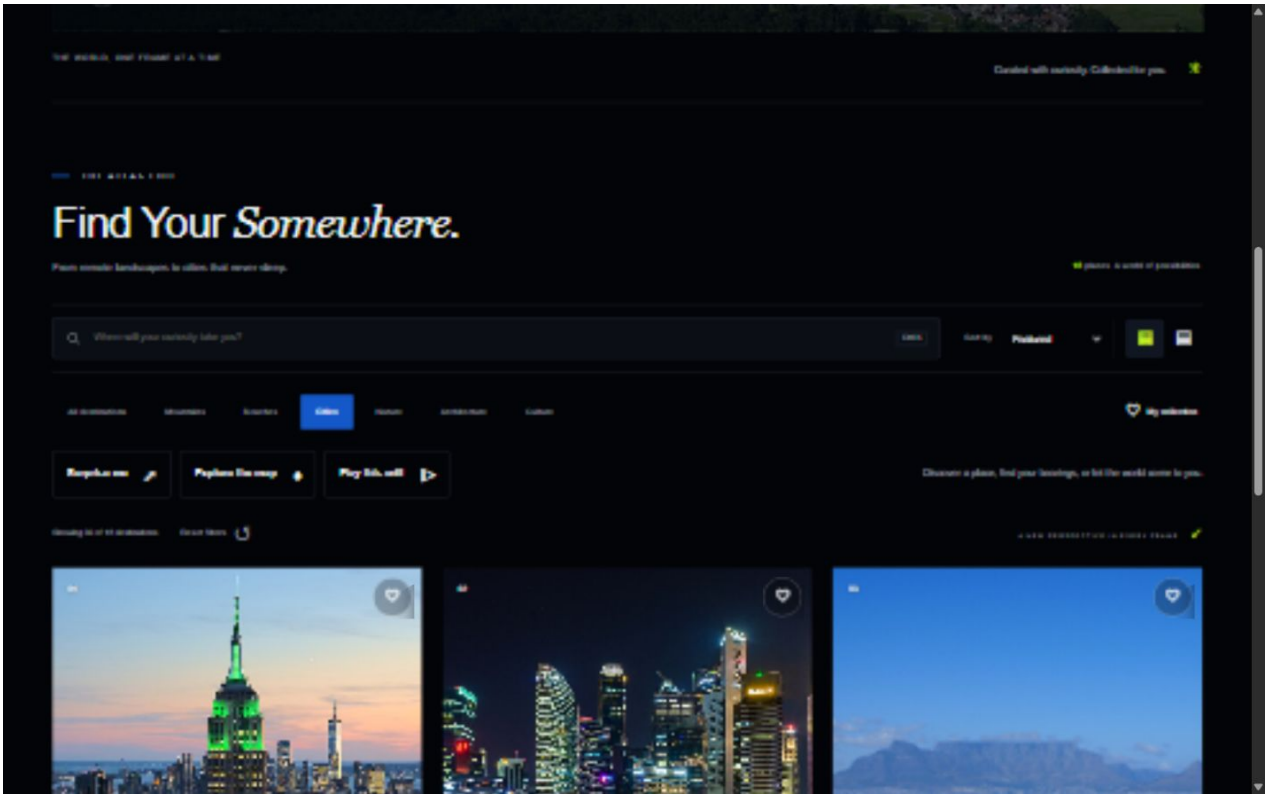


Figure 4.25: Cities: active category and matching gallery [Visitor]

The selected category, live result count and eligible photographs.

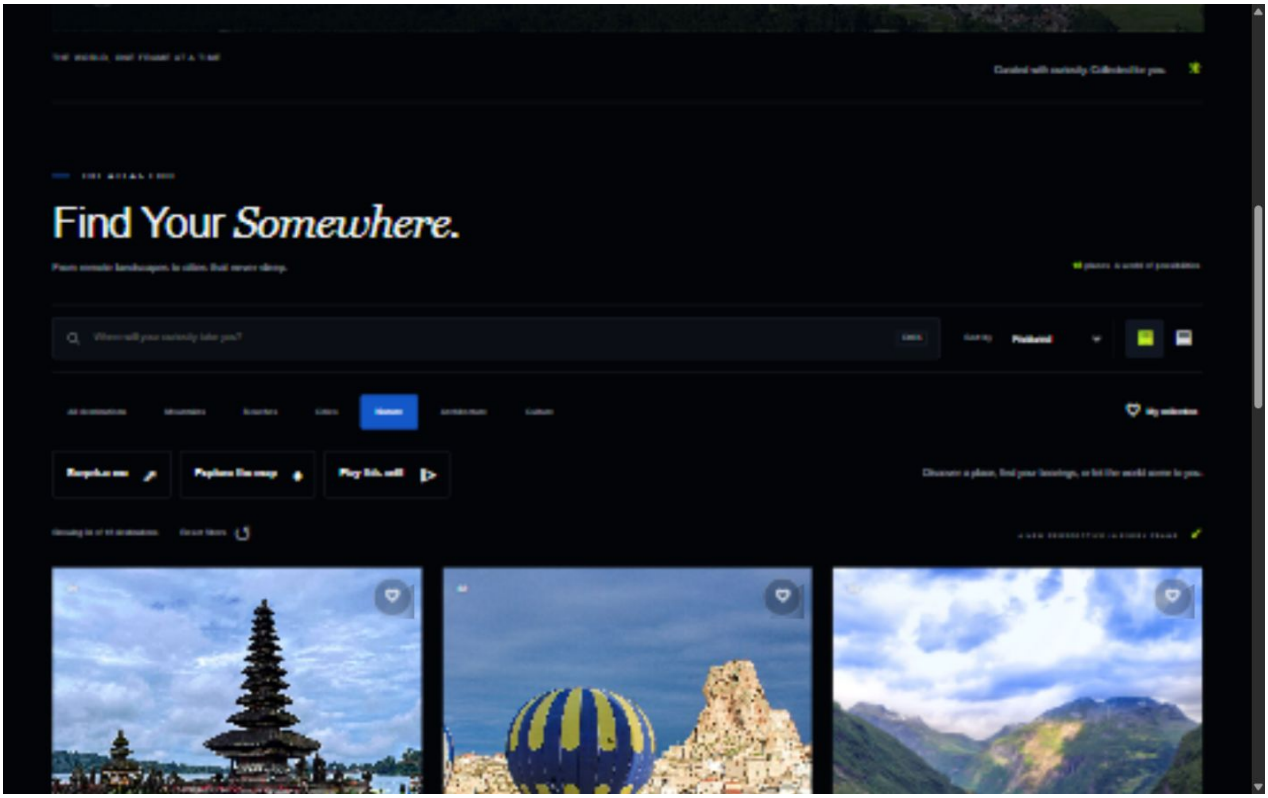


Figure 4.26: Nature: active category and matching gallery [Visitor]

The selected category, live result count and eligible photographs.

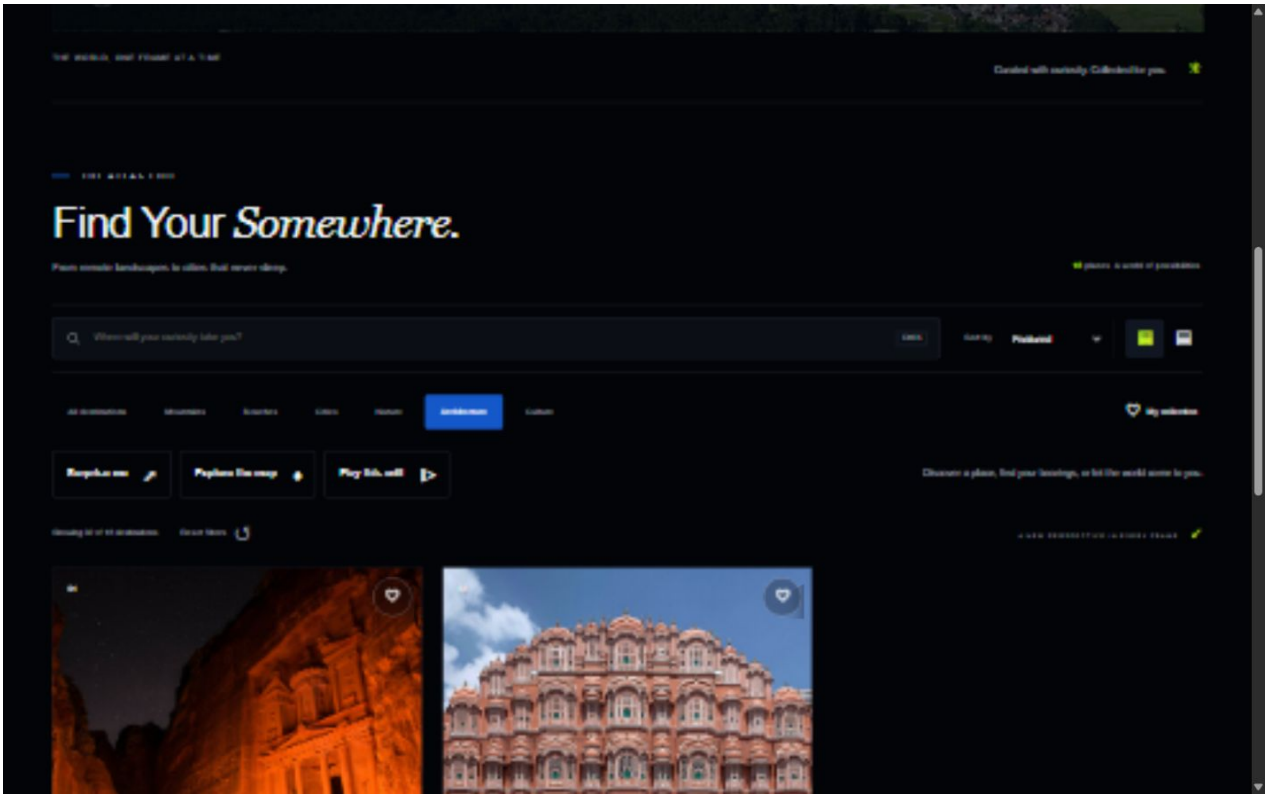


Figure 4.27: Architecture: active category and matching gallery [Visitor]

The selected category, live result count and eligible photographs.

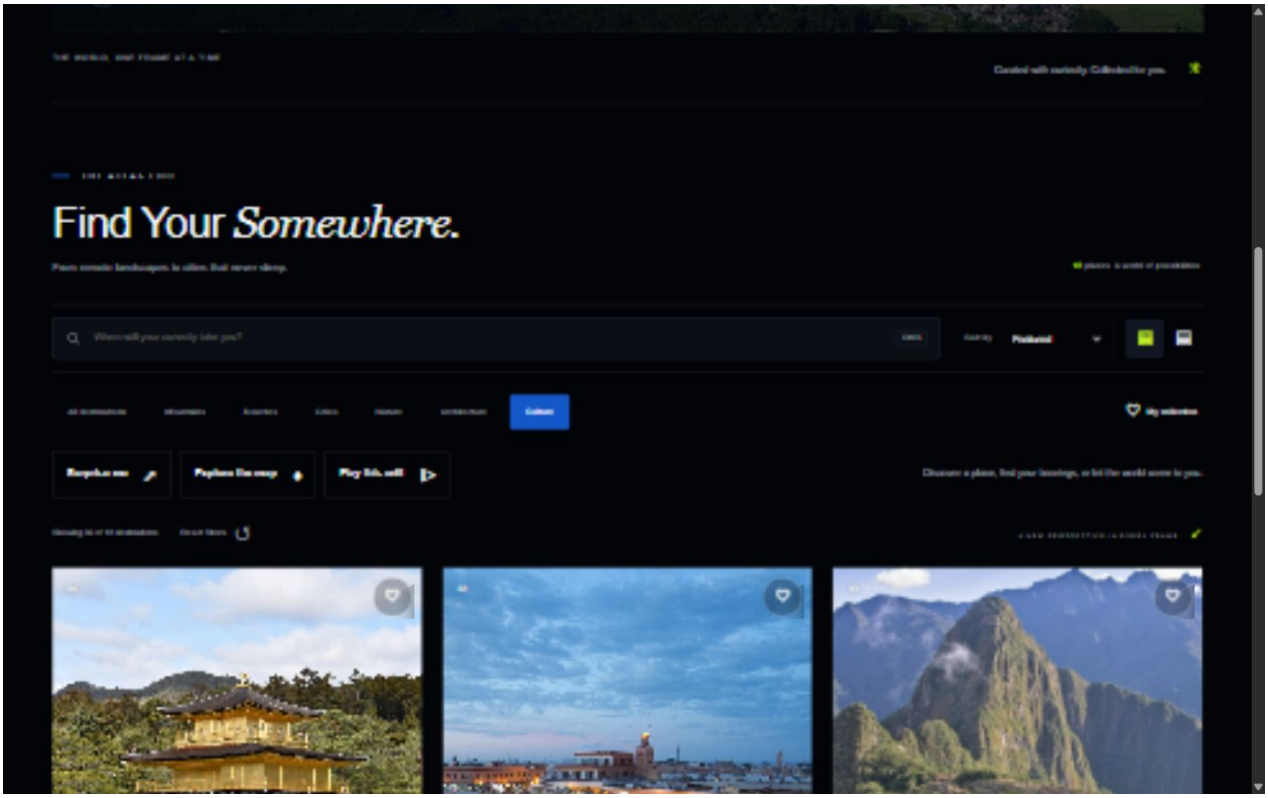


Figure 4.28: Culture: active category and matching gallery [Visitor]

The selected category, live result count and eligible photographs.

4.5 Sorting and Gallery Layout Controls

Featured, destination A-Z and Z-A operate on the eligible collection. Large and compact desktop layouts share the same source records and use responsive grids; layout does not alter filtering or favorites.

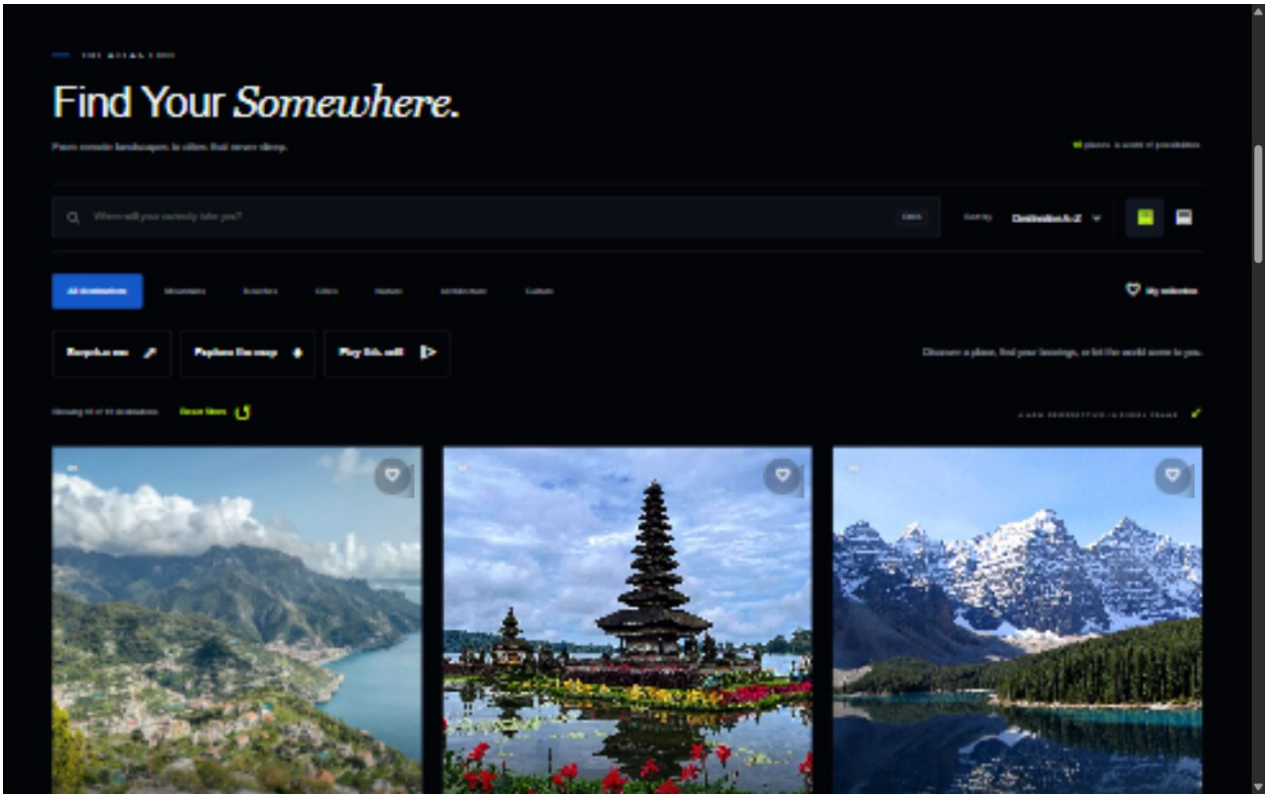


Figure 4.29: Destination sorting: A-Z [Visitor]

Alphabetical ordering of the eligible edit.

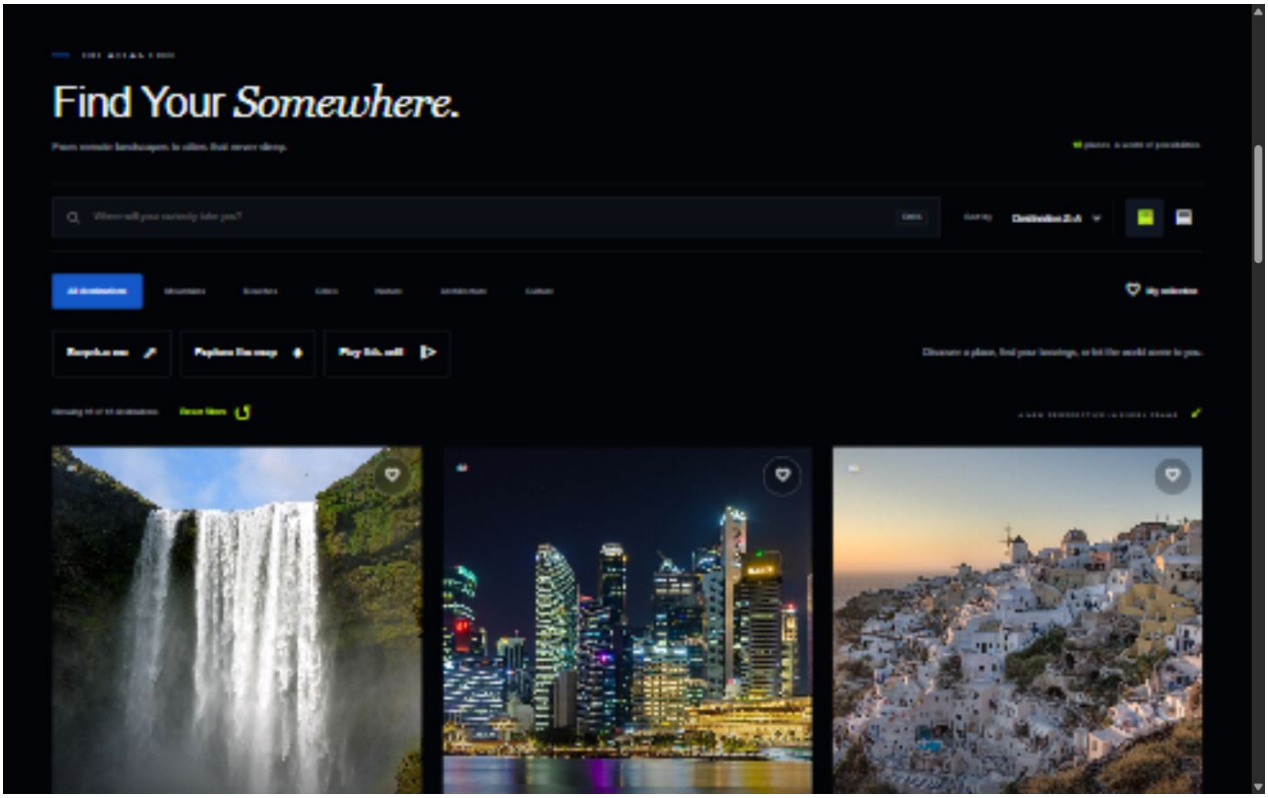


Figure 4.30: Destination sorting: Z-A [Visitor]

Reverse alphabetical ordering without changing filters.

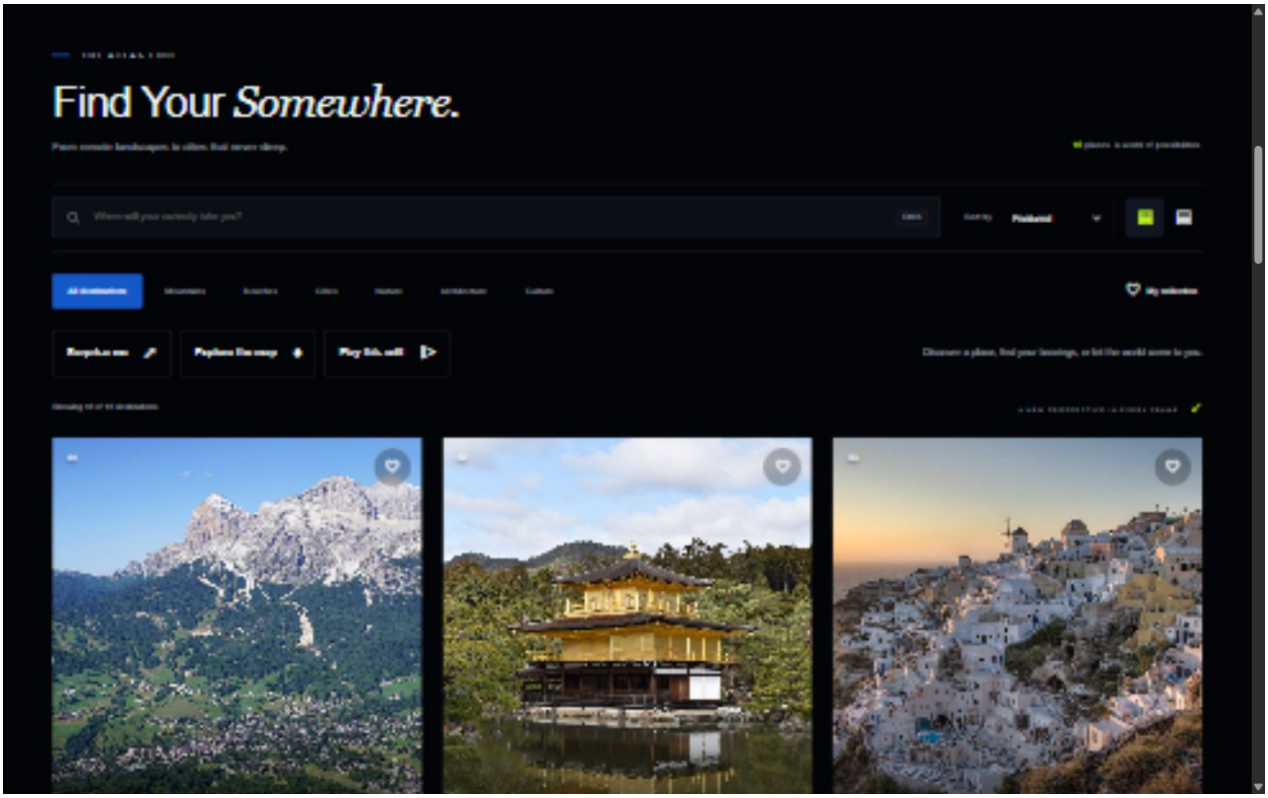


Figure 4.31: Featured ordering restored [Visitor]

The three featured destinations return to the front of the edit.

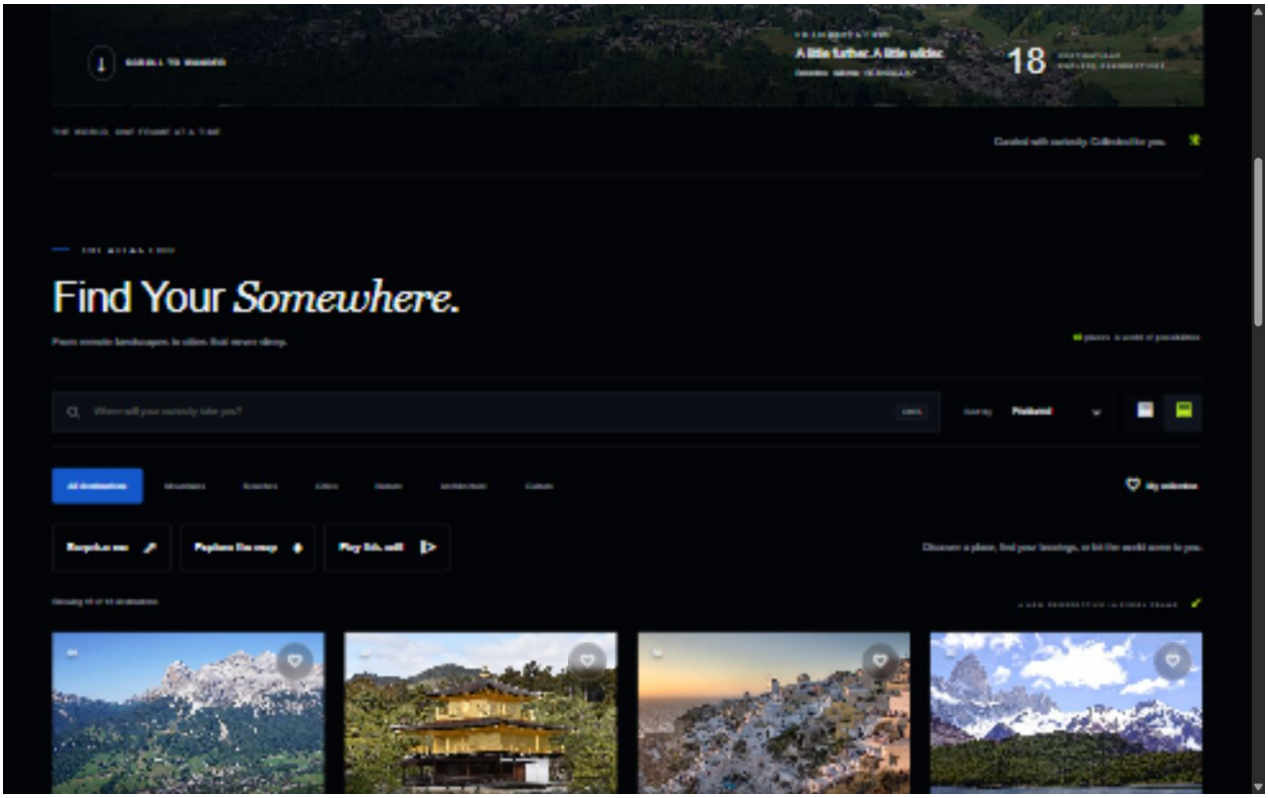


Figure 4.32: Compact desktop gallery layout [Visitor]

A denser grid uses the same destination records.

4.6 Interactive Photograph Viewer

The shared native dialog displays the selected image, country, category, description, creator credit and collection position. Manual arrows stop at the filtered boundaries. Favorite controls remain independent, the background is inert and Escape restores the opening trigger.

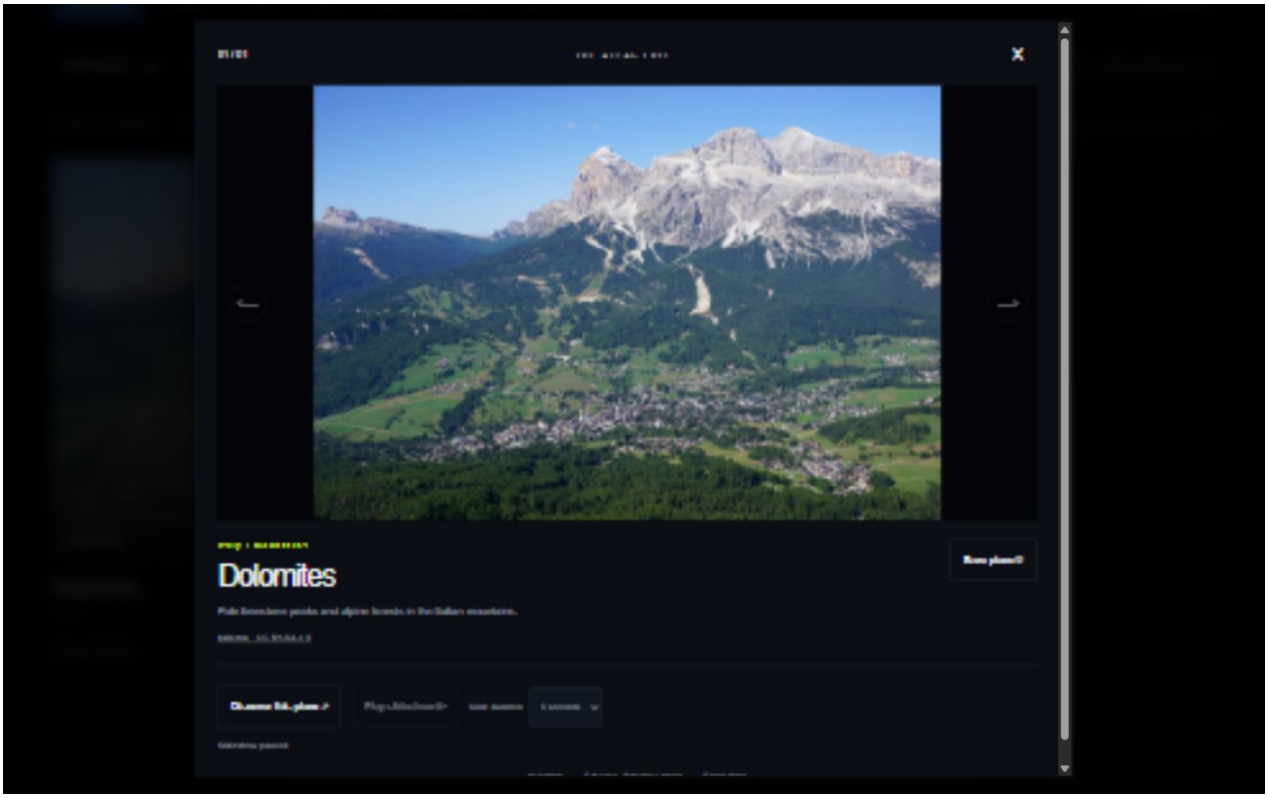


Figure 4.33: Dolomites: photograph and destination metadata [Visitor]

Native modal viewer with image credit, saving and navigation controls.

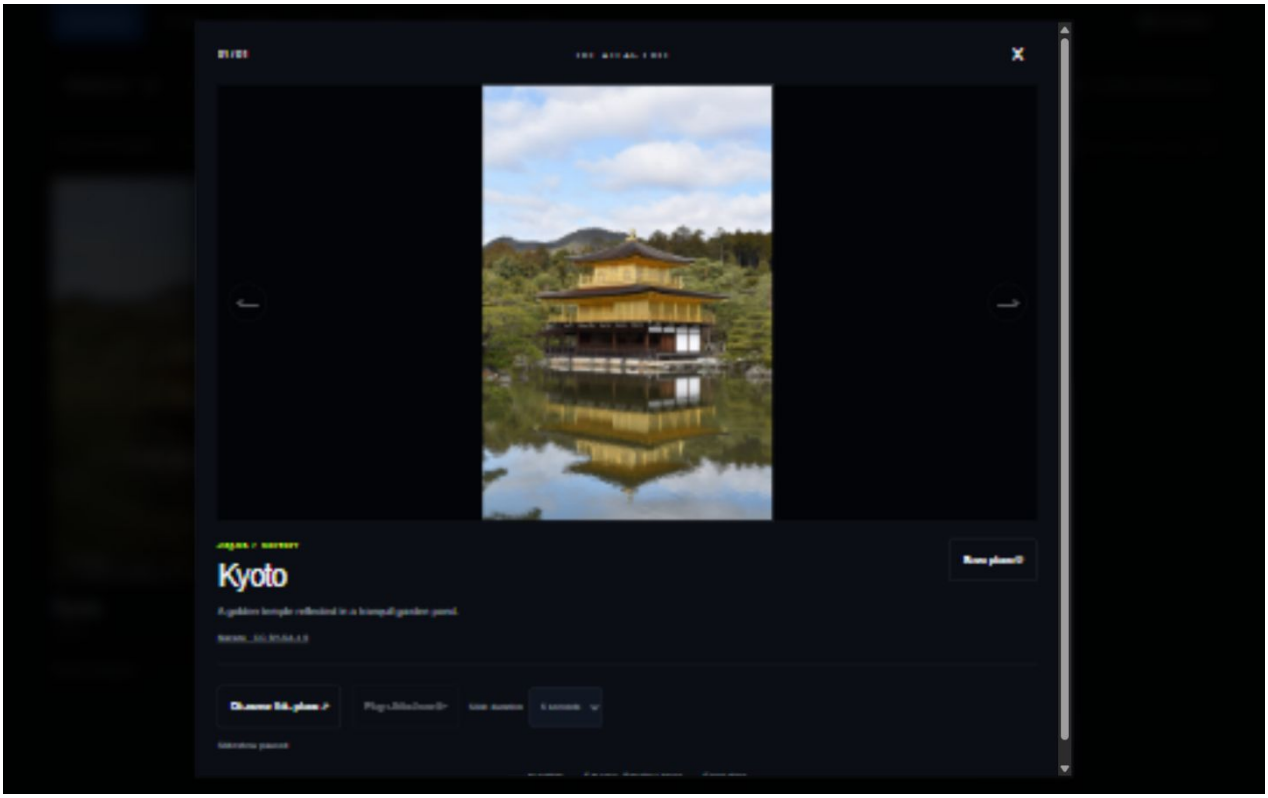


Figure 4.34: Kyoto: photograph and destination metadata [Visitor]

Native modal viewer with image credit, saving and navigation controls.

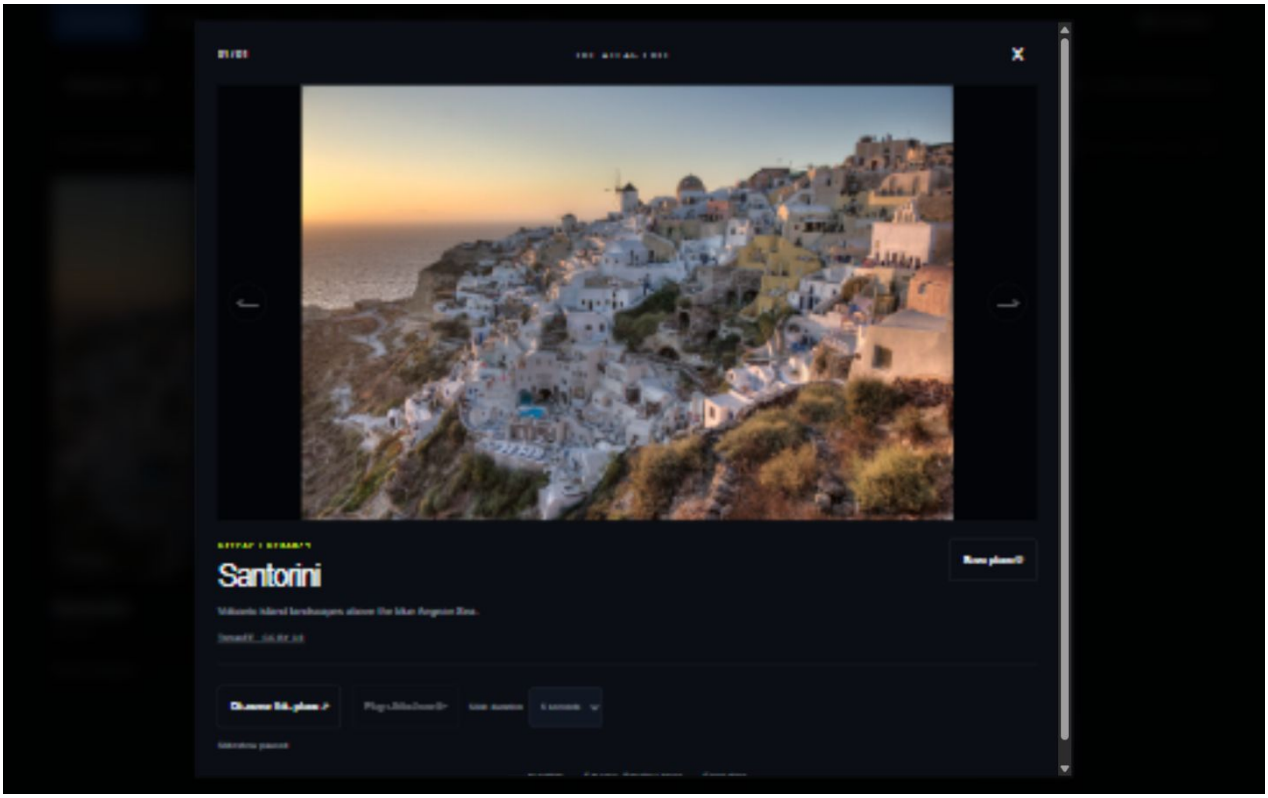


Figure 4.35: Santorini: photograph and destination metadata [Visitor]

Native modal viewer with image credit, saving and navigation controls.



Figure 4.36: Patagonia: photograph and destination metadata [Visitor]

Native modal viewer with image credit, saving and navigation controls.

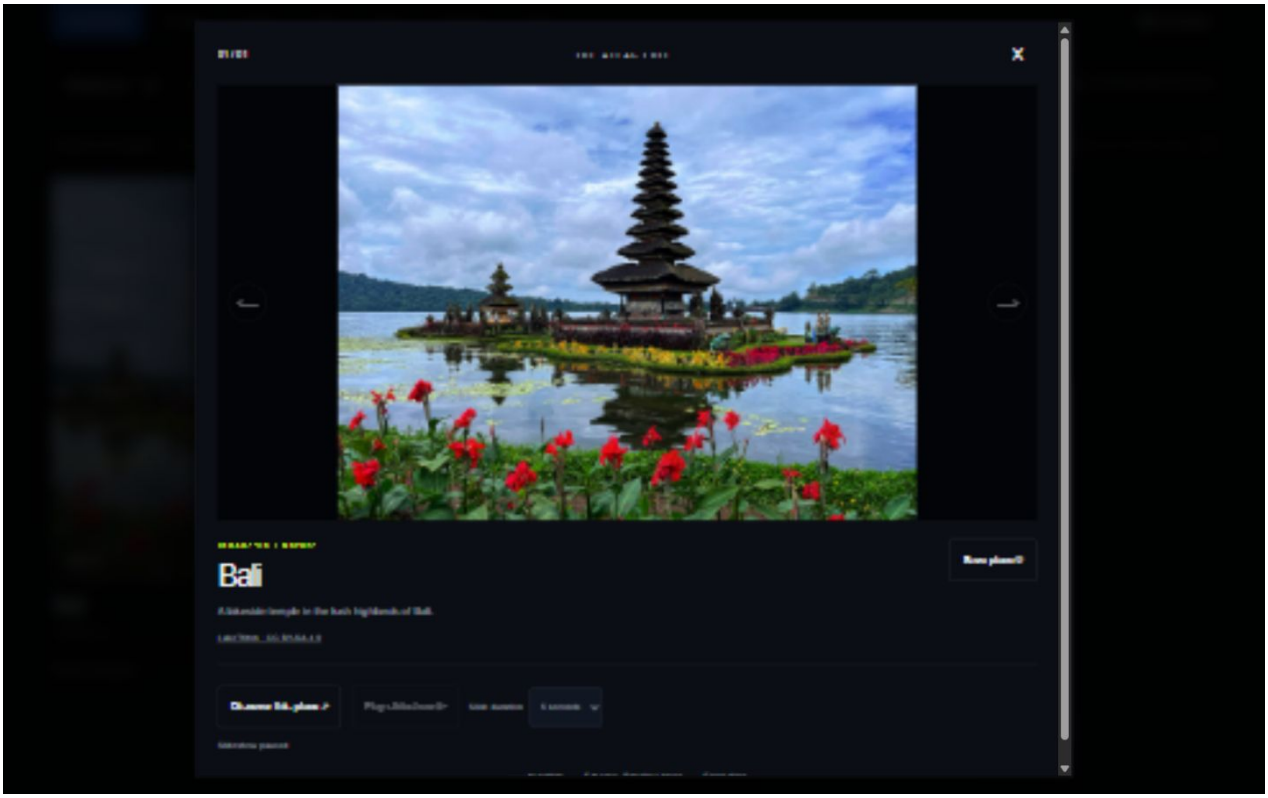


Figure 4.37: Bali: photograph and destination metadata [Visitor]

Native modal viewer with image credit, saving and navigation controls.



Figure 4.38: New York City: photograph and destination metadata [Visitor]

Native modal viewer with image credit, saving and navigation controls.

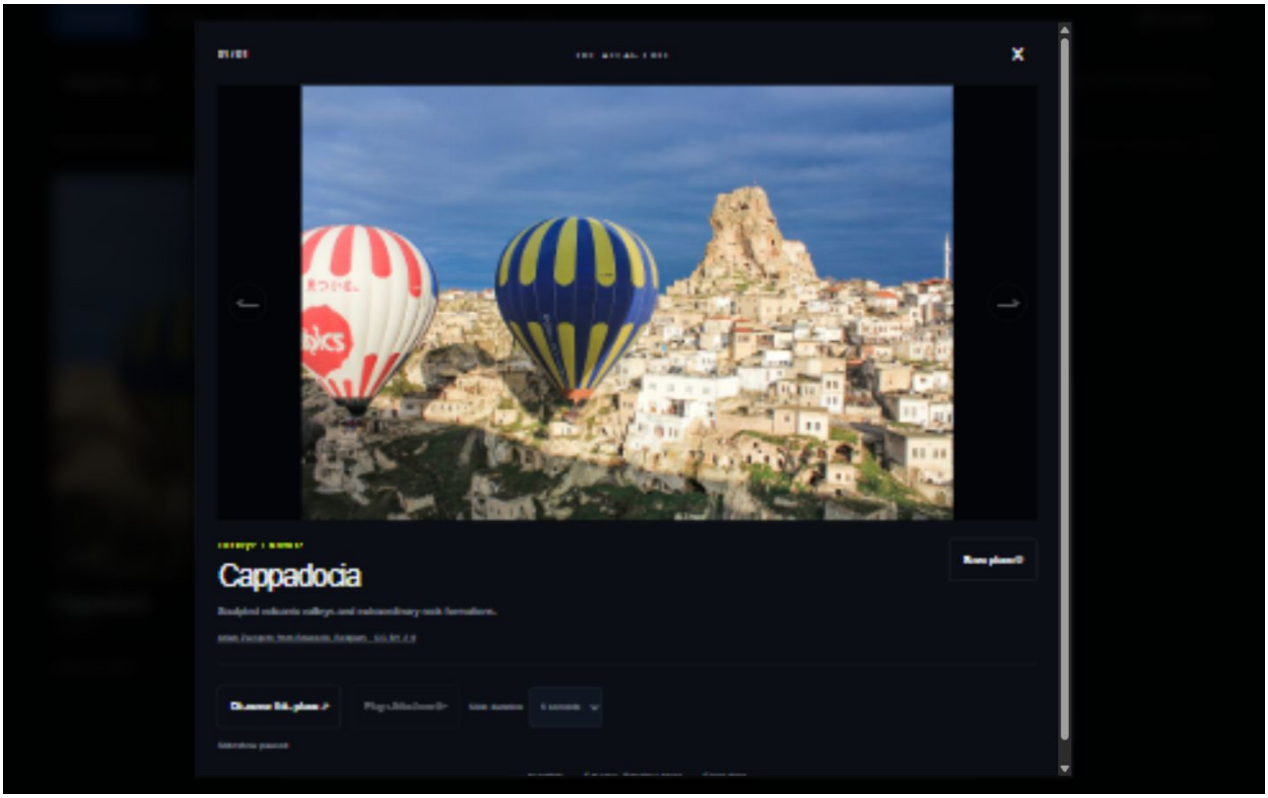


Figure 4.39: Cappadocia: photograph and destination metadata [Visitor]

A source-credited photograph in the native viewer.

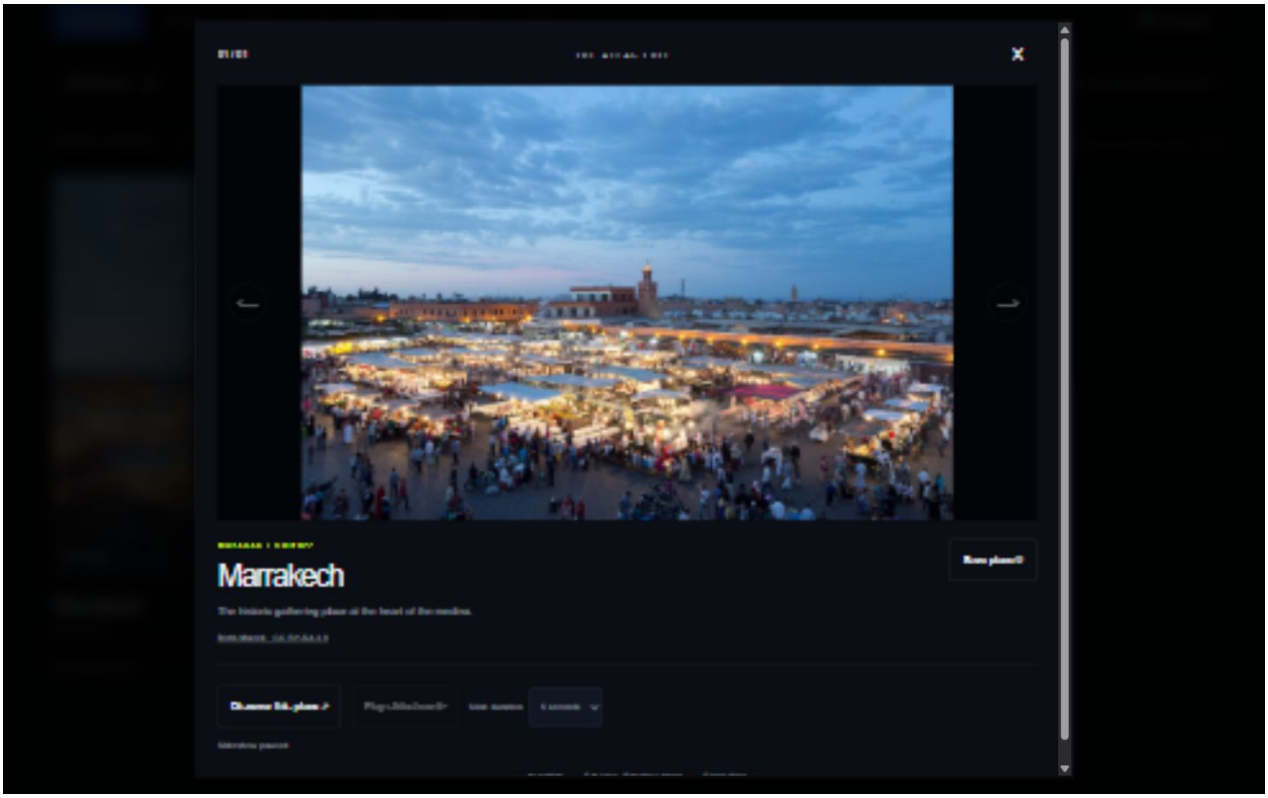


Figure 4.40: Marrakech: photograph and destination metadata [Visitor]

A source-credited photograph in the native viewer.

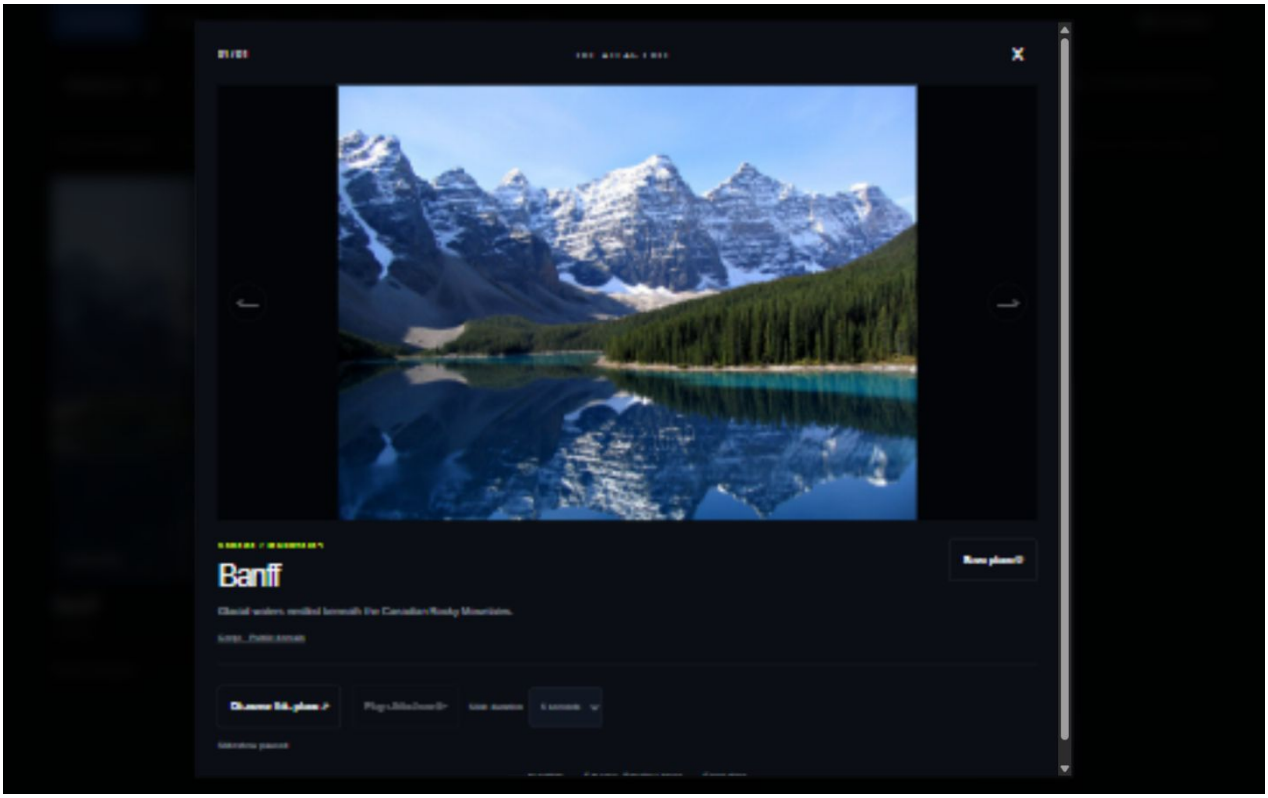


Figure 4.41: Banff: photograph and destination metadata [Visitor]

A source-credited photograph in the native viewer.

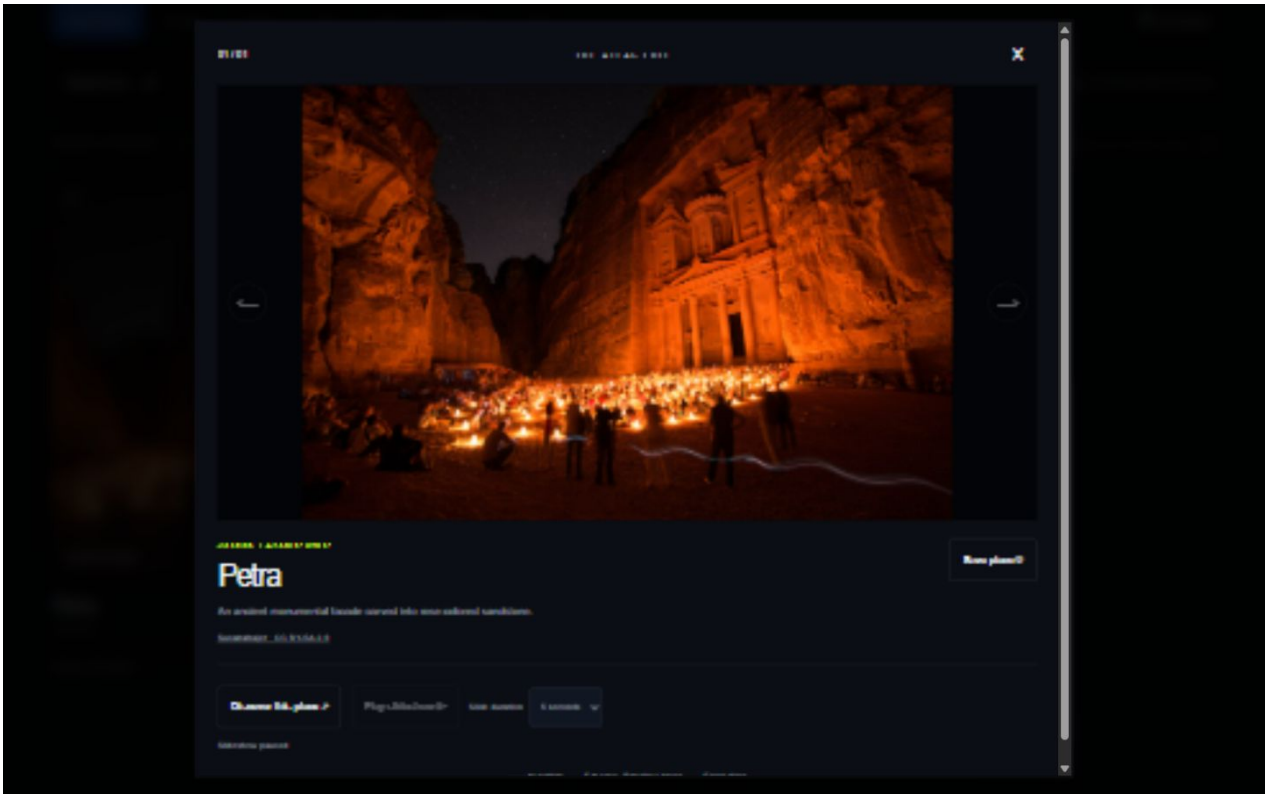


Figure 4.42: Petra: photograph and destination metadata [Visitor]

A source-credited photograph in the native viewer.

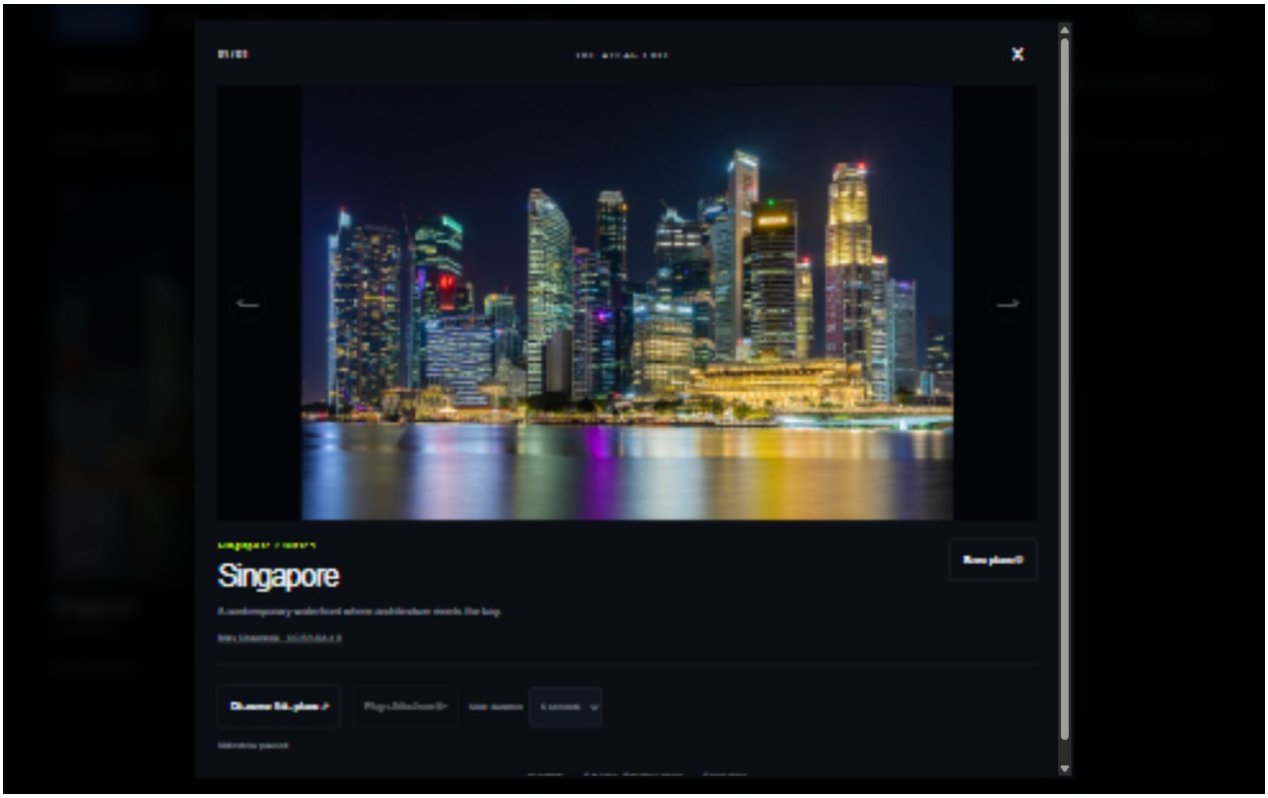


Figure 4.43: Singapore: photograph and destination metadata [Visitor]

A source-credited photograph in the native viewer.

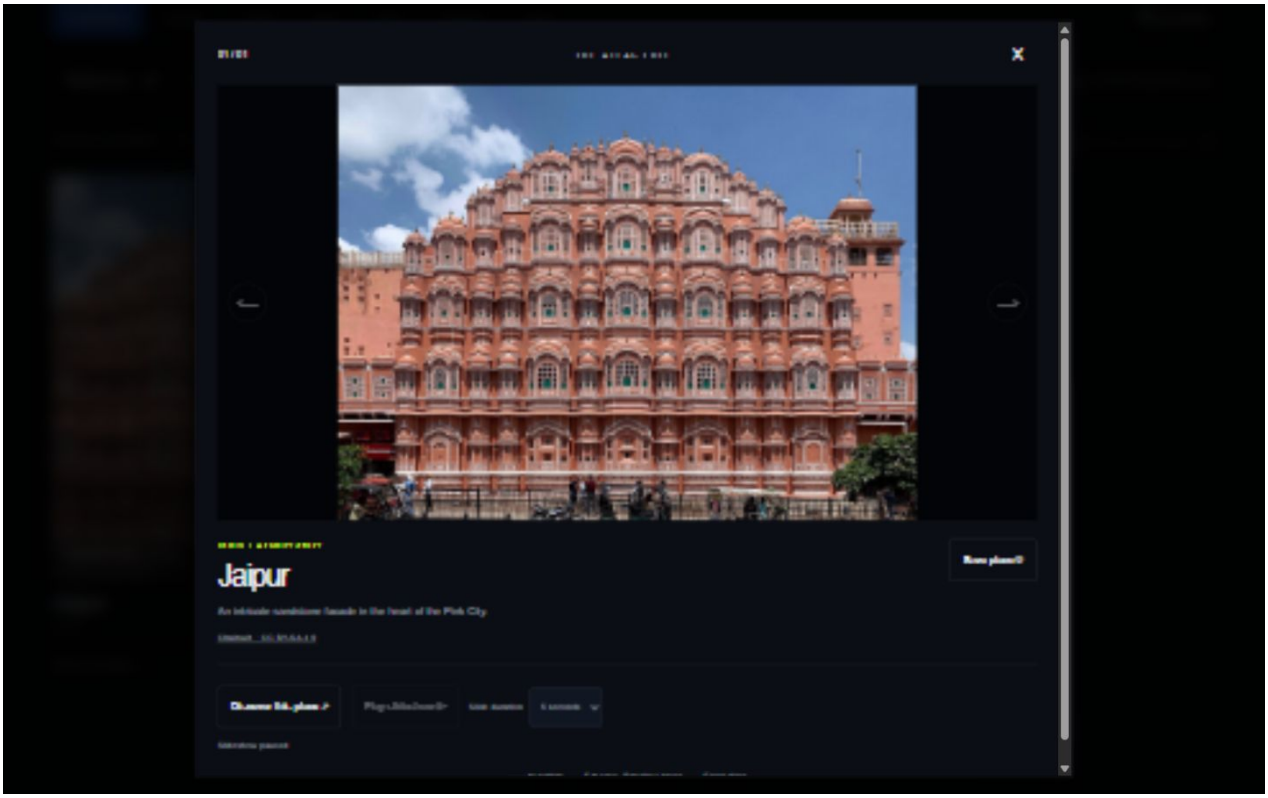


Figure 4.44: Jaipur: photograph and destination metadata [Visitor]

A source-credited photograph in the native viewer.

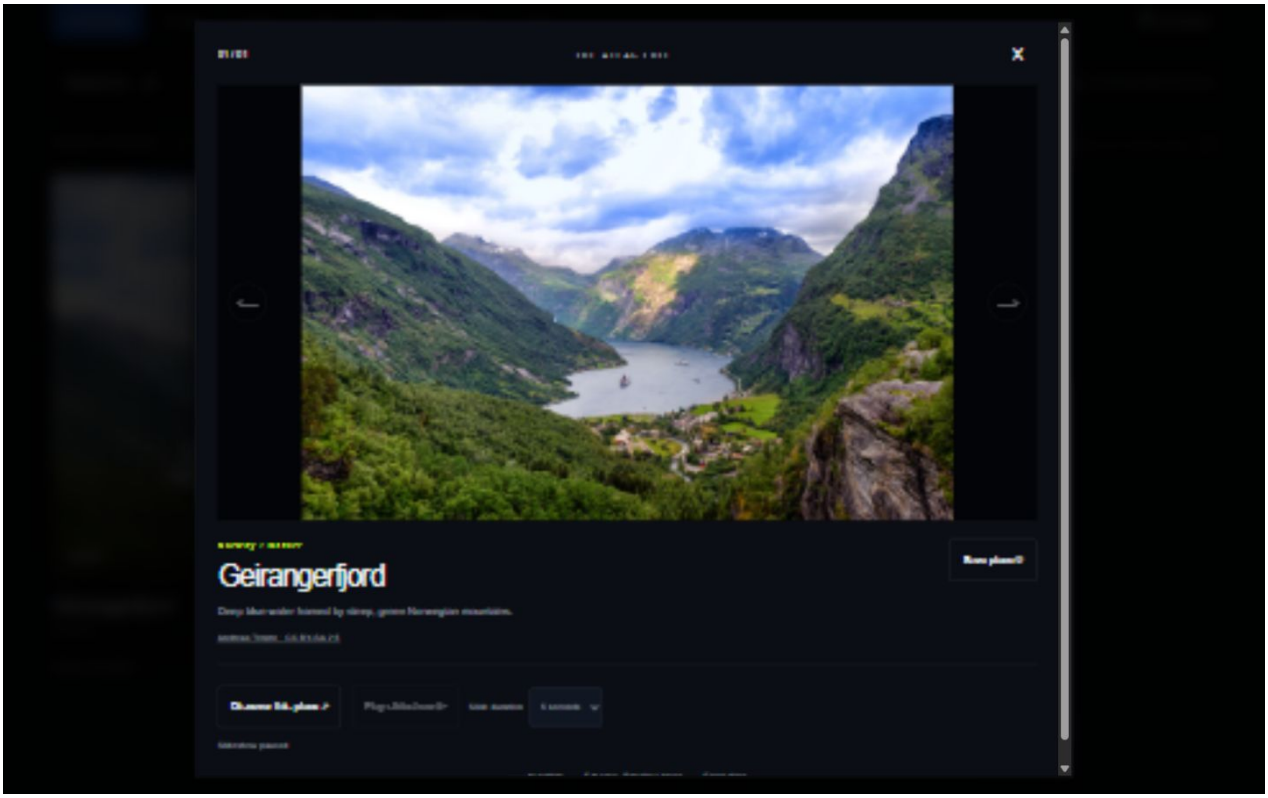


Figure 4.45: Geirangerfjord: photograph and destination metadata [Visitor]

Photography, location, category and creator attribution.

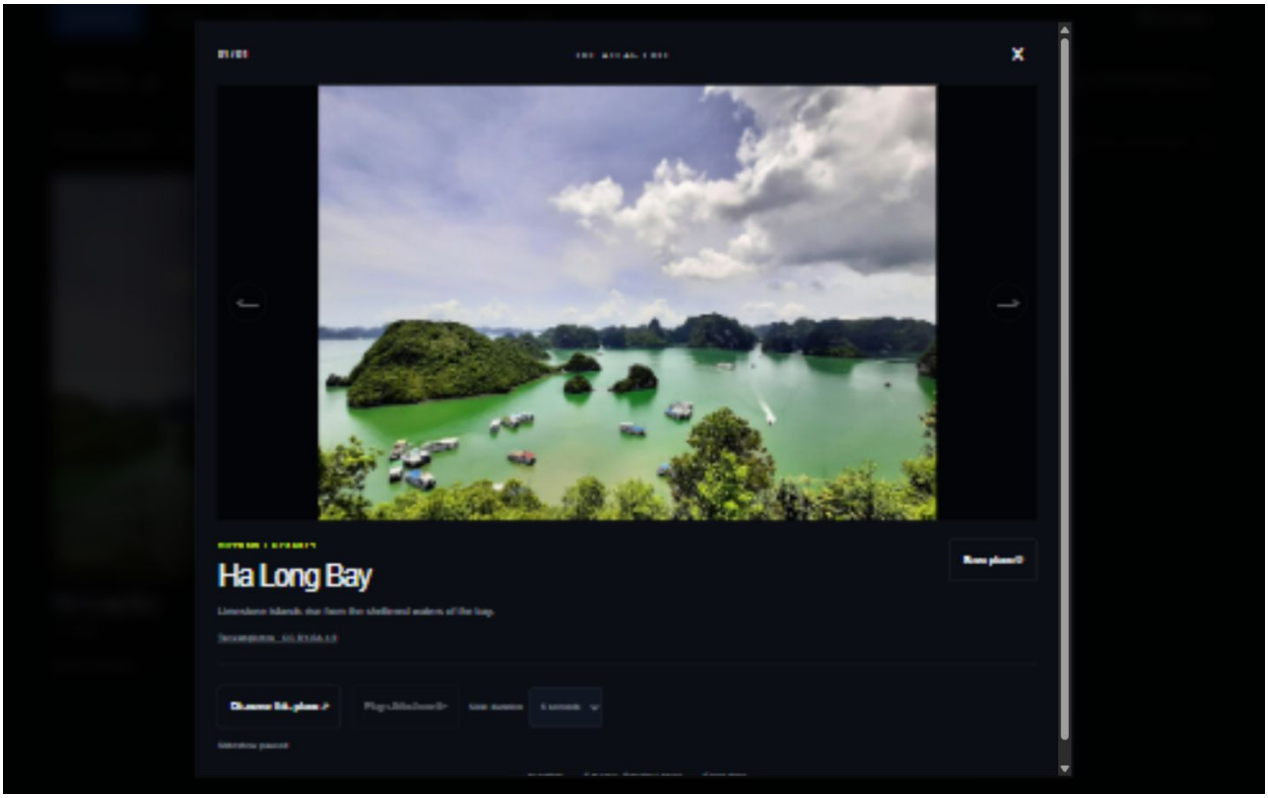


Figure 4.46: Ha Long Bay: photograph and destination metadata [Visitor]

Photography, location, category and creator attribution.

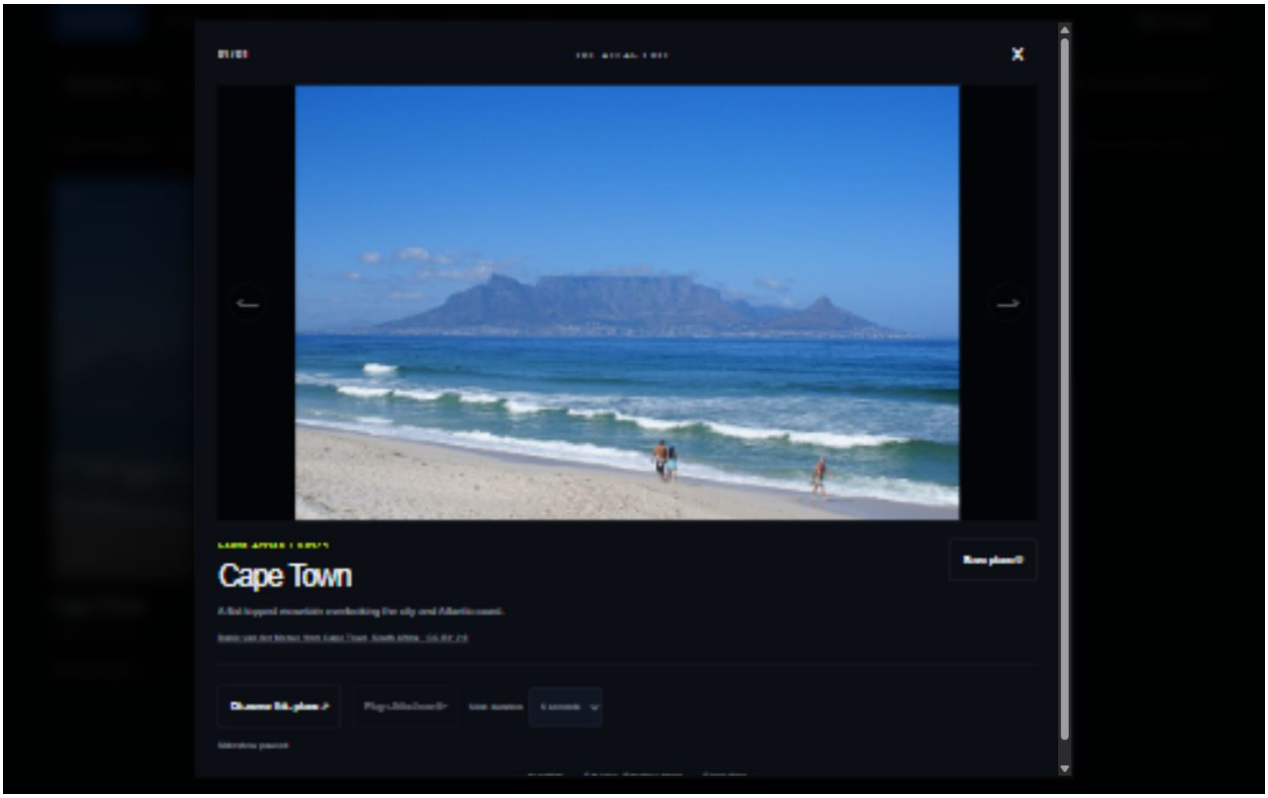


Figure 4.47: Cape Town: photograph and destination metadata [Visitor]

Photography, location, category and creator attribution.

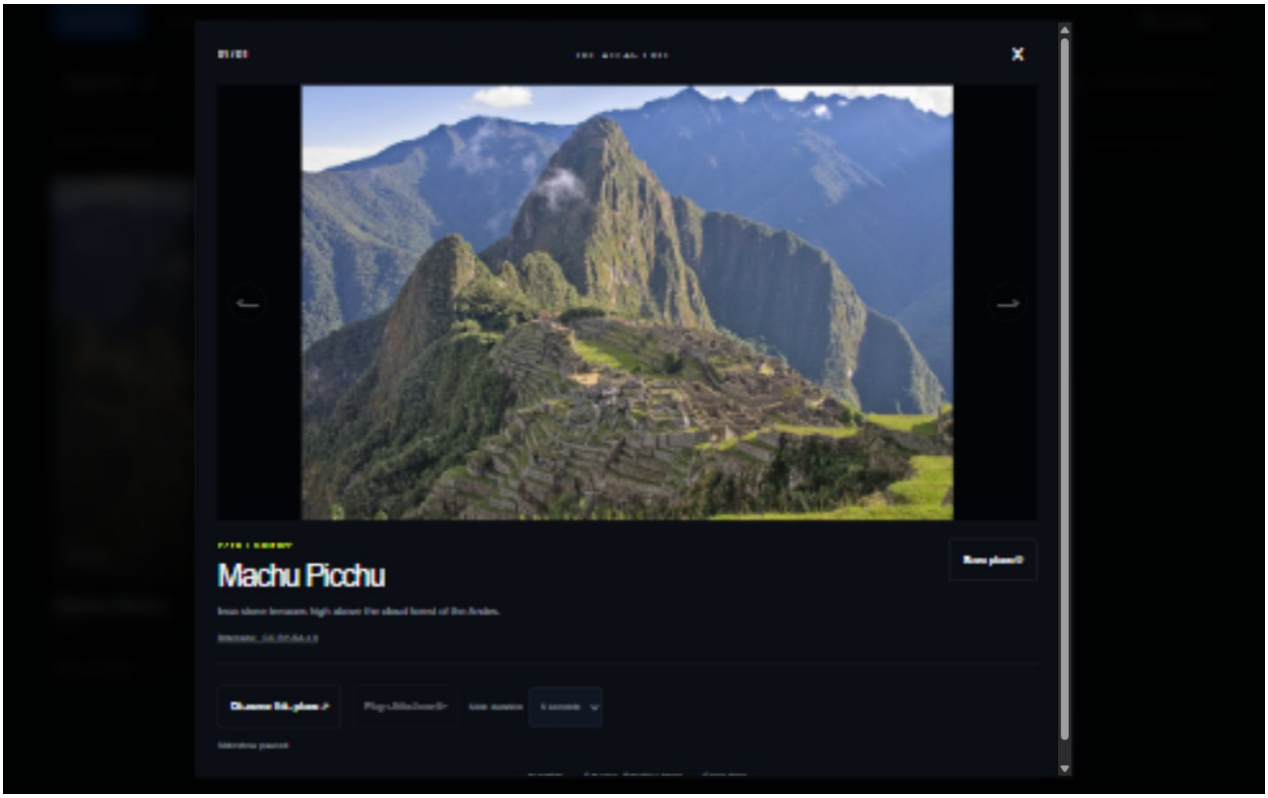


Figure 4.48: Machu Picchu: photograph and destination metadata [Visitor]

Photography, location, category and creator attribution.

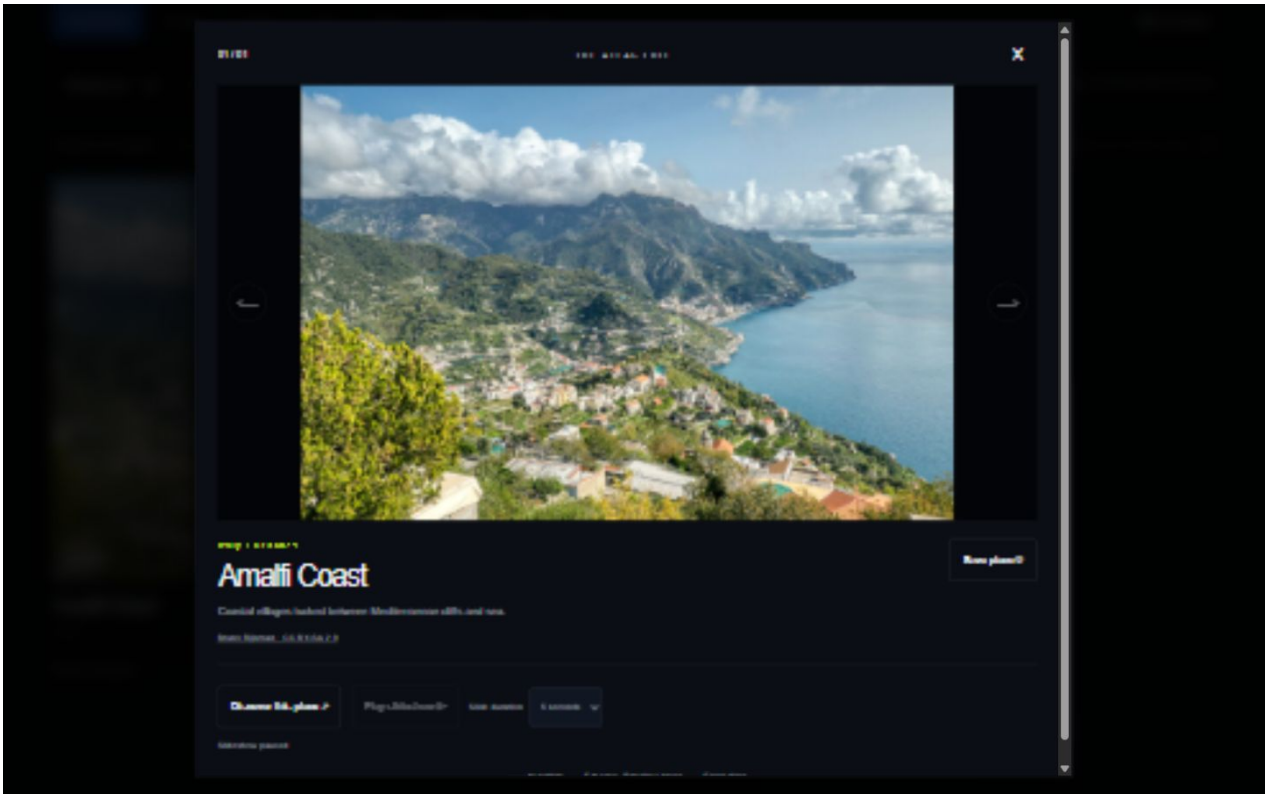


Figure 4.49: Amalfi Coast: photograph and destination metadata [Visitor]

Photography, location, category and creator attribution.

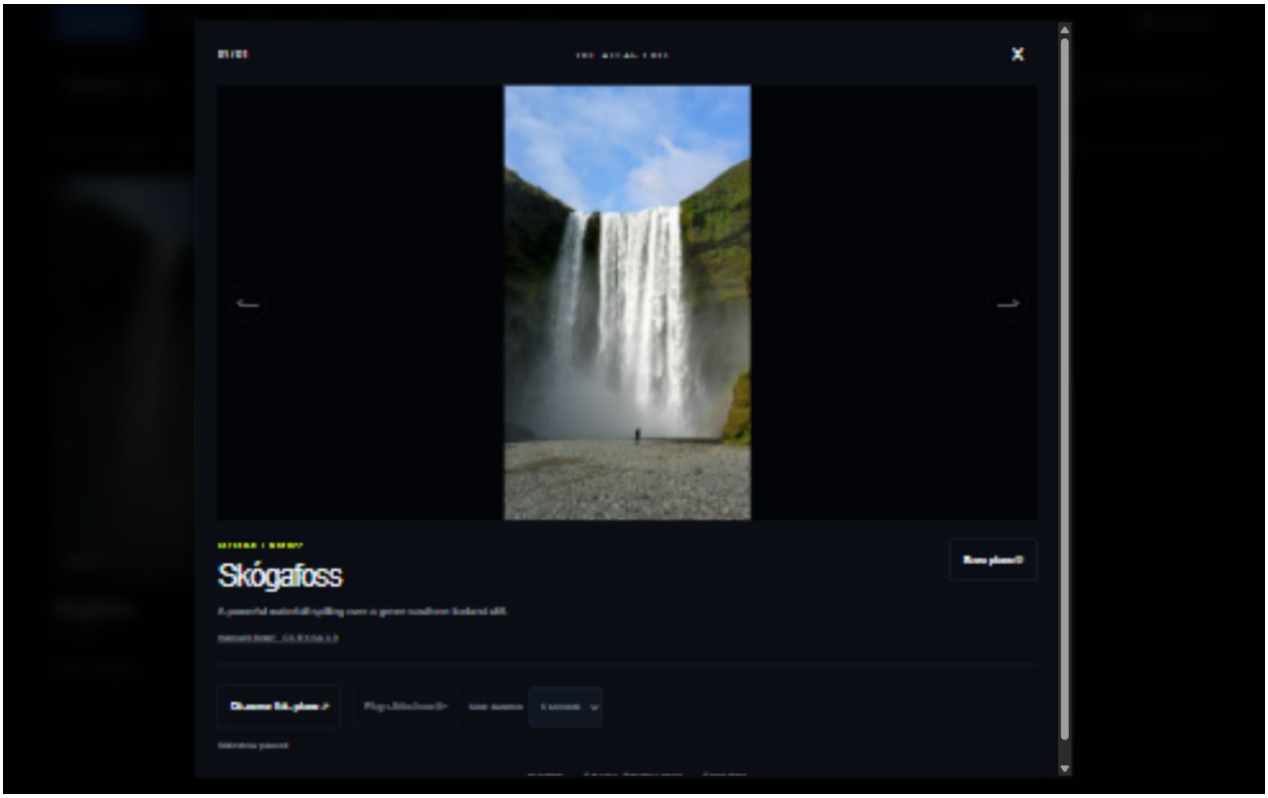


Figure 4.50: Skógafoss: photograph and destination metadata [Visitor]

Photography, location, category and creator attribution.

4.7 Detailed Destination Guides

Each place has an original overview, three highlights, three interests, cultural context, a planning suggestion and geographic/map references. The guide lives inside the viewer, can be collapsed, and pauses slideshow playback. Tips are general guidance rather than live schedules or travel advisories.

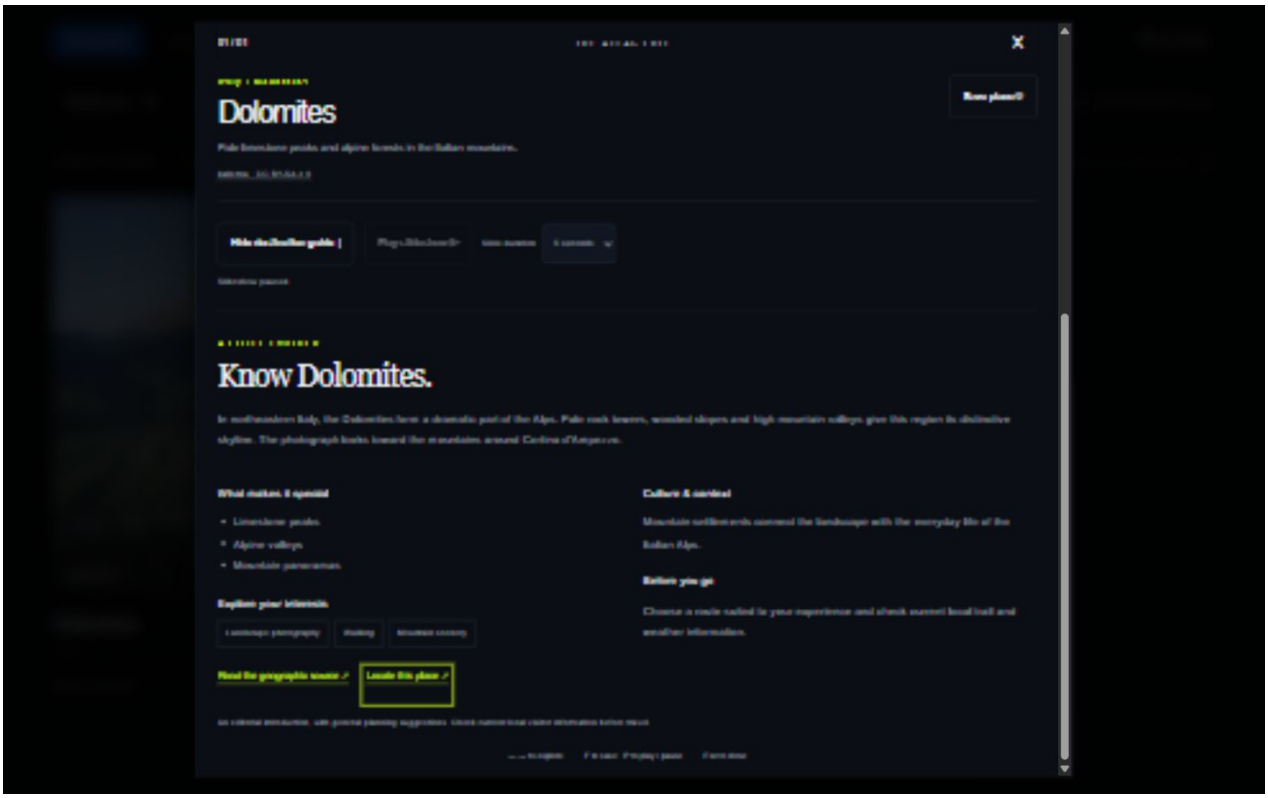


Figure 4.51: Dolomites: detailed destination guide [Visitor]

Overview, highlights, interests, culture, planning note and source links.

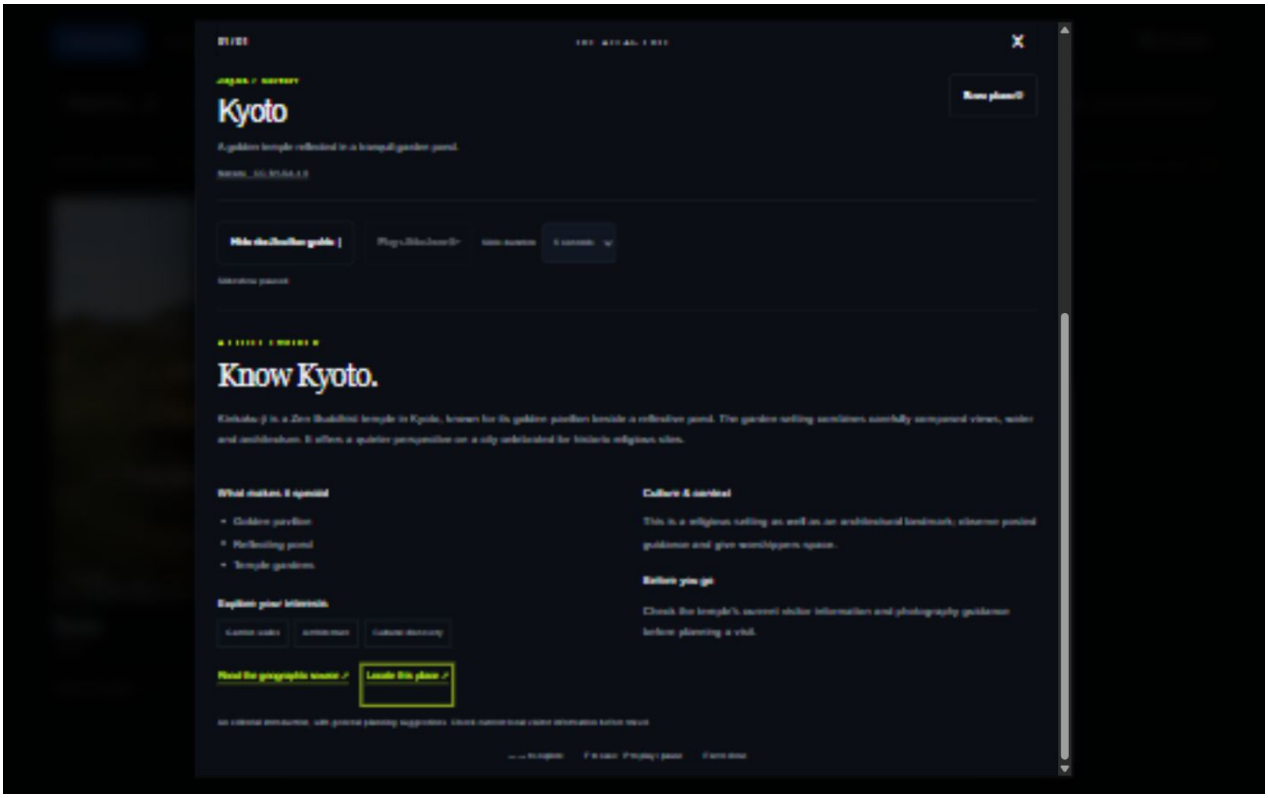


Figure 4.52: Kyoto: detailed destination guide [Visitor]

Overview, highlights, interests, culture, planning note and source links.

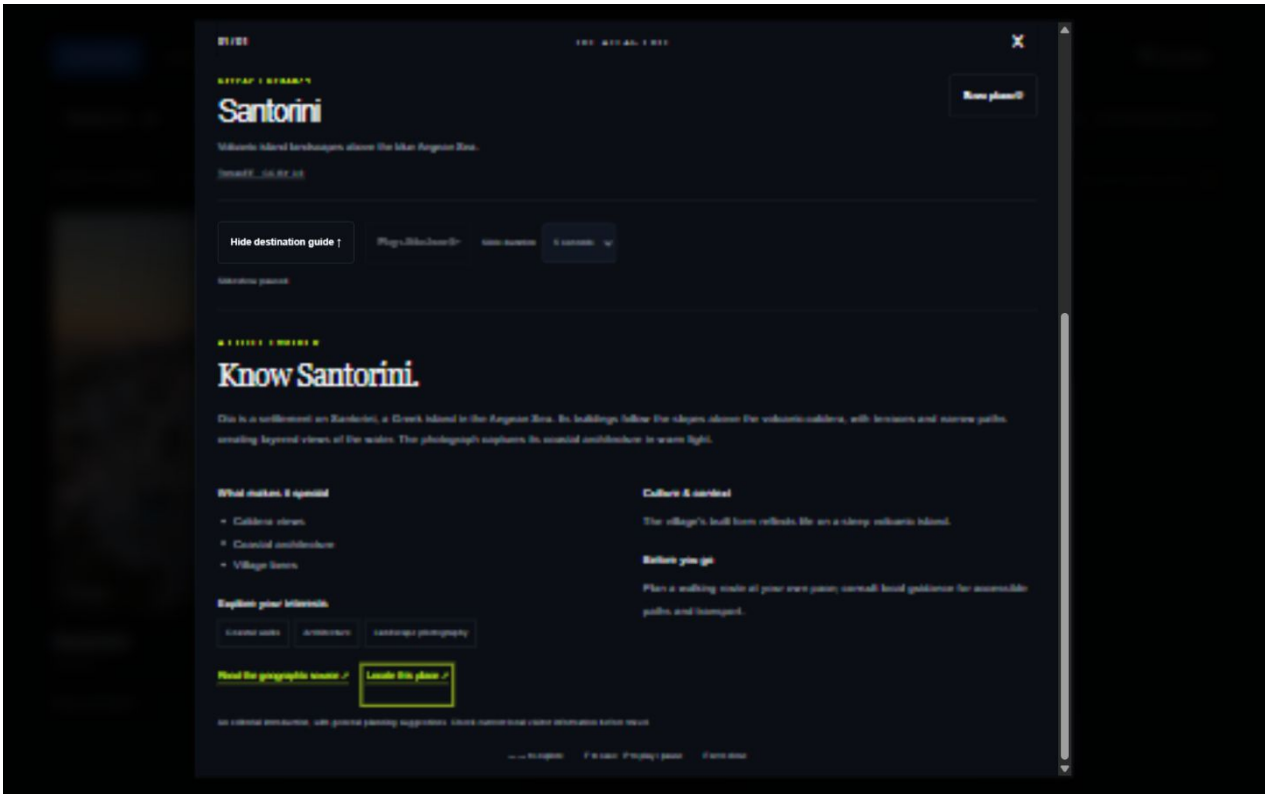


Figure 4.53: Santorini: detailed destination guide [Visitor]

Overview, highlights, interests, culture, planning note and source links.

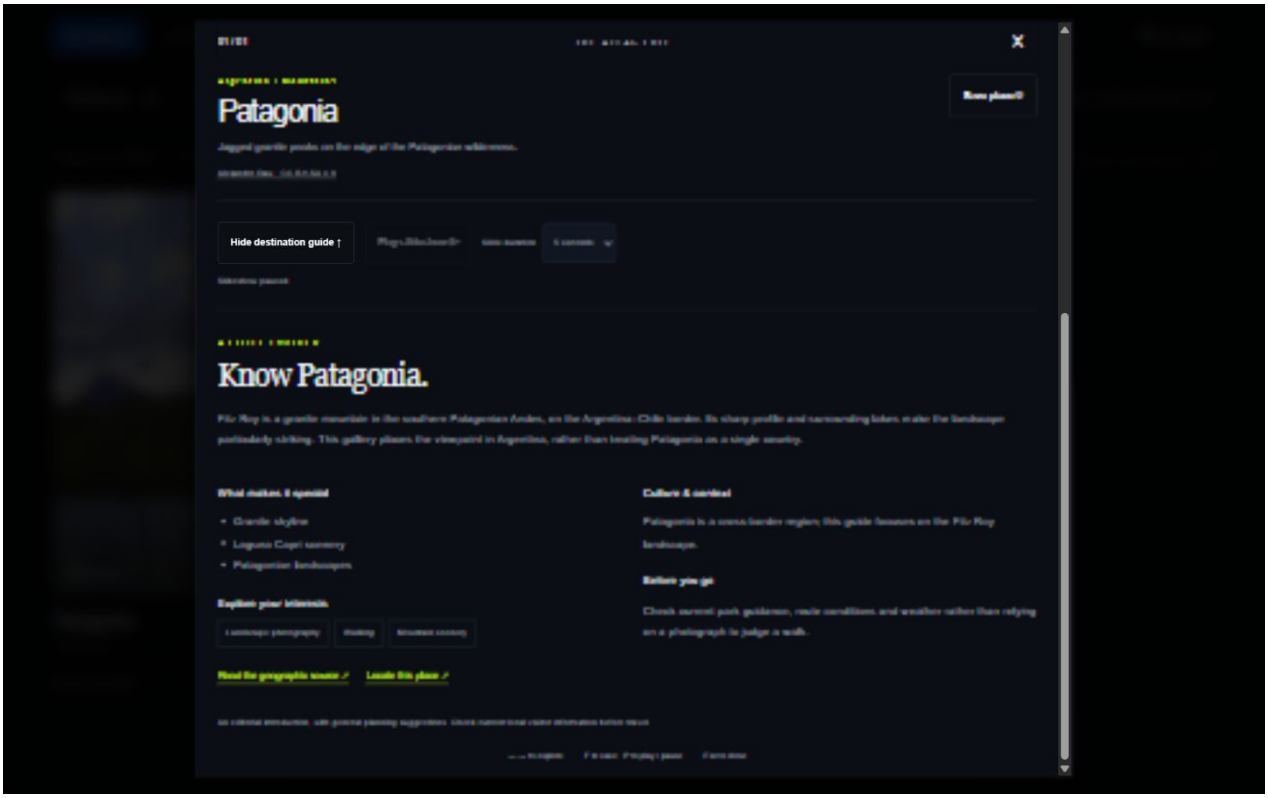


Figure 4.54: Patagonia: detailed destination guide [Visitor]

Overview, highlights, interests, culture, planning note and source links.

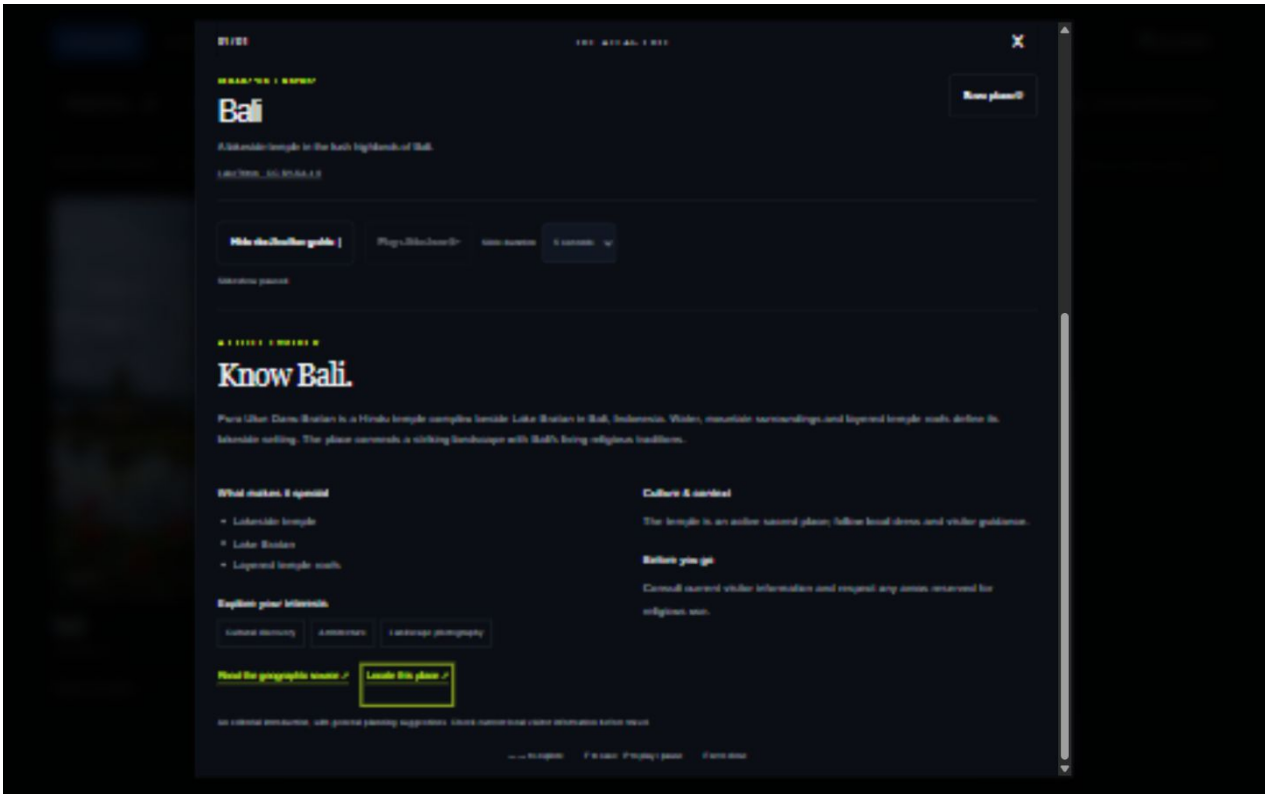


Figure 4.55: Bali: detailed destination guide [Visitor]

Overview, highlights, interests, culture, planning note and source links.

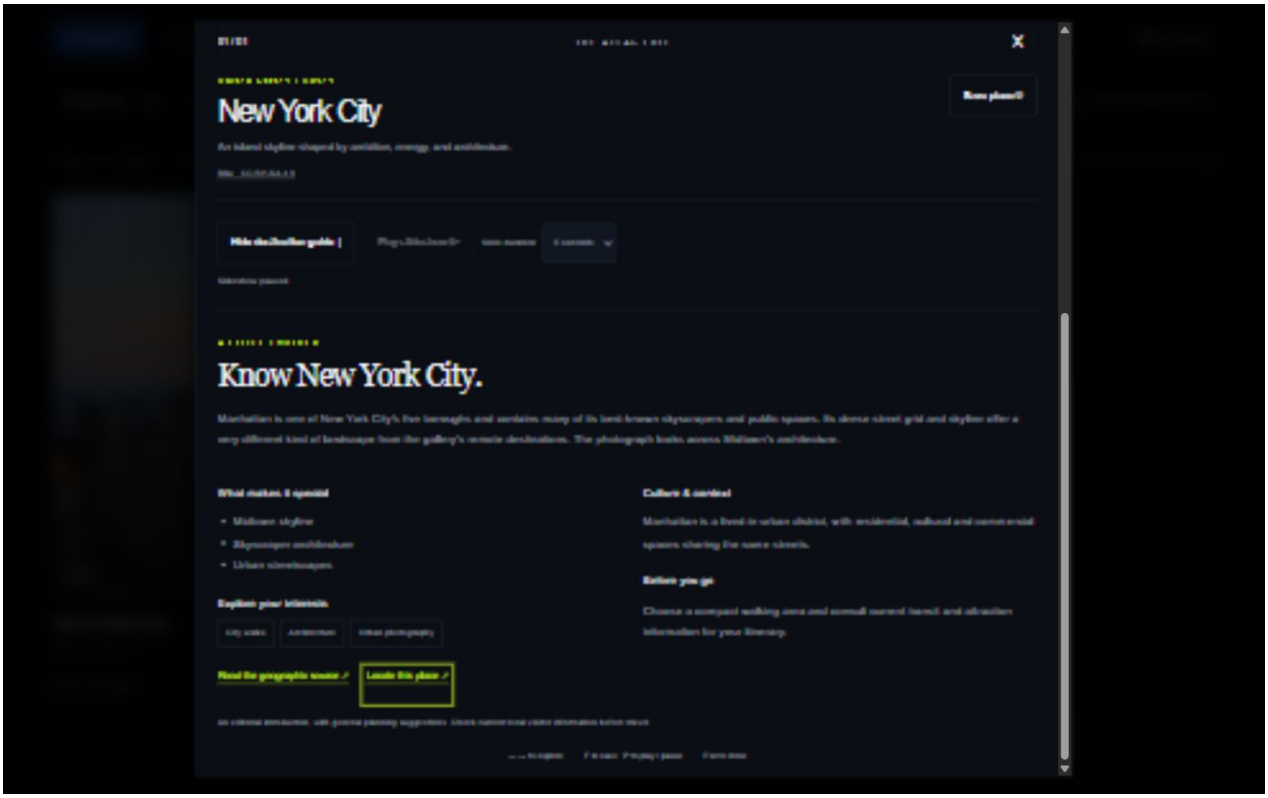


Figure 4.56: New York City: detailed destination guide [Visitor]

Overview, highlights, interests, culture, planning note and source links.

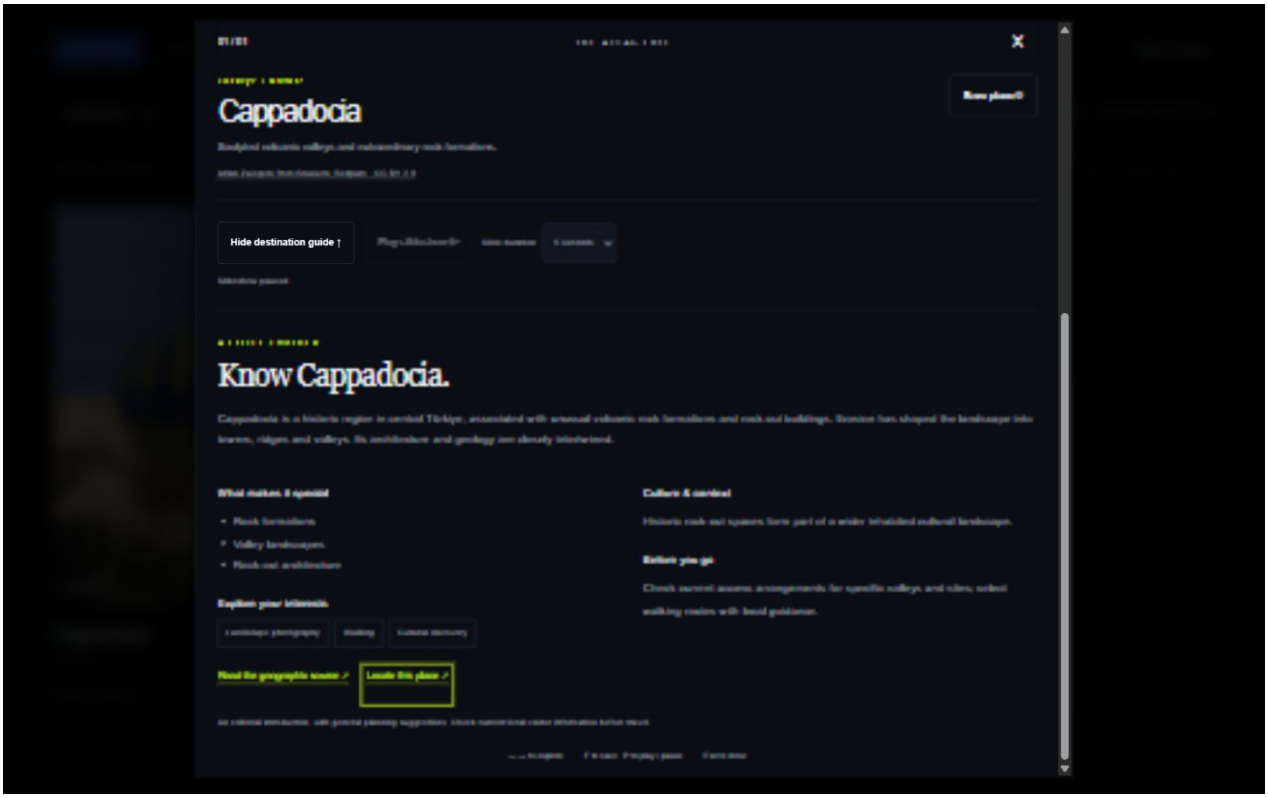


Figure 4.57: Cappadocia: detailed destination guide [Visitor]

Original editorial context, interests and practical planning suggestions.

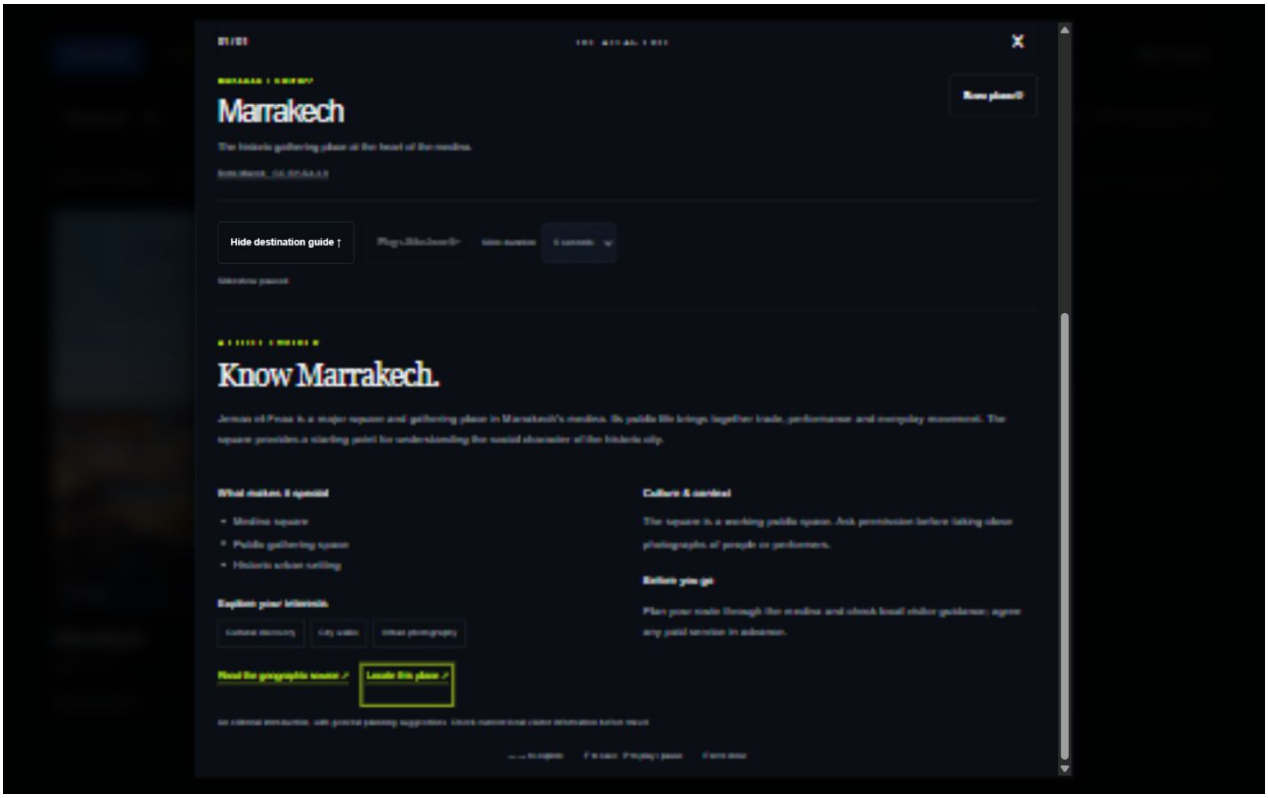


Figure 4.58: Marrakech: detailed destination guide [Visitor]

Original editorial context, interests and practical planning suggestions.

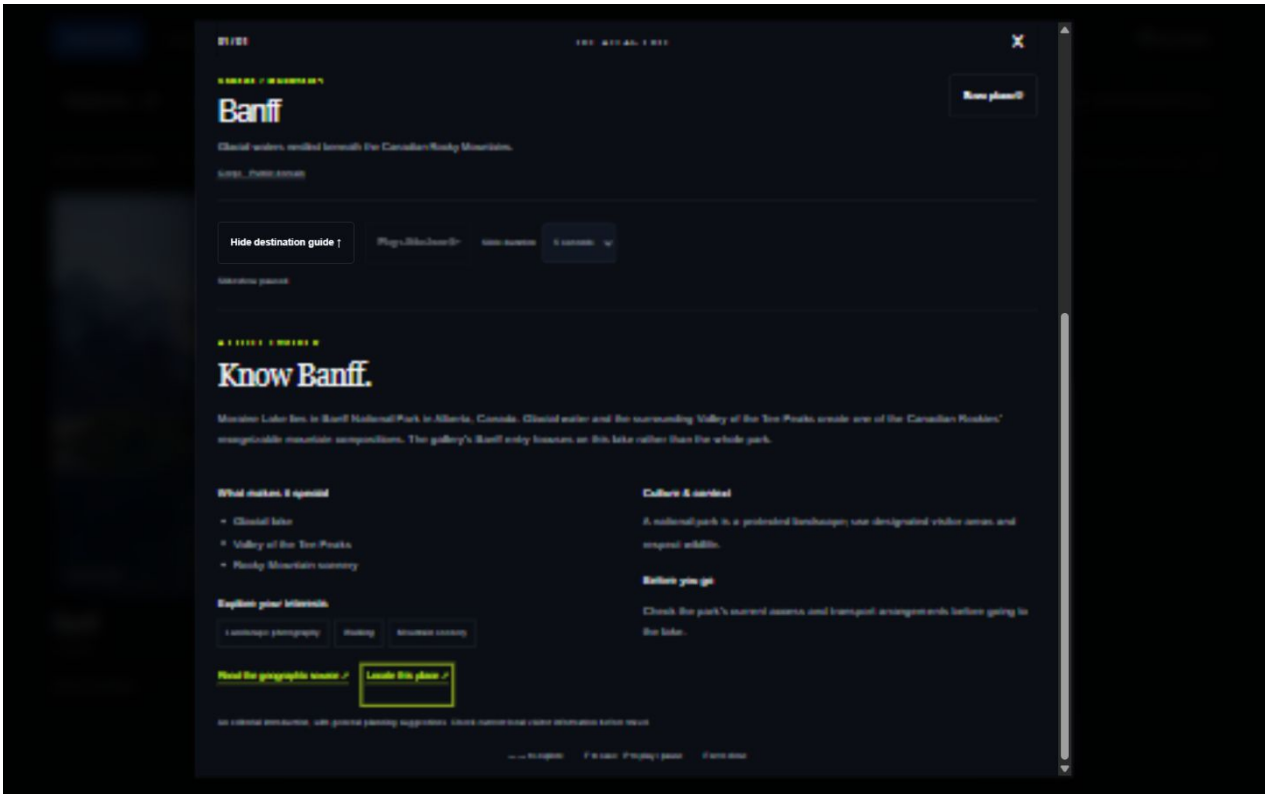


Figure 4.59: Banff: detailed destination guide [Visitor]

Original editorial context, interests and practical planning suggestions.

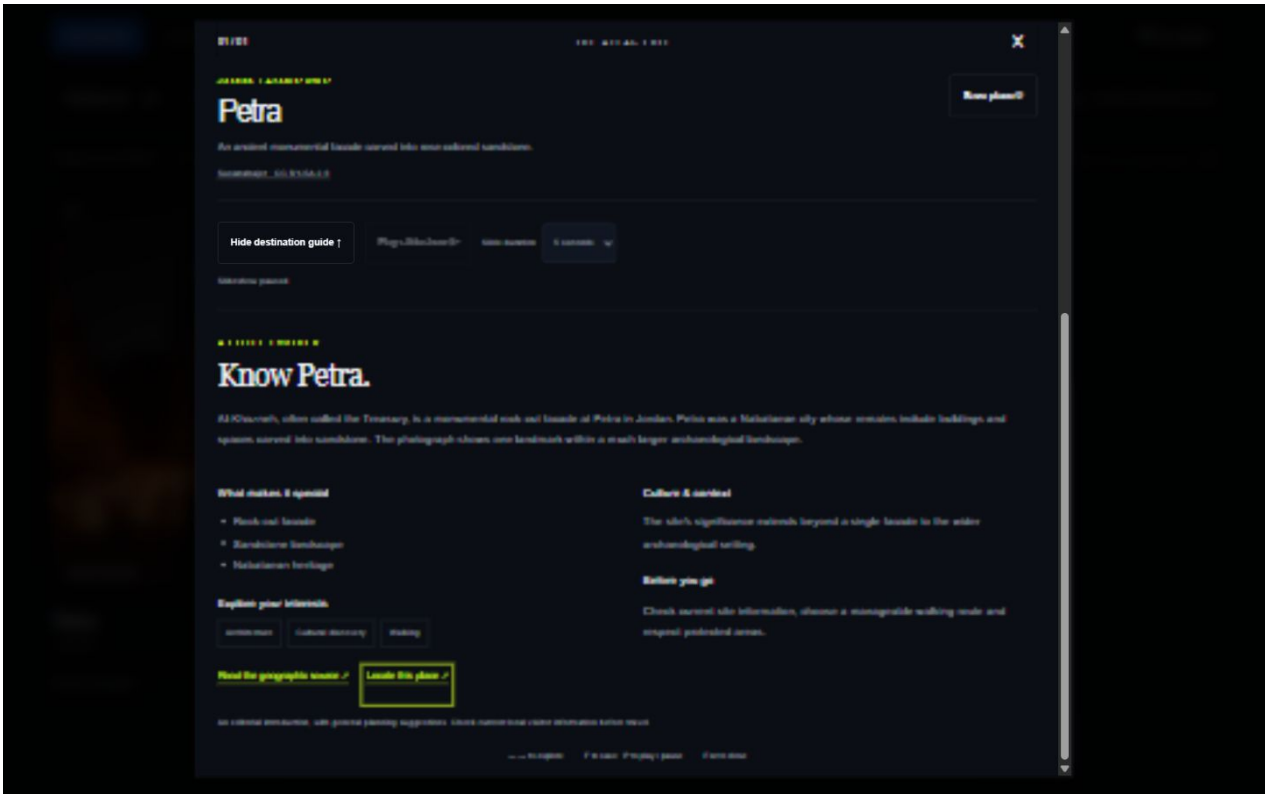


Figure 4.60: Petra: detailed destination guide [Visitor]

Original editorial context, interests and practical planning suggestions.

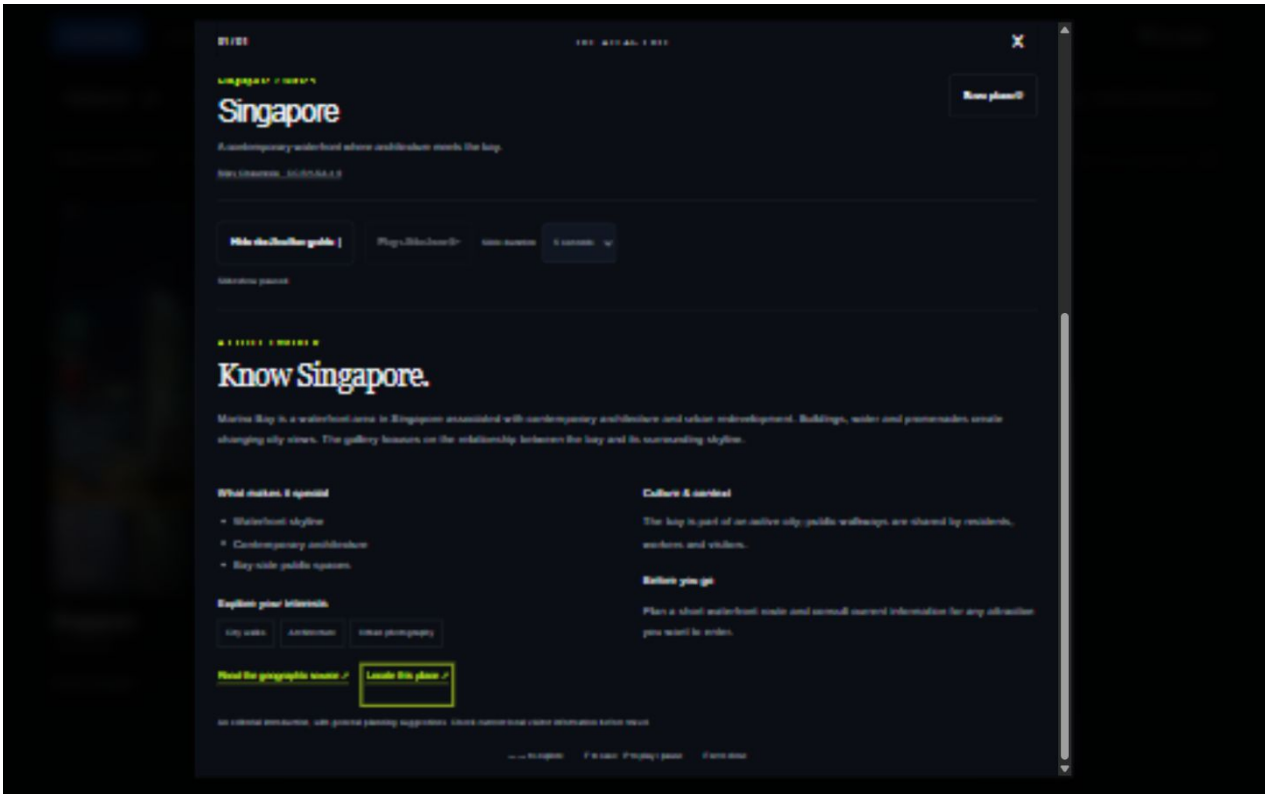


Figure 4.61: Singapore: detailed destination guide [Visitor]

Original editorial context, interests and practical planning suggestions.

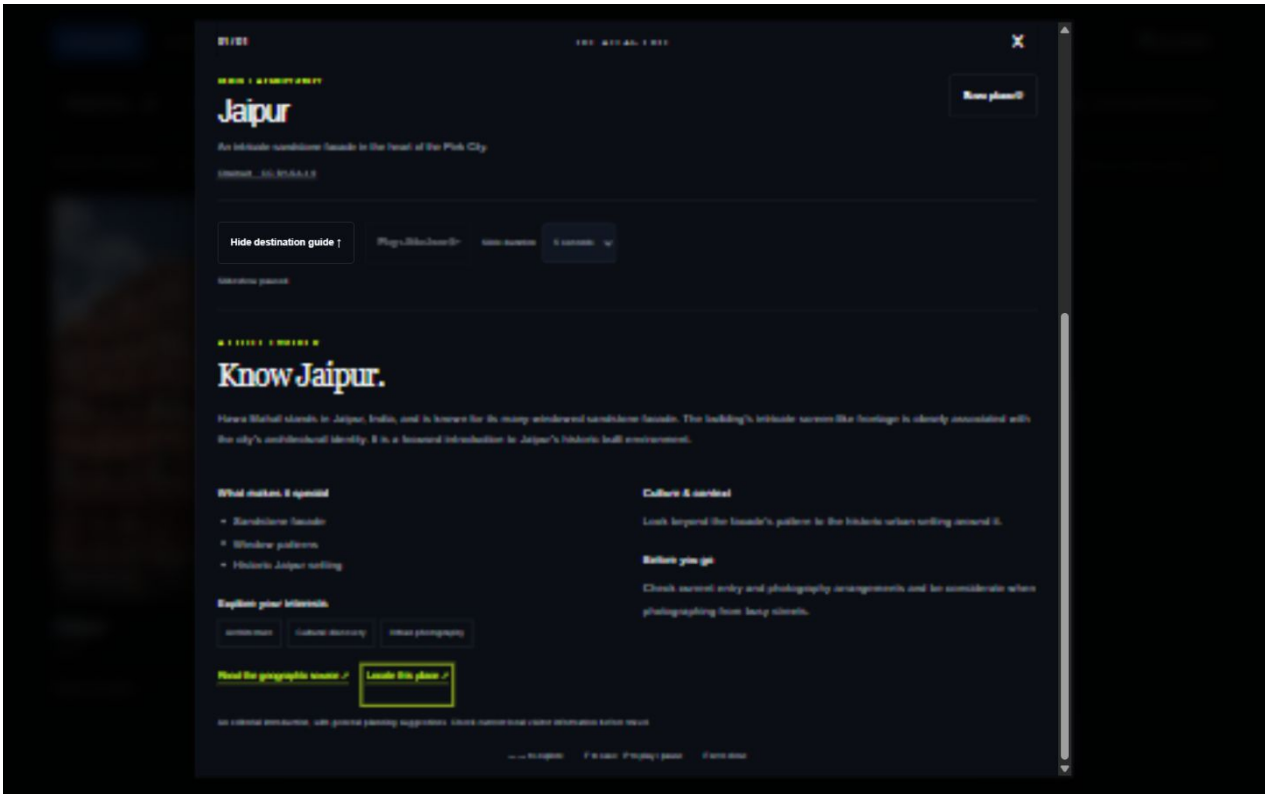


Figure 4.62: Jaipur: detailed destination guide [Visitor]

Original editorial context, interests and practical planning suggestions.

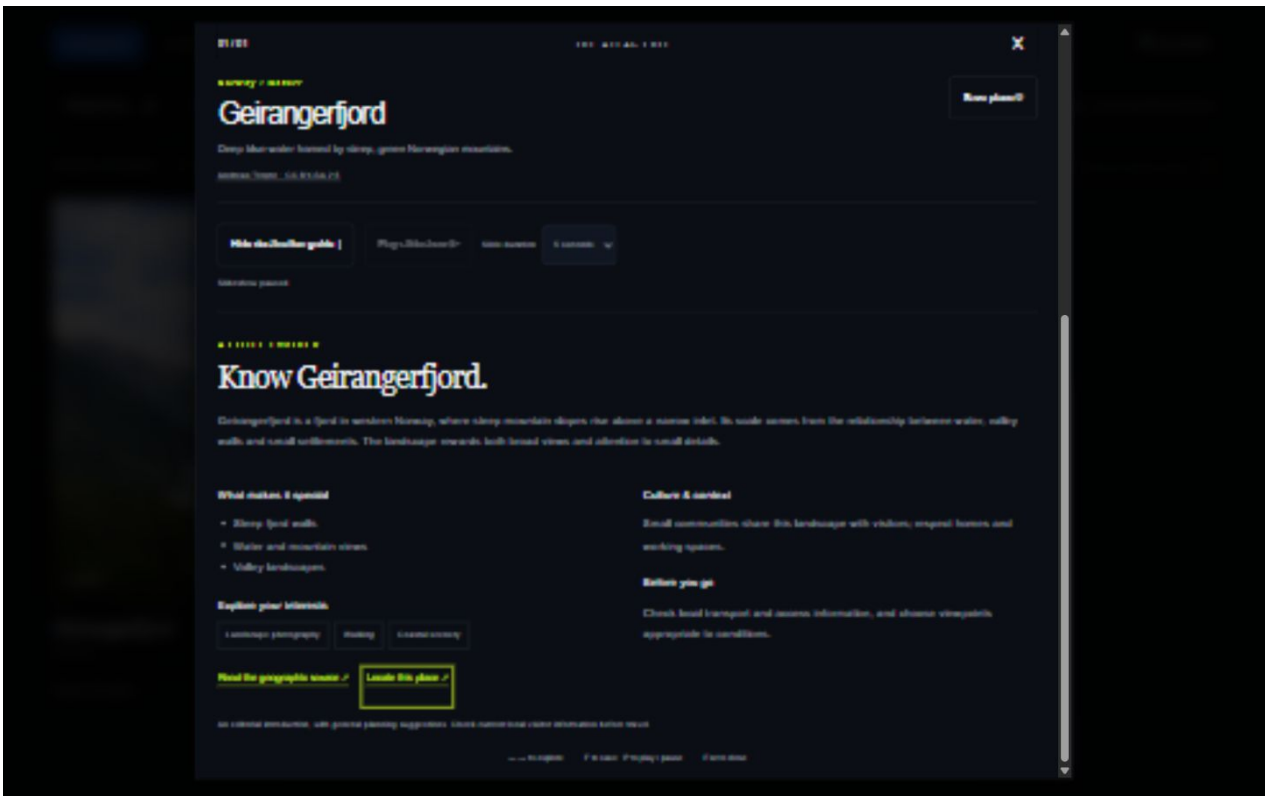


Figure 4.63: Geirangerfjord: detailed destination guide [Visitor]

Sourced context, highlights, interests and general planning information.

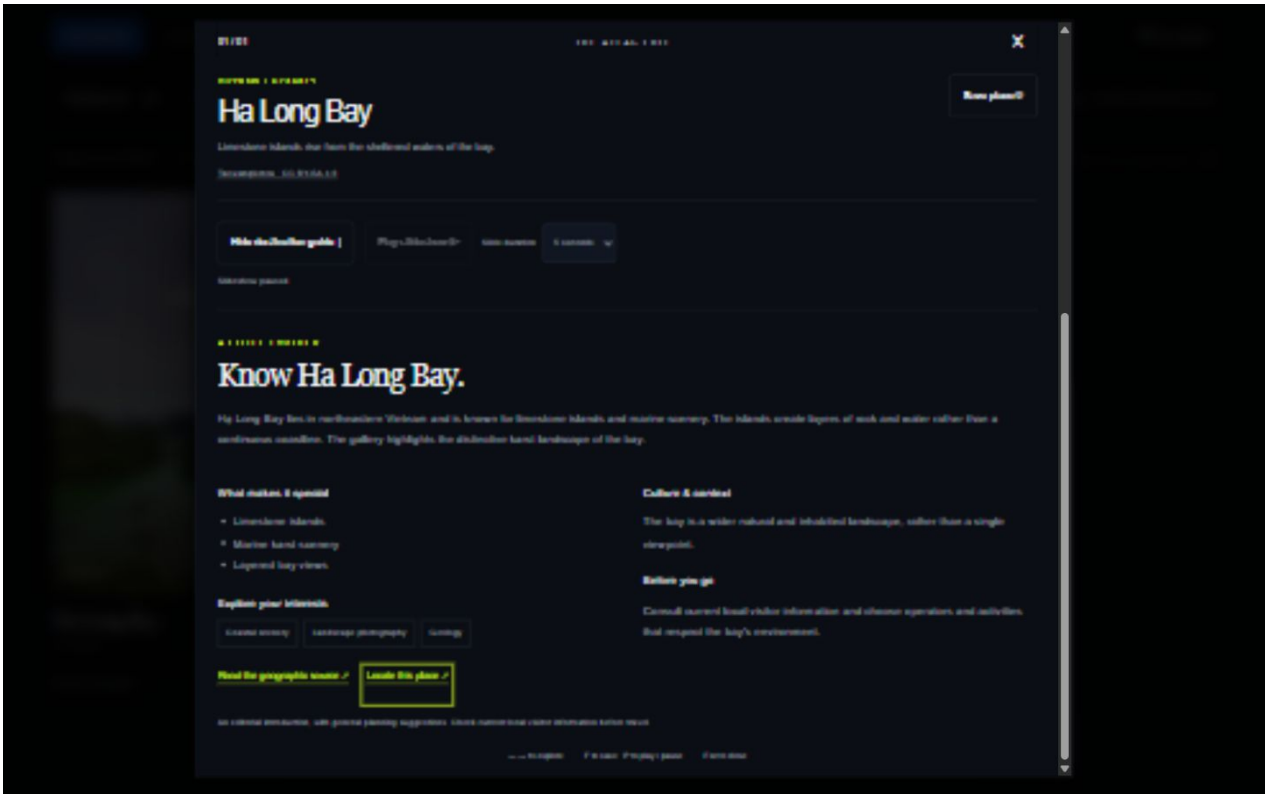


Figure 4.64: Ha Long Bay: detailed destination guide [Visitor]

Sourced context, highlights, interests and general planning information.

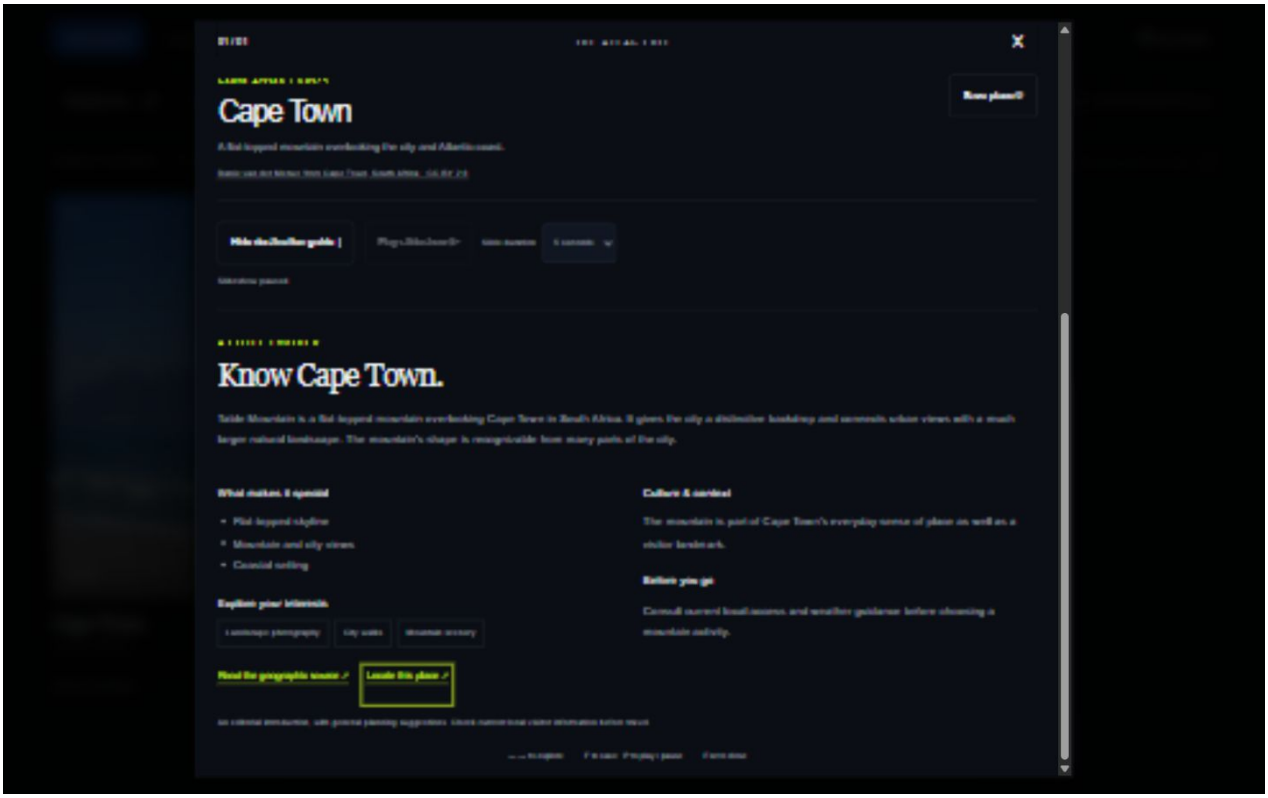


Figure 4.65: Cape Town: detailed destination guide [Visitor]

Sourced context, highlights, interests and general planning information.

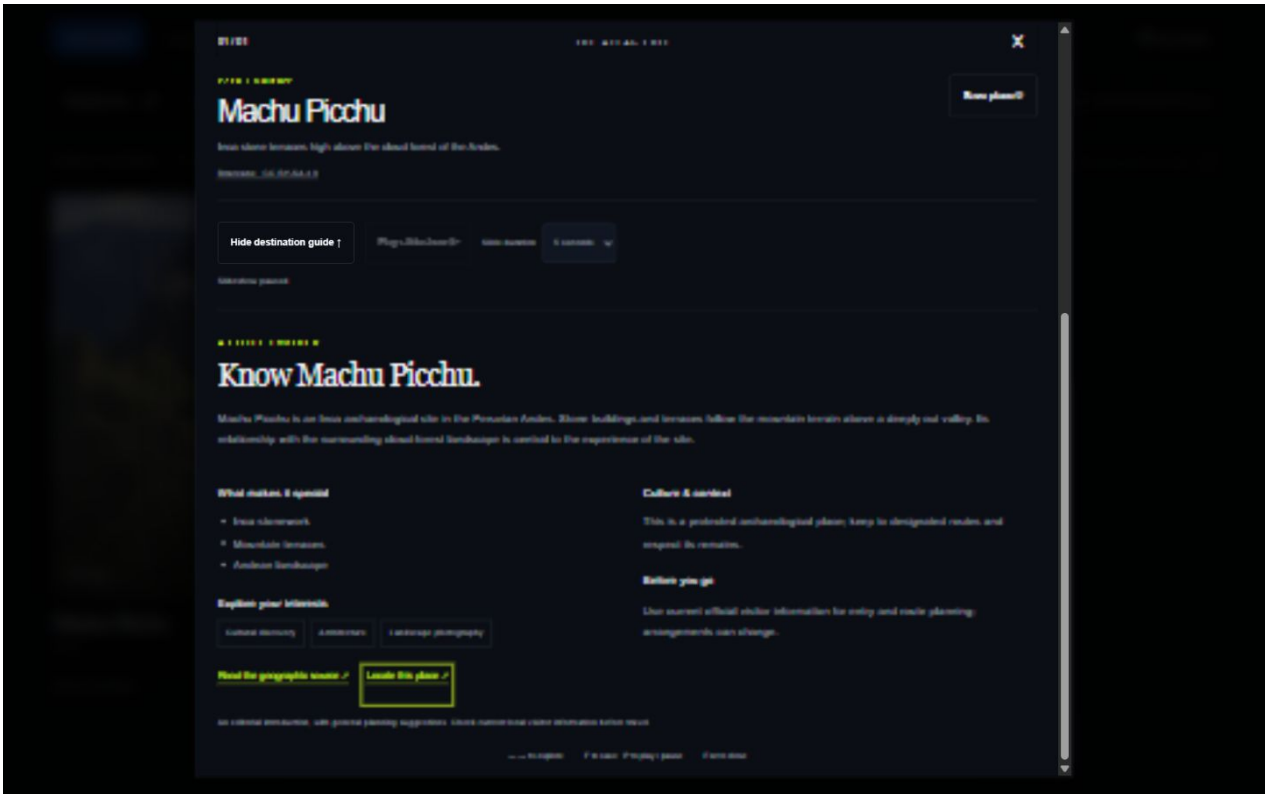


Figure 4.66: Machu Picchu: detailed destination guide [Visitor]

Sourced context, highlights, interests and general planning information.

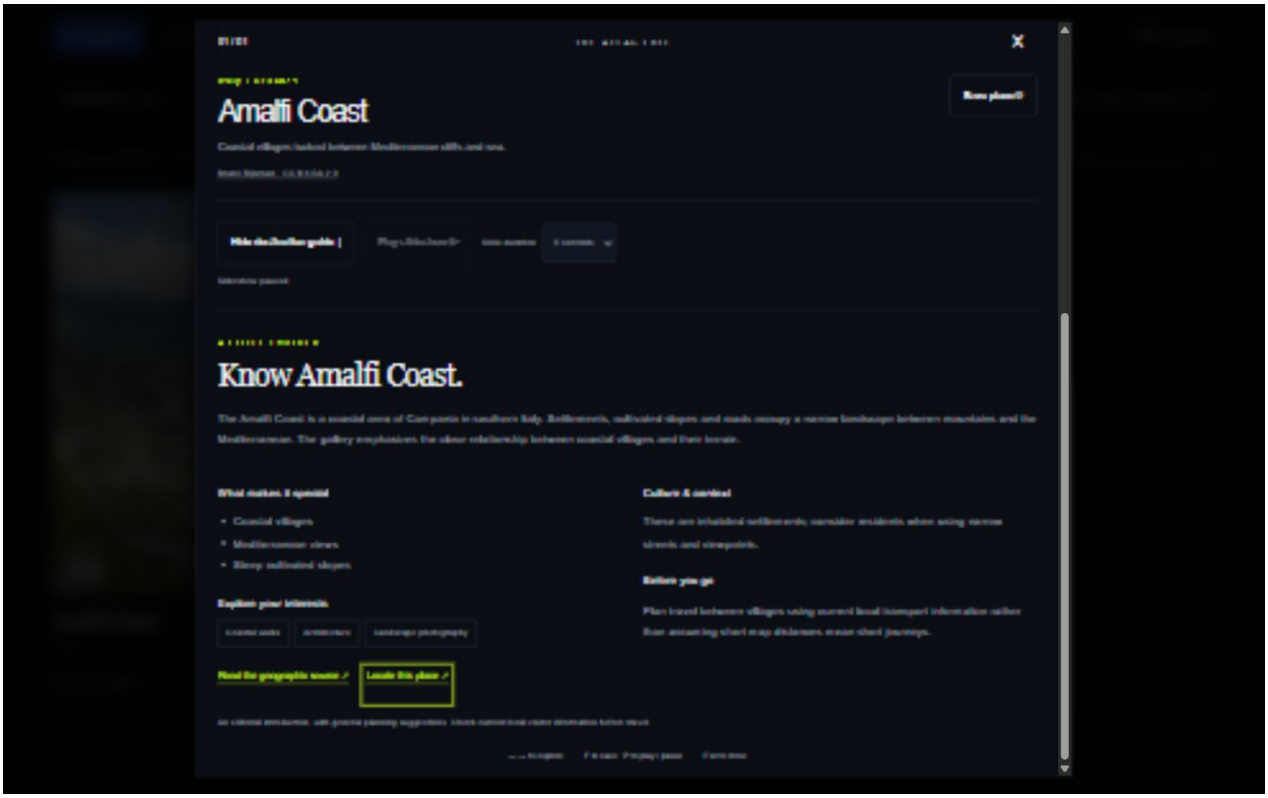


Figure 4.67: Amalfi Coast: detailed destination guide [Visitor]

Sourced context, highlights, interests and general planning information.

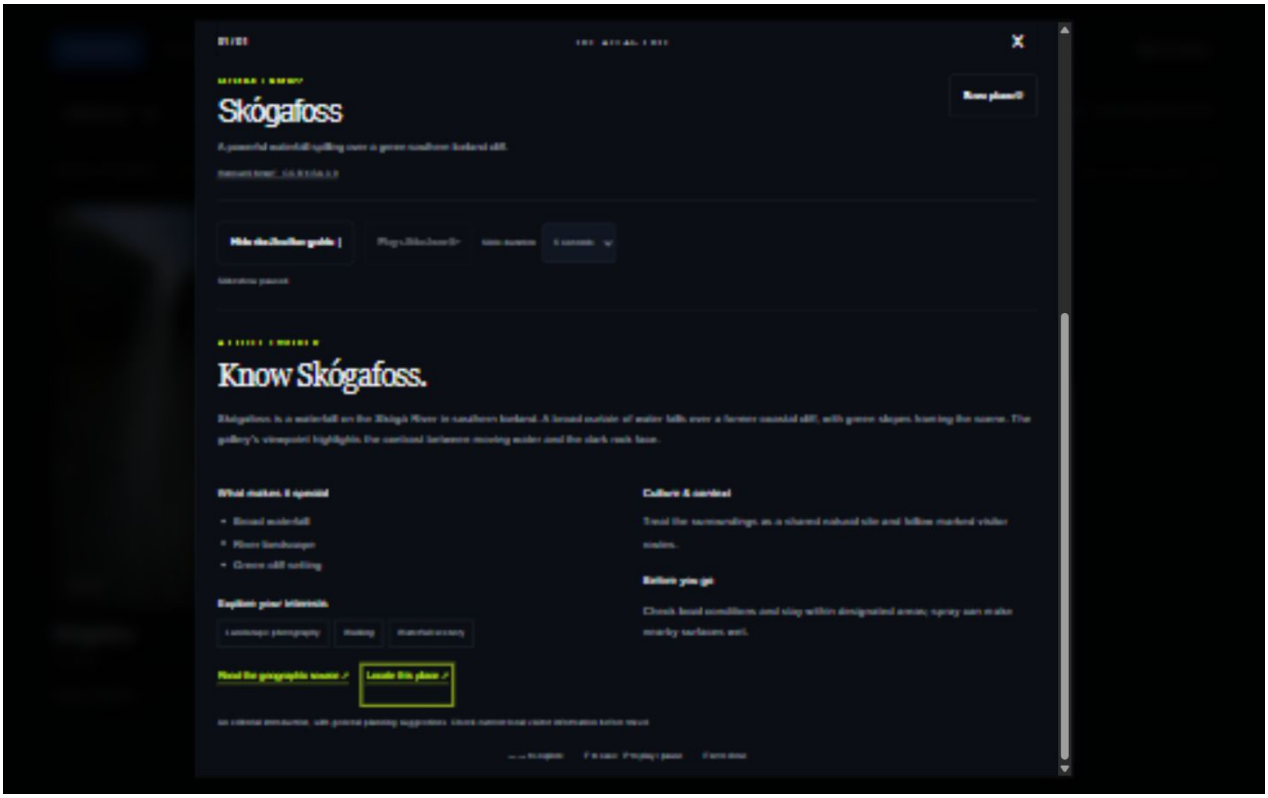


Figure 4.68: Skógafoss: detailed destination guide [Visitor]

Sourced context, highlights, interests and general planning information.

4.8 Favorites, Saved Collections and Persistence

Cards and the viewer share one Set of valid favorite IDs. Hearts and the header count update together. localStorage preserves favorites after refresh; the saved-only view combines with categories and search. Comparison selection is deliberately separate.

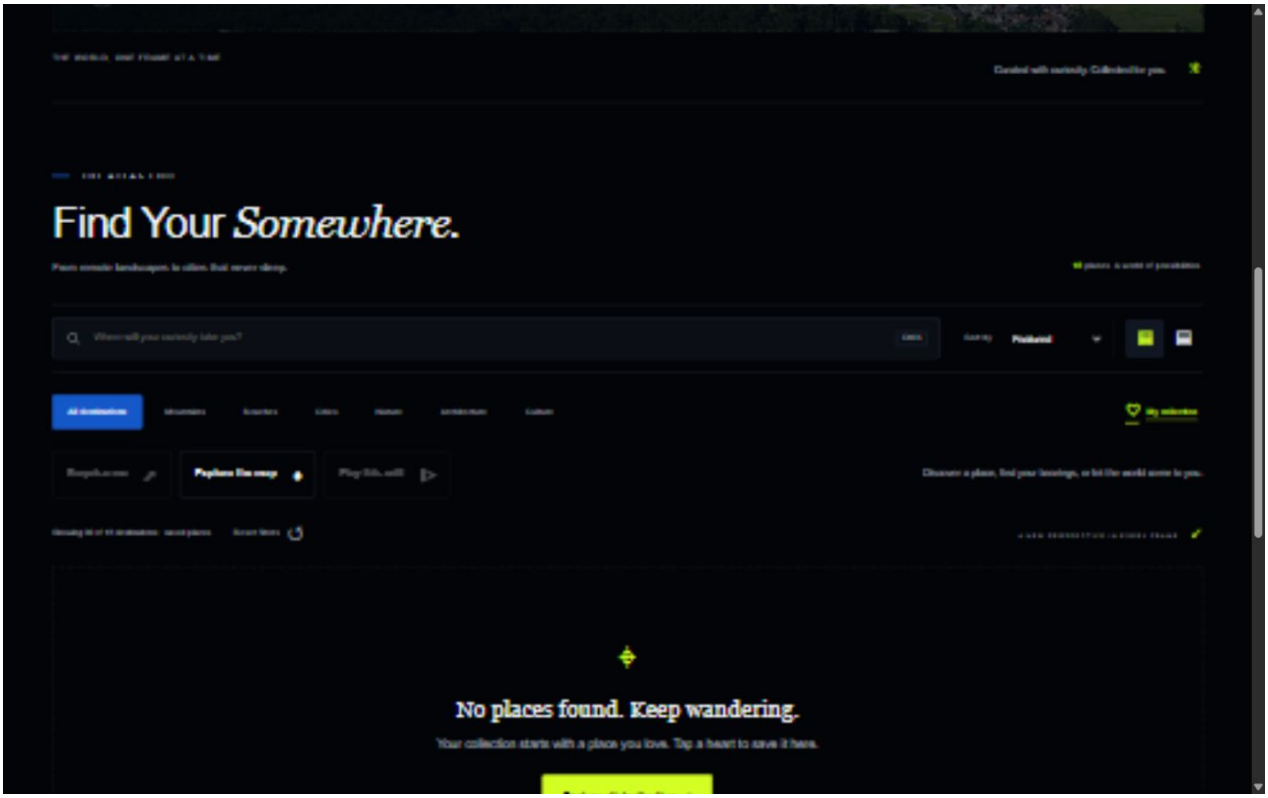


Figure 4.69: Empty saved-places collection [Visitor]

Useful guidance explains how to start a collection.

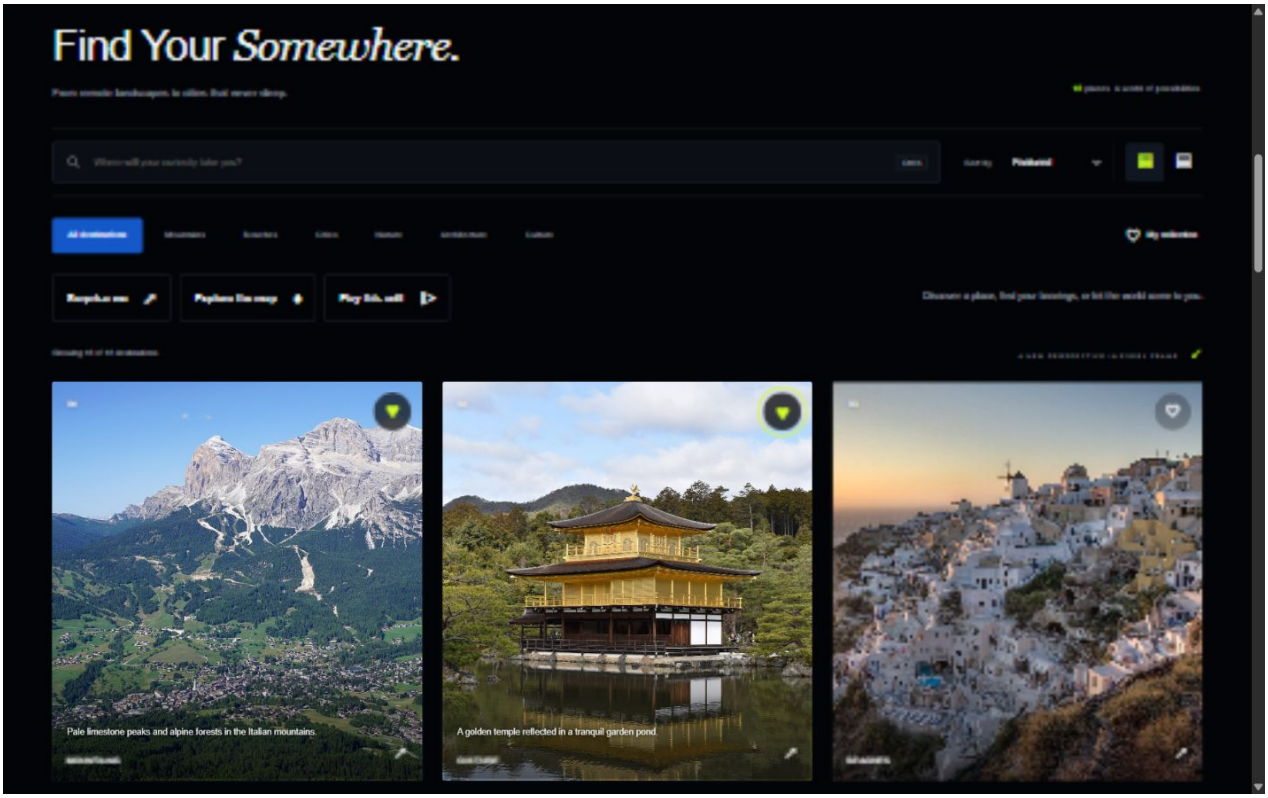


Figure 4.70: Two saved destinations and synchronized hearts [Visitor]

The header counter updates immediately after mouse/keyboard activation.

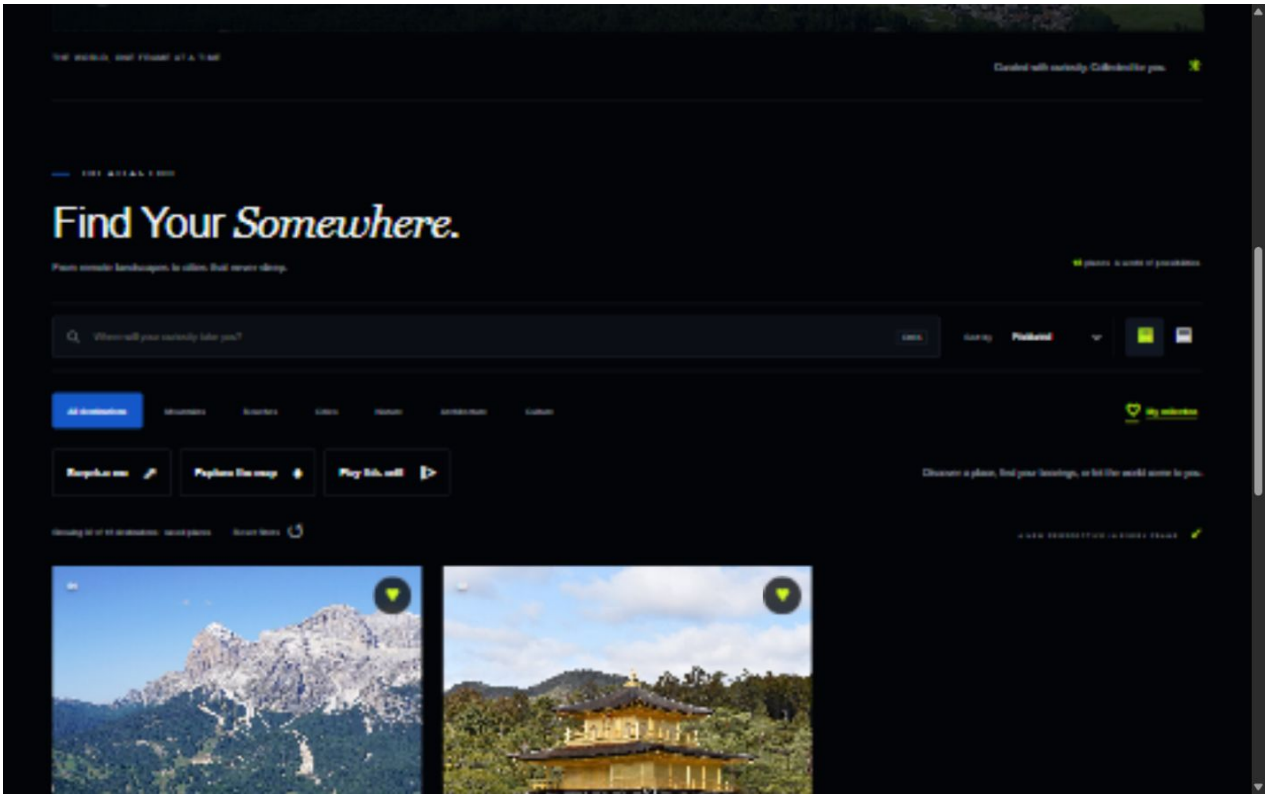


Figure 4.71: Saved-only gallery after refresh [Visitor]

localStorage persists favorite identifiers in the same browser origin.

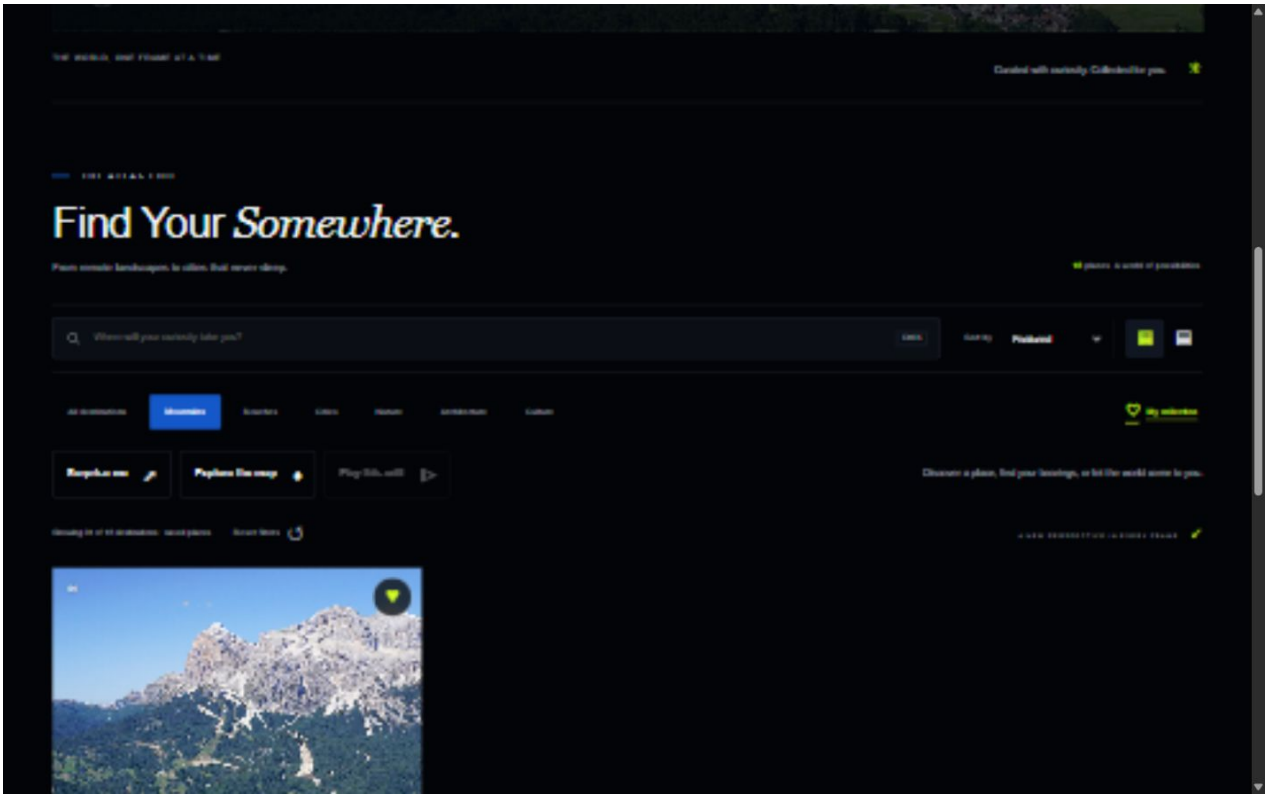


Figure 4.72: Favorites intersected with Mountains [Visitor]

The saved collection narrows to Dolomites.

4.9 Keyboard Navigation and Focus Handling

Native activation, visible focus indicators, explicit modal wrapping and return focus support keyboard operation. The figure shows a visible focus state. This targeted evidence is not a complete screen-reader or WCAG audit.

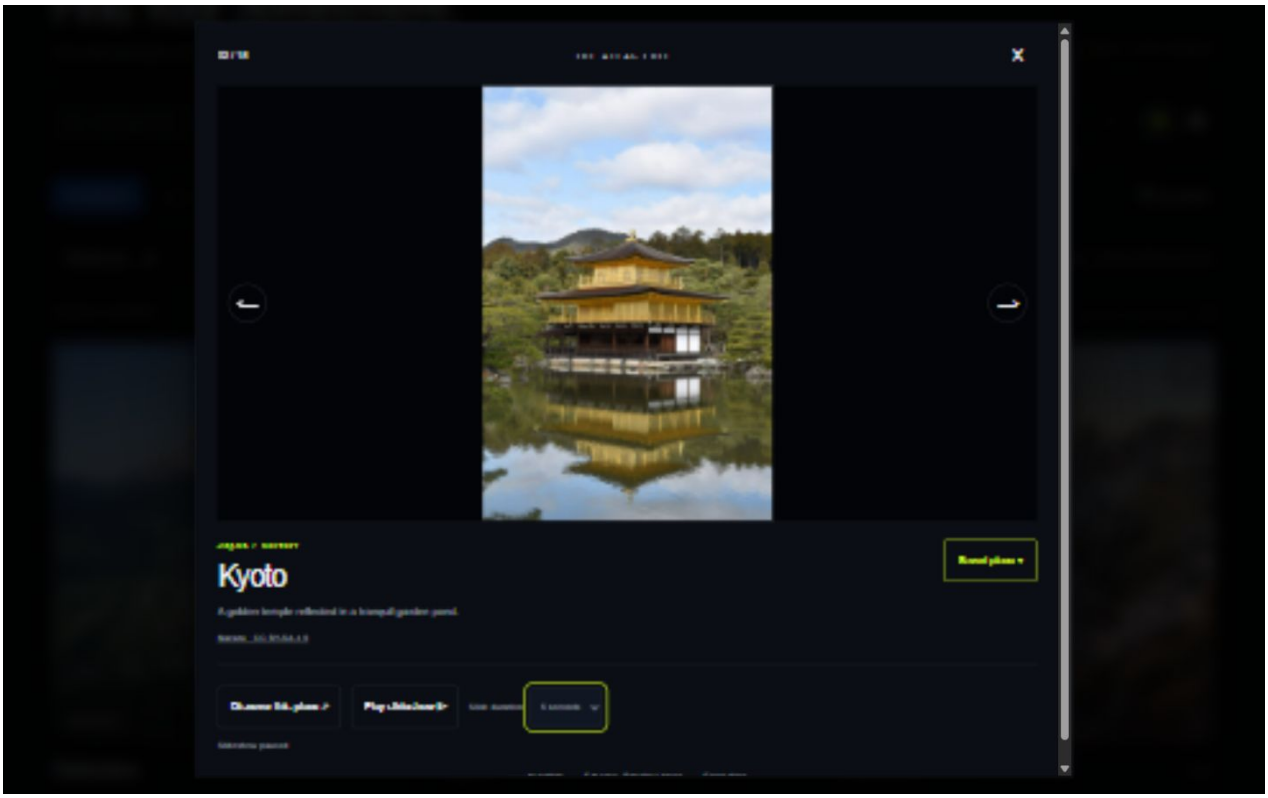


Figure 4.73: Visible keyboard focus within the photograph viewer [Visitor]

The focus ring identifies the slideshow duration control at the dialog boundary.

4.10 Two/Three-Destination Comparison

A selection tray accepts up to three destinations, requires two before comparison and provides removal/clear actions. The separate native comparison dialog presents landscape, highlights, interests and context. Hidden-by-filter selections remain available, and their guides open with the selected comparison order.

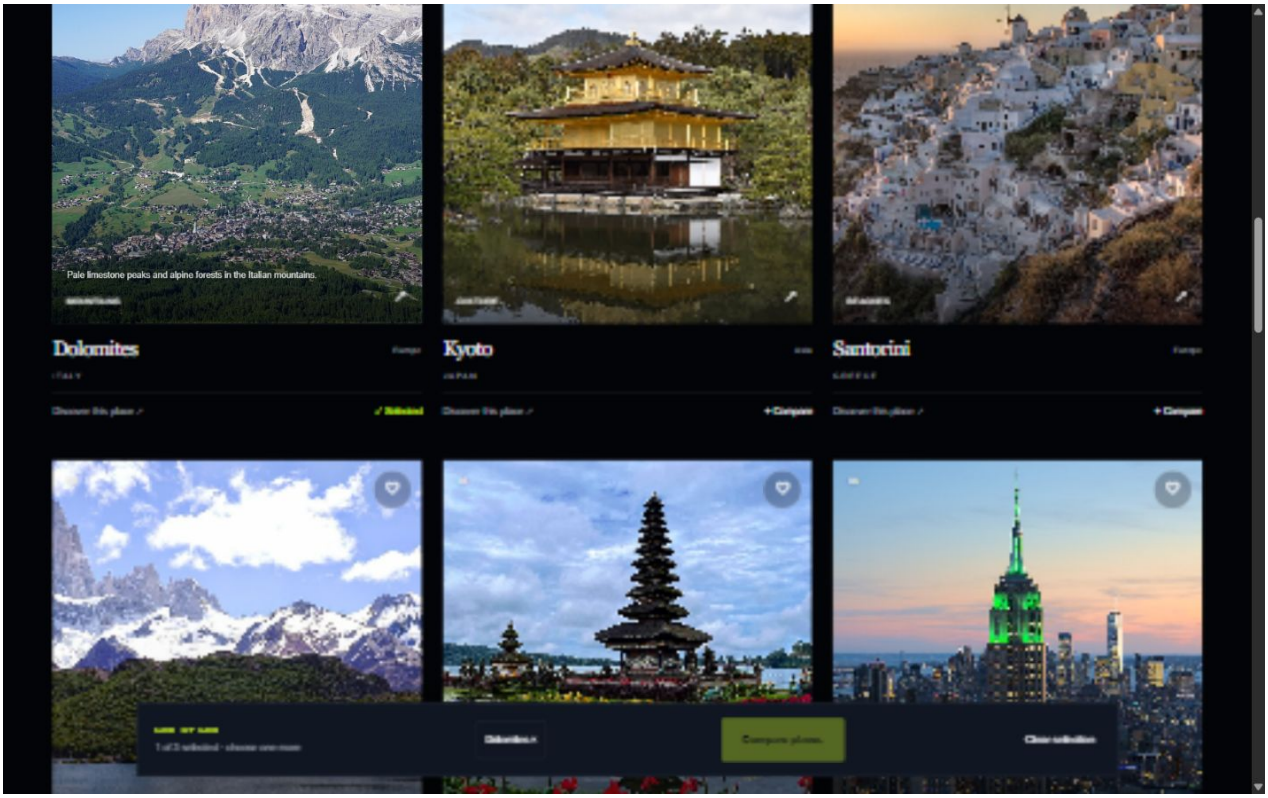


Figure 4.74: One selected destination: comparison is disabled [Visitor]

The tray asks for one more place.

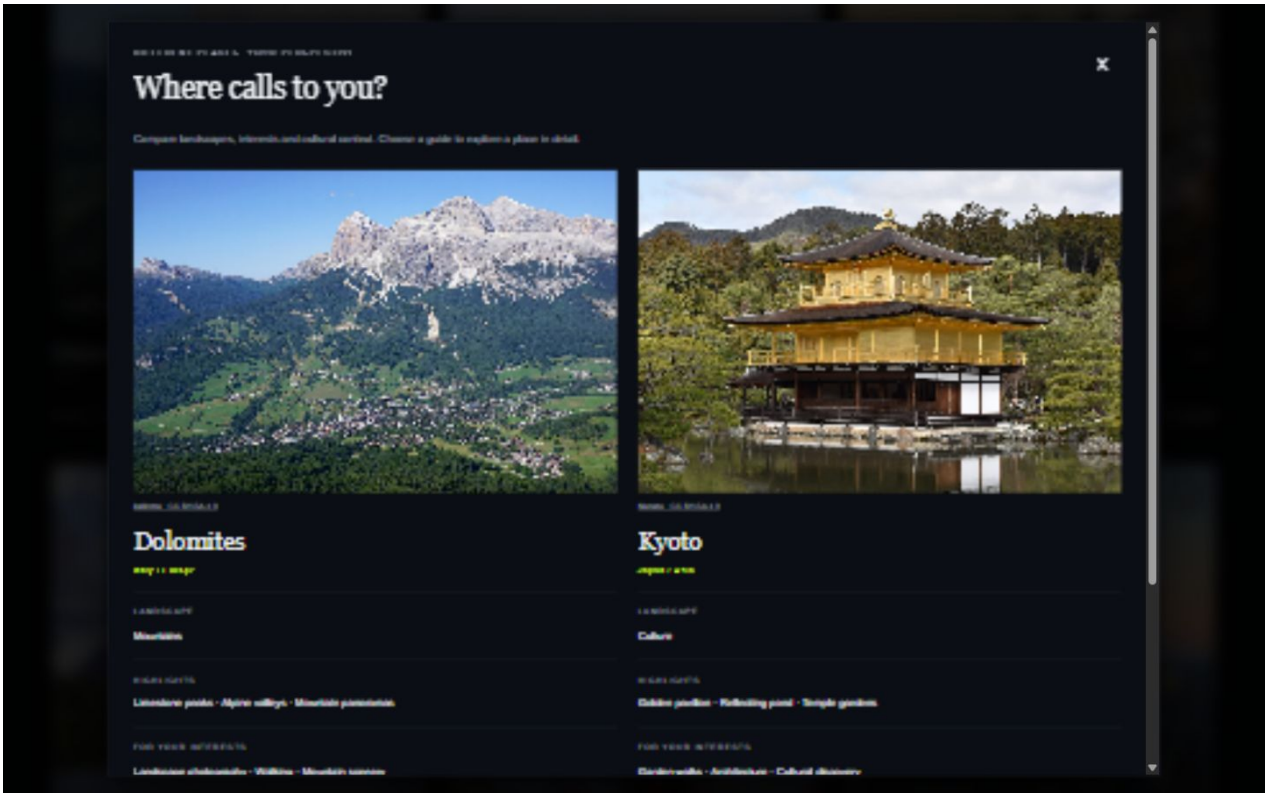


Figure 4.75: Two-place comparison: Dolomites and Kyoto [Visitor]

Qualitative details appear together with links to each guide.

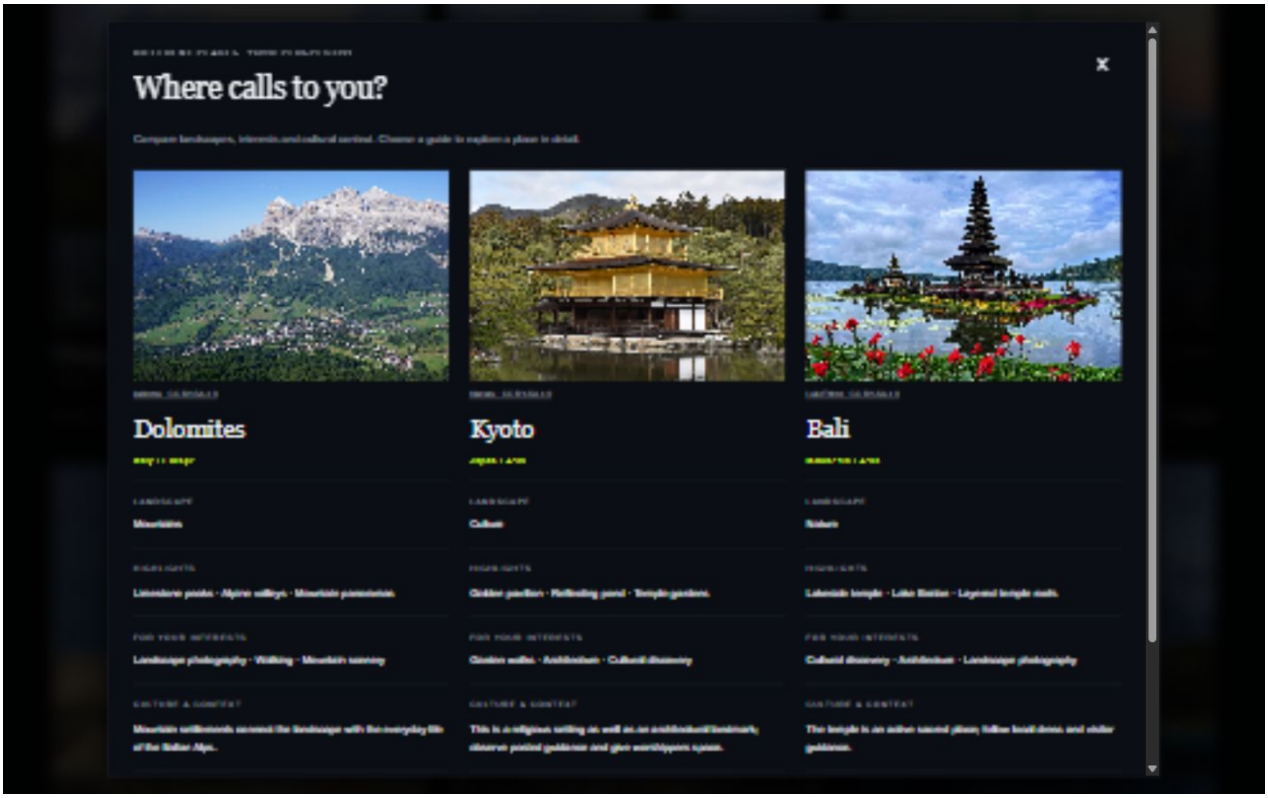


Figure 4.76: Three-place comparison: Dolomites, Kyoto and Bali [Visitor]

The supported maximum is three selected destinations.

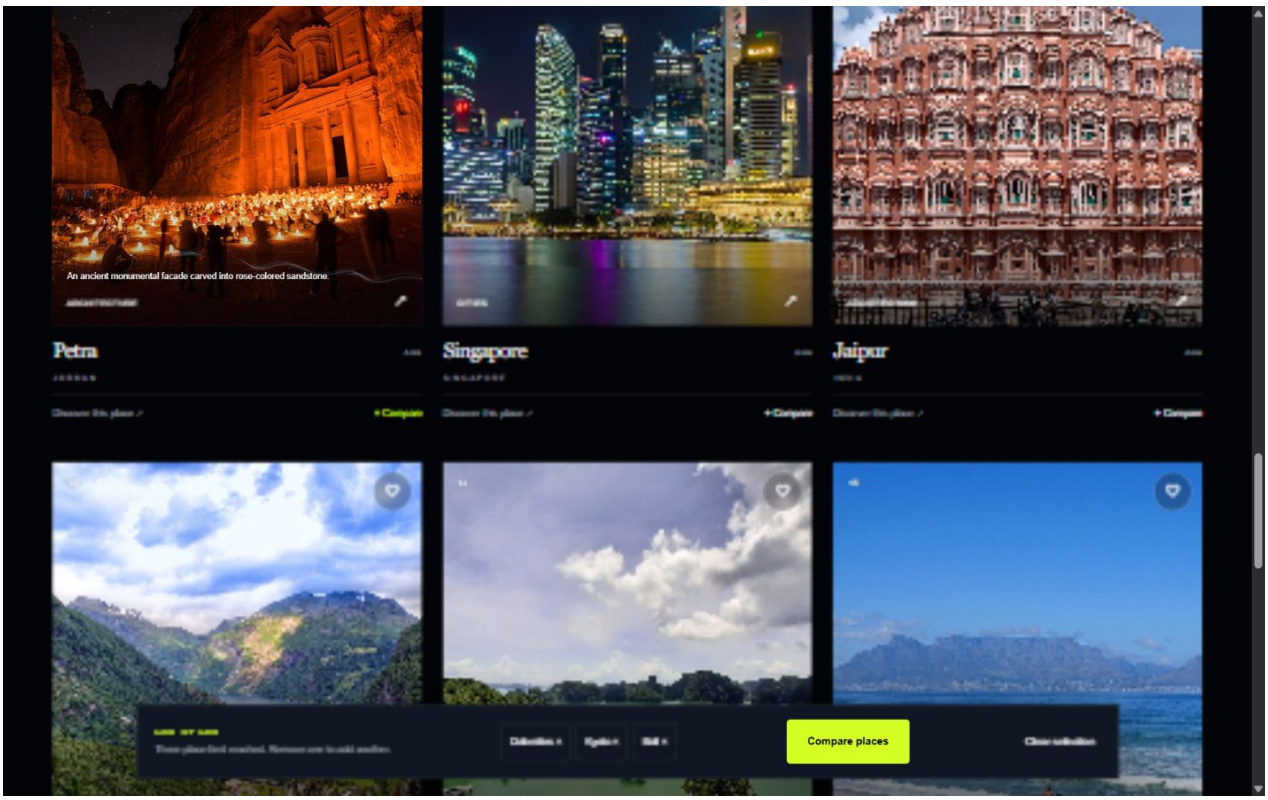


Figure 4.77: Fourth-selection limit feedback [Visitor]

A visible message asks the user to remove an existing choice.

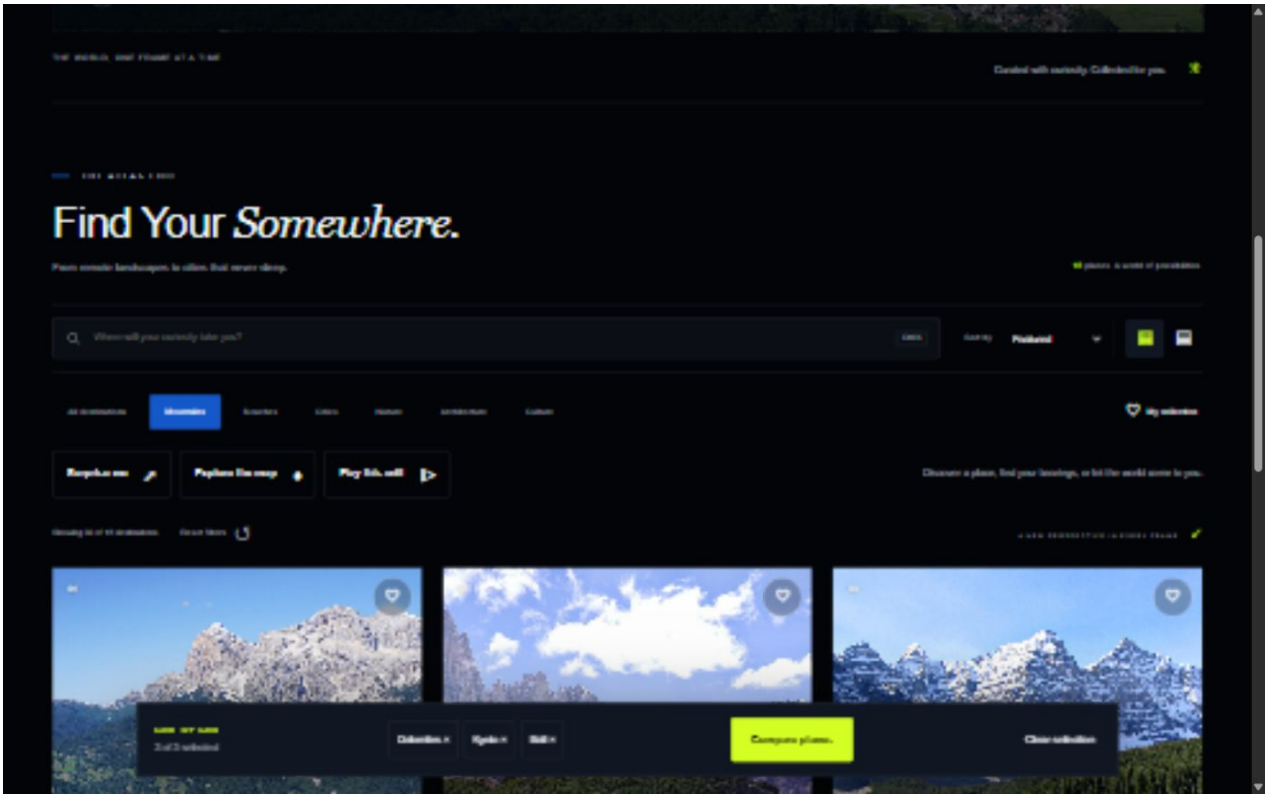


Figure 4.78: Comparison choices retained across gallery filtering [Visitor]

Kyoto and Bali remain selected even when Mountains is active.

4.11 Interactive Destination Maps and Nearby Exploration

The map chooser reflects the currently eligible destinations and loads a representative OpenStreetMap pin. Native map controls support zoom and pan. A larger-map link and explicit attribution remain available. Each location below was captured after the embedded map reflected the selected longitude, avoiding screenshots of the previous location.

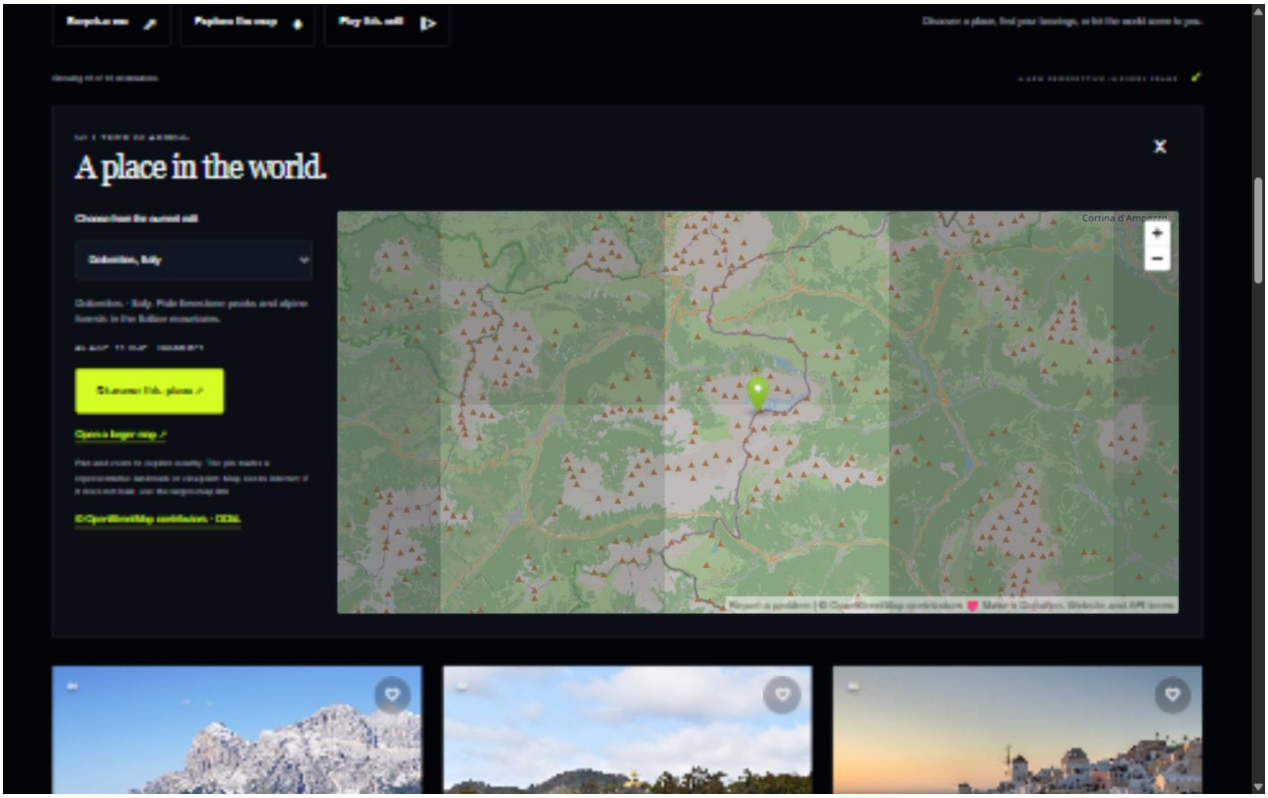


Figure 4.79: Dolomites: representative location on OpenStreetMap [Visitor]

A sourced representative pin, nearby map context and visible contributor attribution.

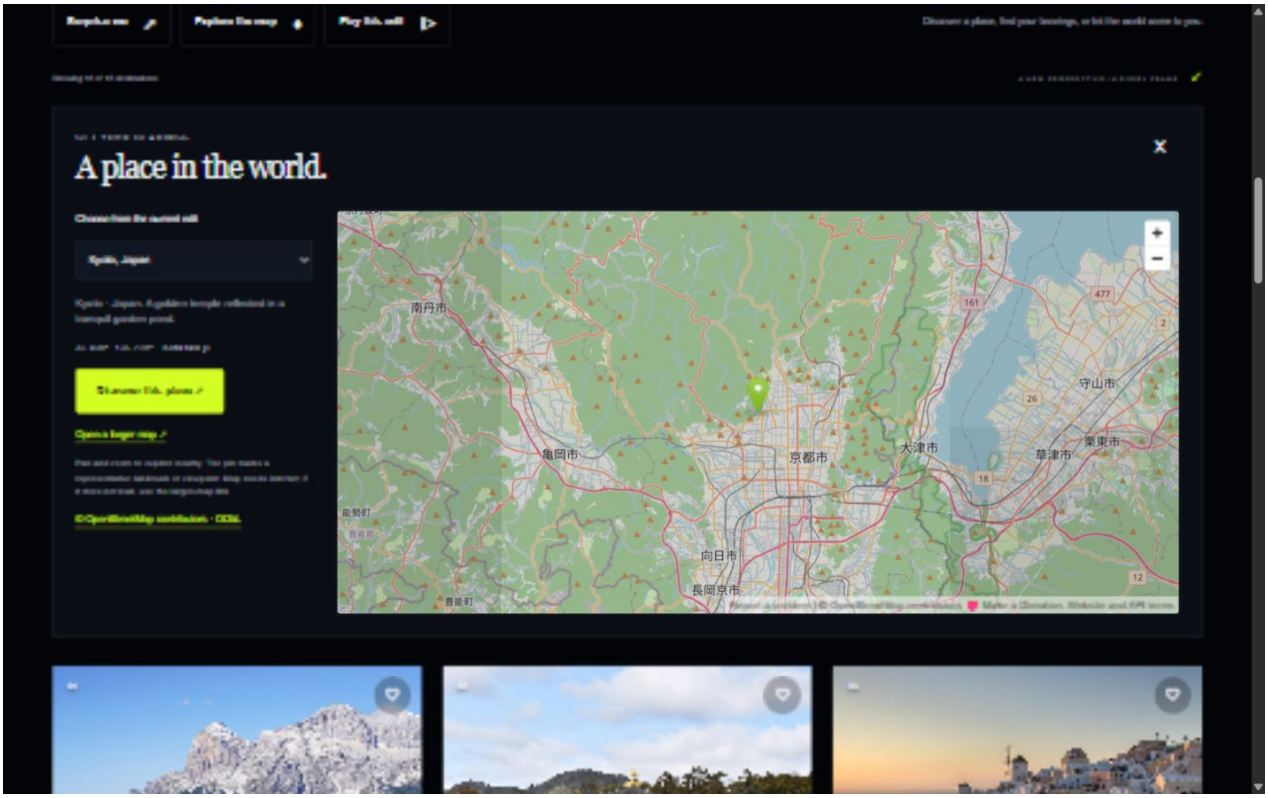


Figure 4.80: Kyoto: representative location on OpenStreetMap [Visitor]

A sourced representative pin, nearby map context and visible contributor attribution.

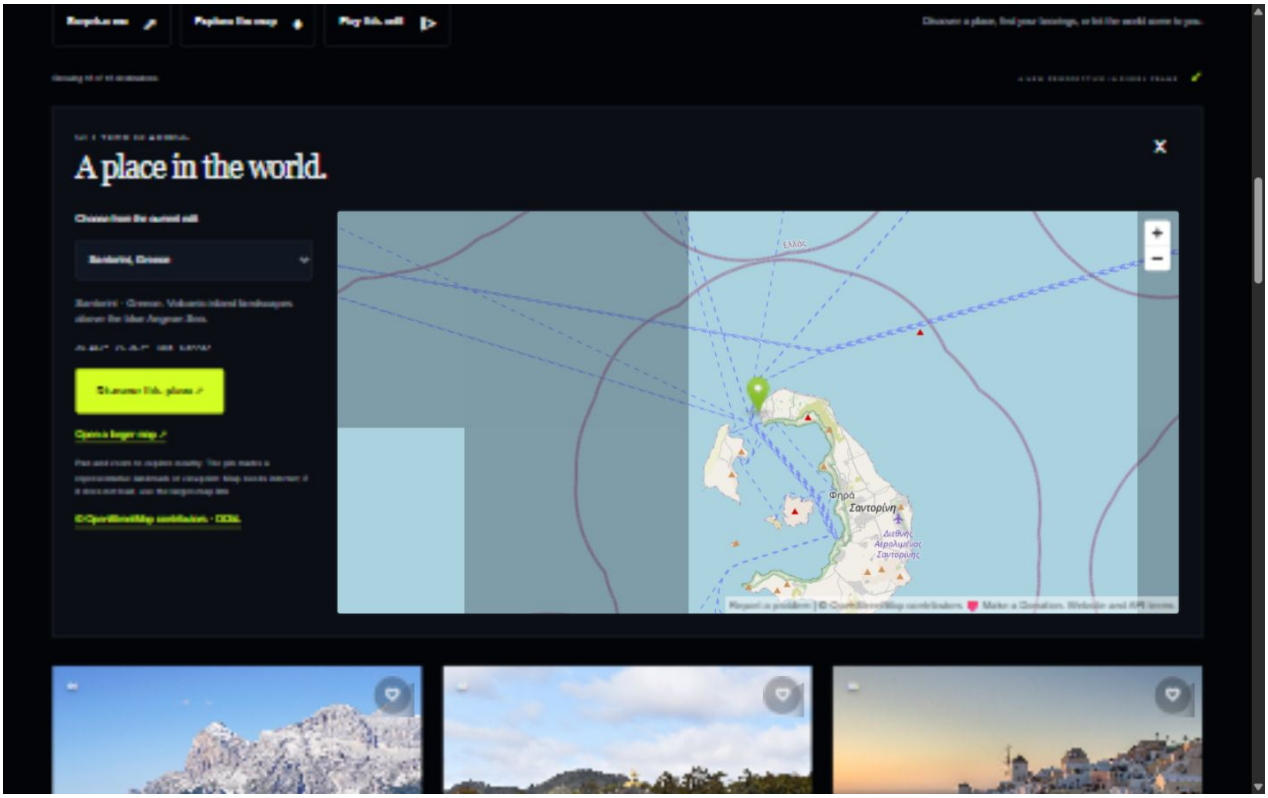


Figure 4.81: Santorini: representative location on OpenStreetMap [Visitor]

A sourced representative pin, nearby map context and visible contributor attribution.

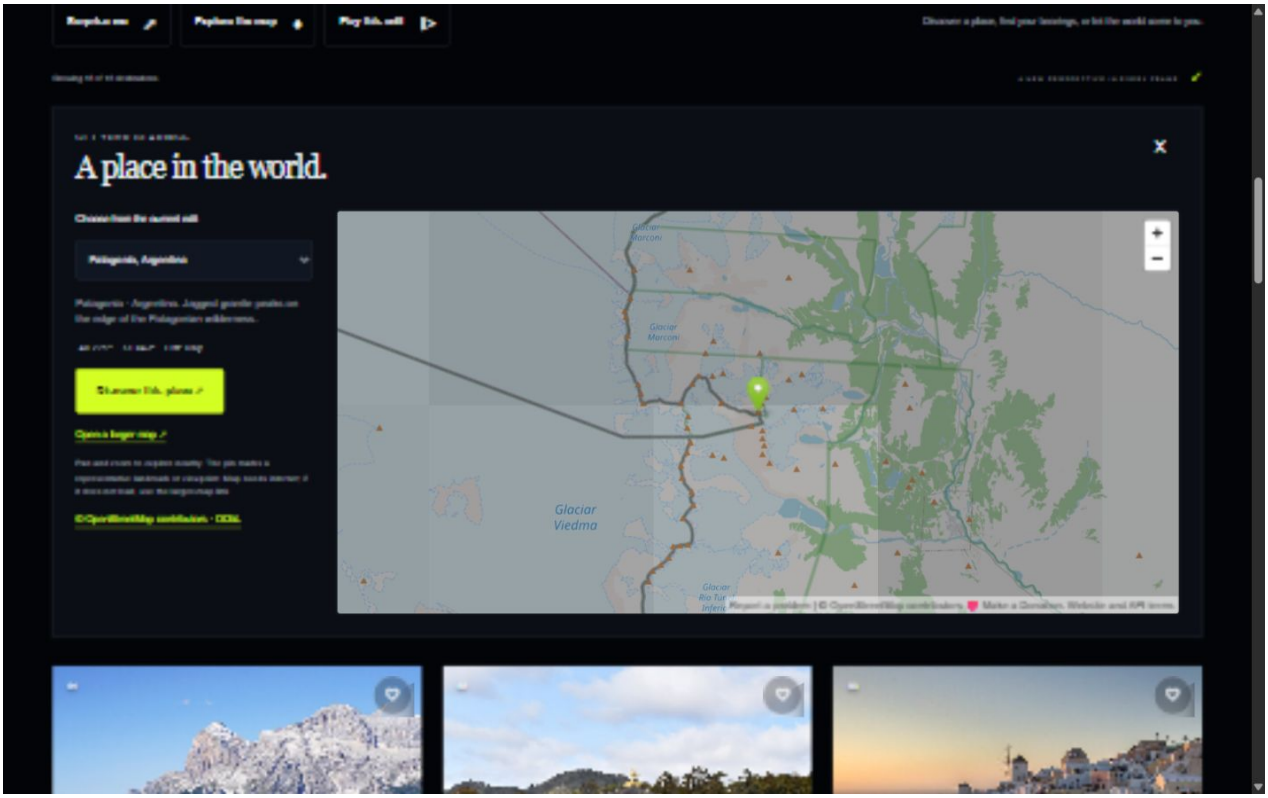


Figure 4.82: Patagonia: representative location on OpenStreetMap [Visitor]

A sourced representative pin, nearby map context and visible contributor attribution.

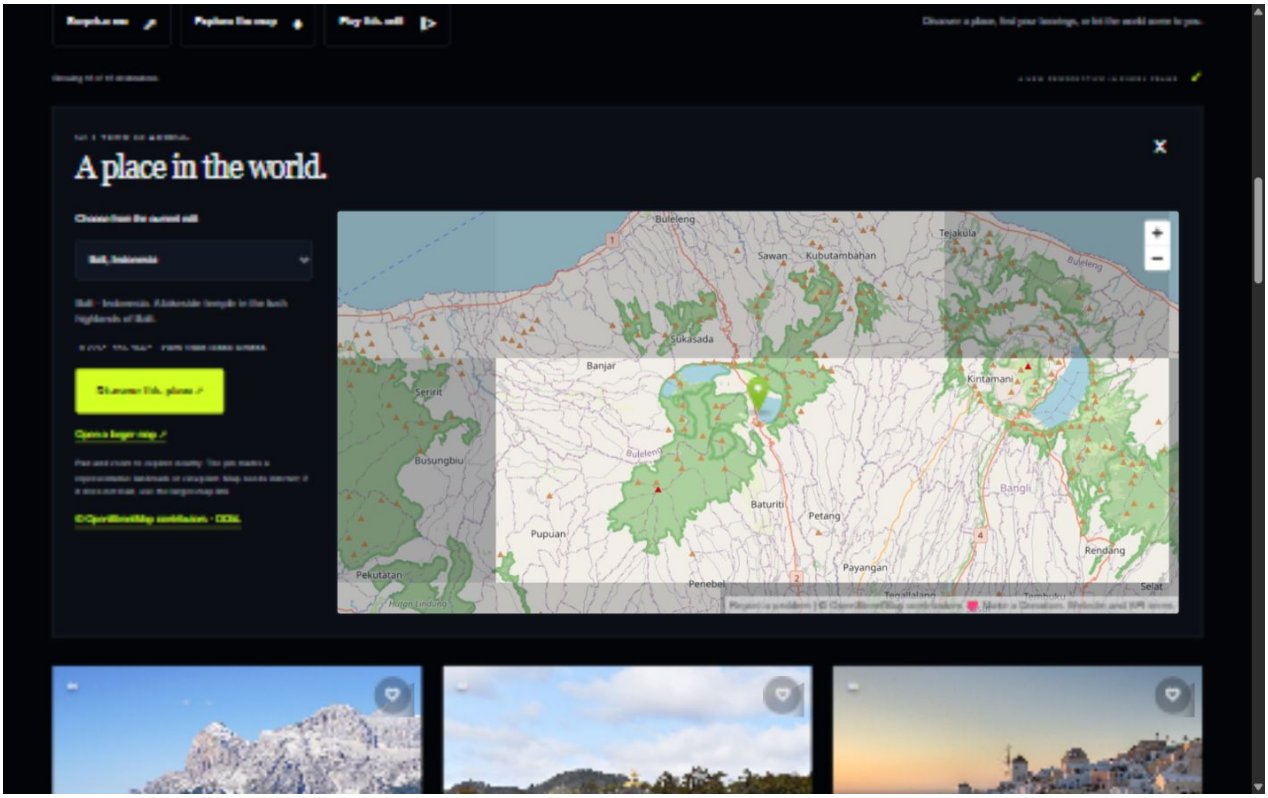


Figure 4.83: Bali: representative location on OpenStreetMap [Visitor]

A sourced representative pin, nearby map context and visible contributor attribution.

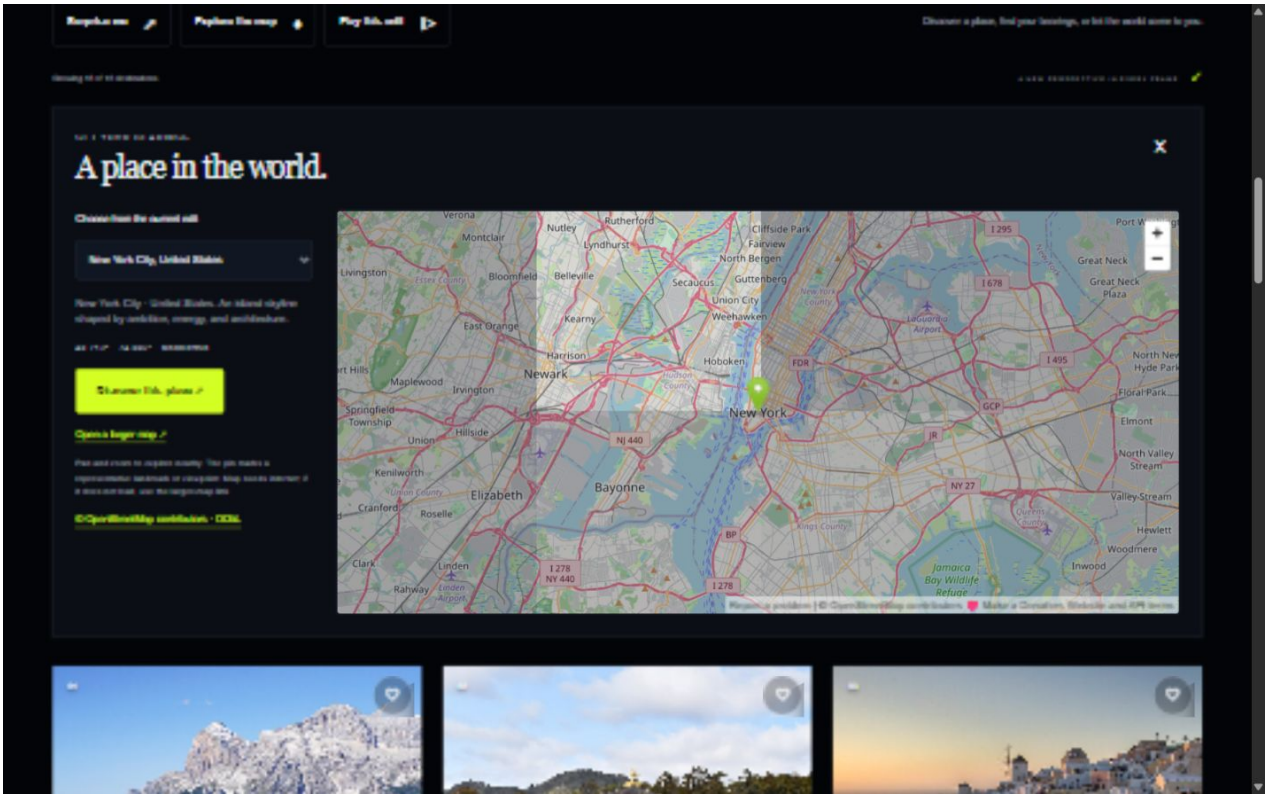


Figure 4.84: New York City: representative location on OpenStreetMap [Visitor]

A sourced representative pin, nearby map context and visible contributor attribution.

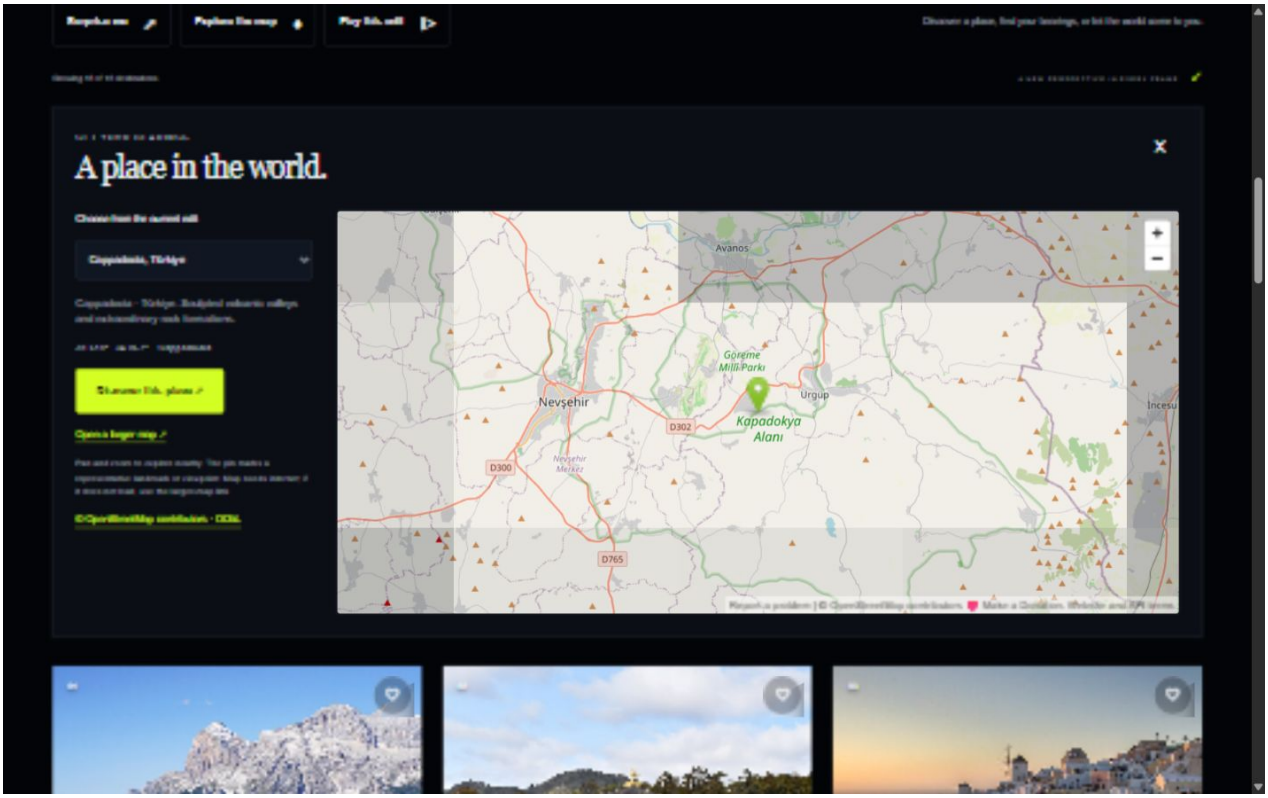


Figure 4.85: Cappadocia: representative location on OpenStreetMap [Visitor]

The selected landmark/viewpoint is mapped; the guide remains available beside it.

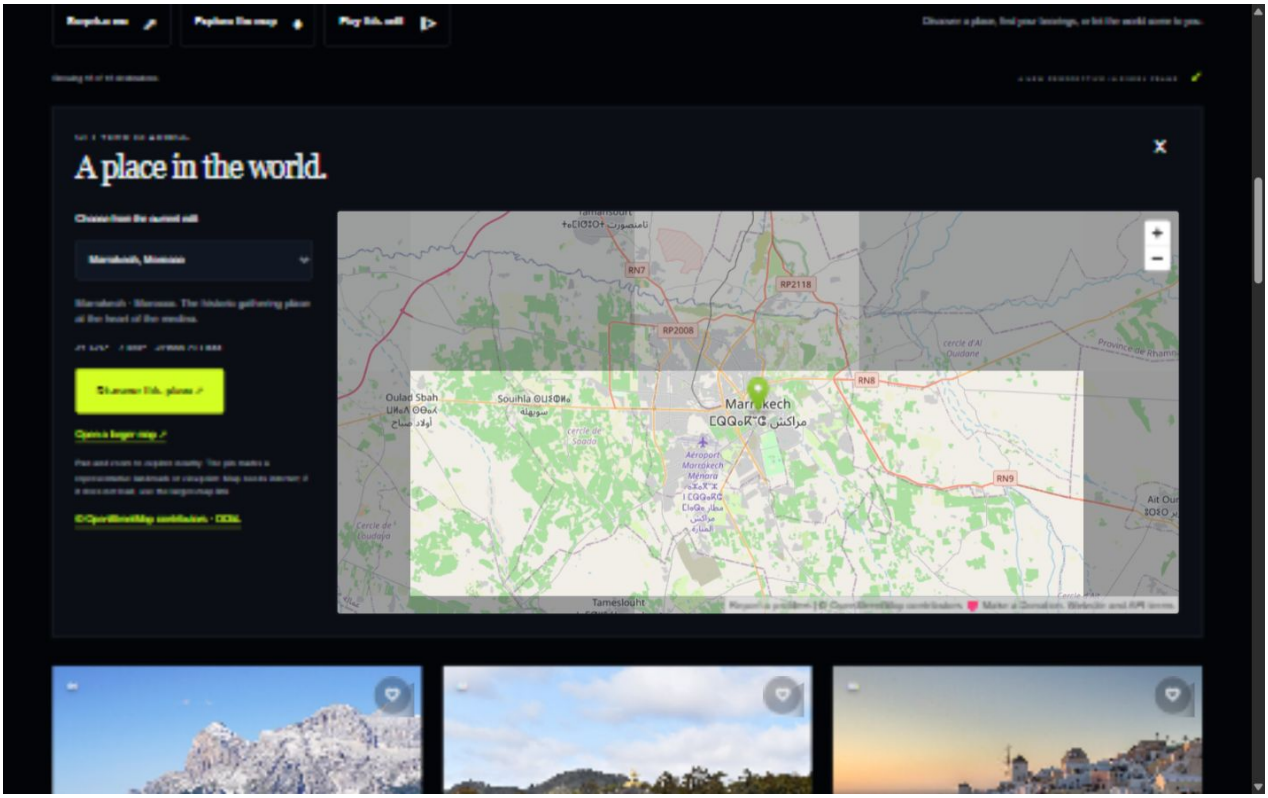


Figure 4.86: Marrakech: representative location on OpenStreetMap [Visitor]

The selected landmark/viewpoint is mapped; the guide remains available beside it.

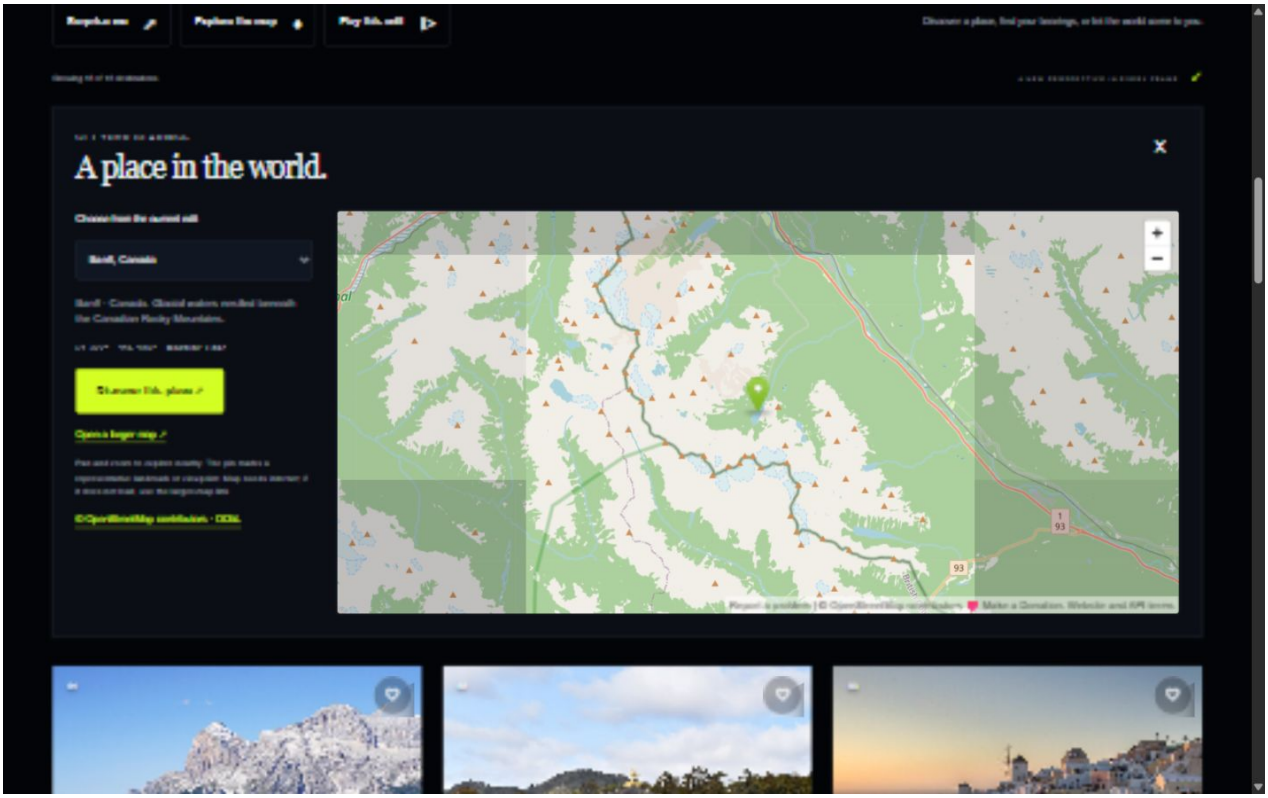


Figure 4.87: Banff: representative location on OpenStreetMap [Visitor]

The selected landmark/viewpoint is mapped; the guide remains available beside it.

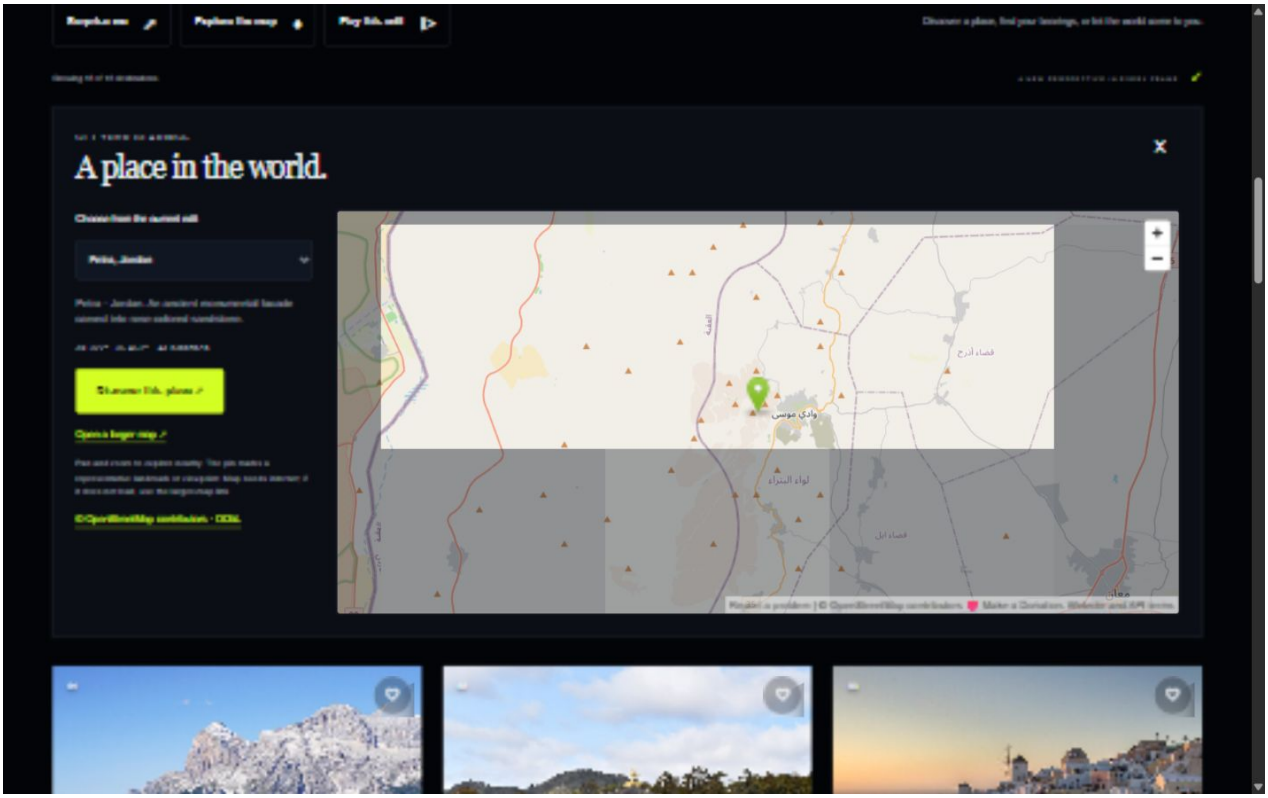


Figure 4.88: Petra: representative location on OpenStreetMap [Visitor]

The selected landmark/viewpoint is mapped; the guide remains available beside it.

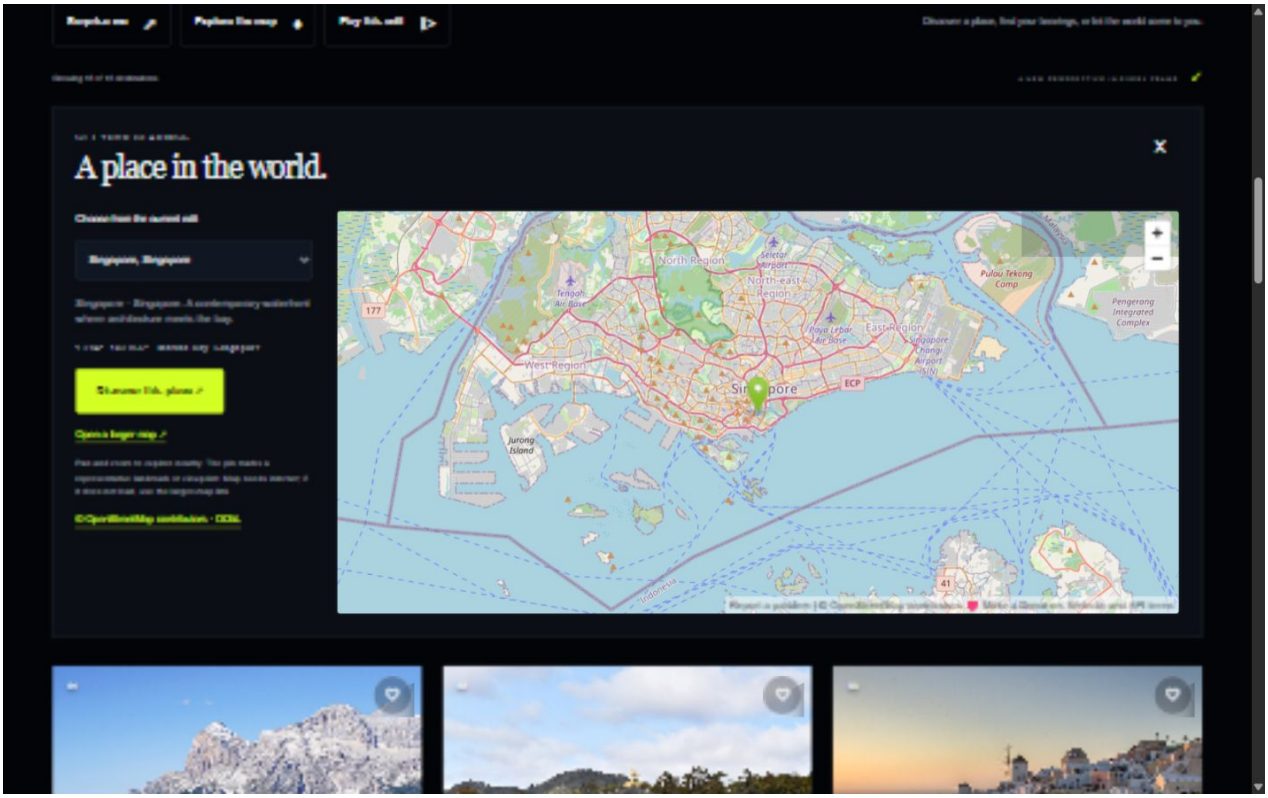


Figure 4.89: Singapore: representative location on OpenStreetMap [Visitor]

The selected landmark/viewpoint is mapped; the guide remains available beside it.

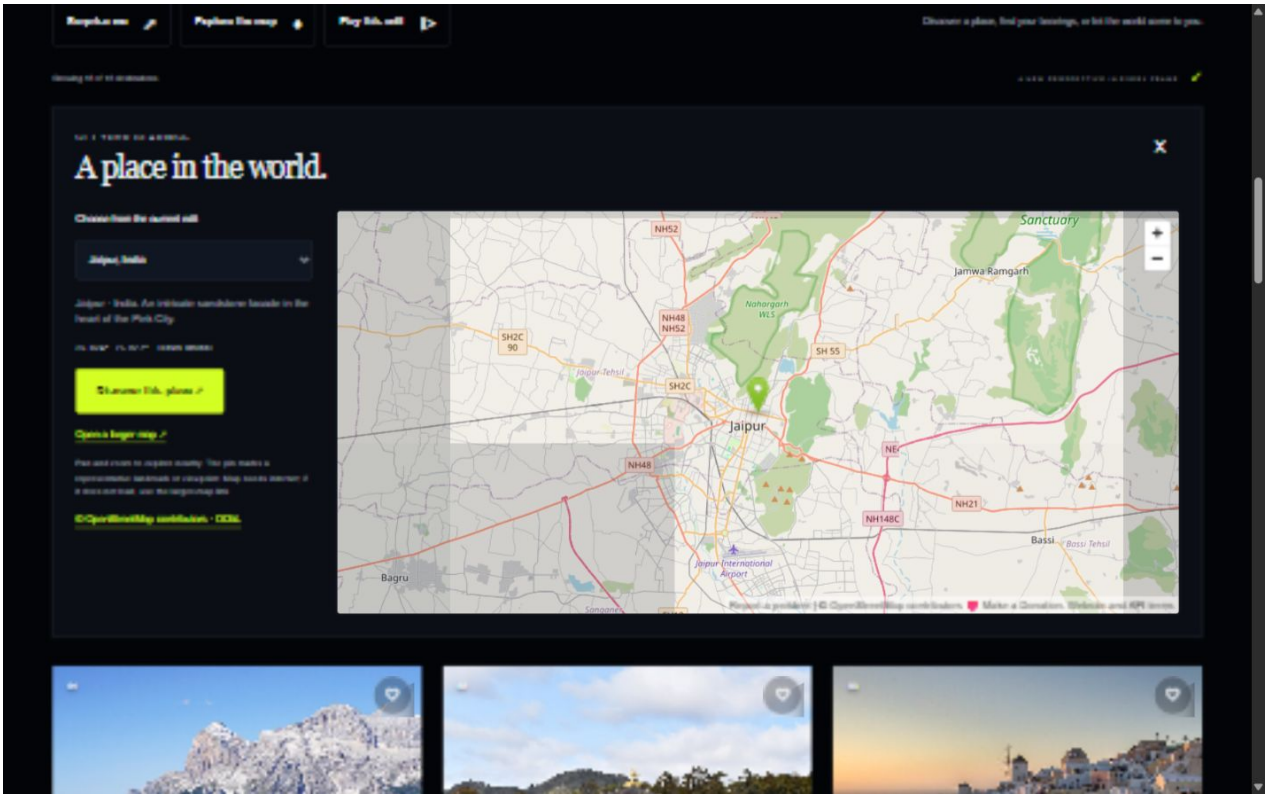


Figure 4.90: Jaipur: representative location on OpenStreetMap [Visitor]

The selected landmark/viewpoint is mapped; the guide remains available beside it.

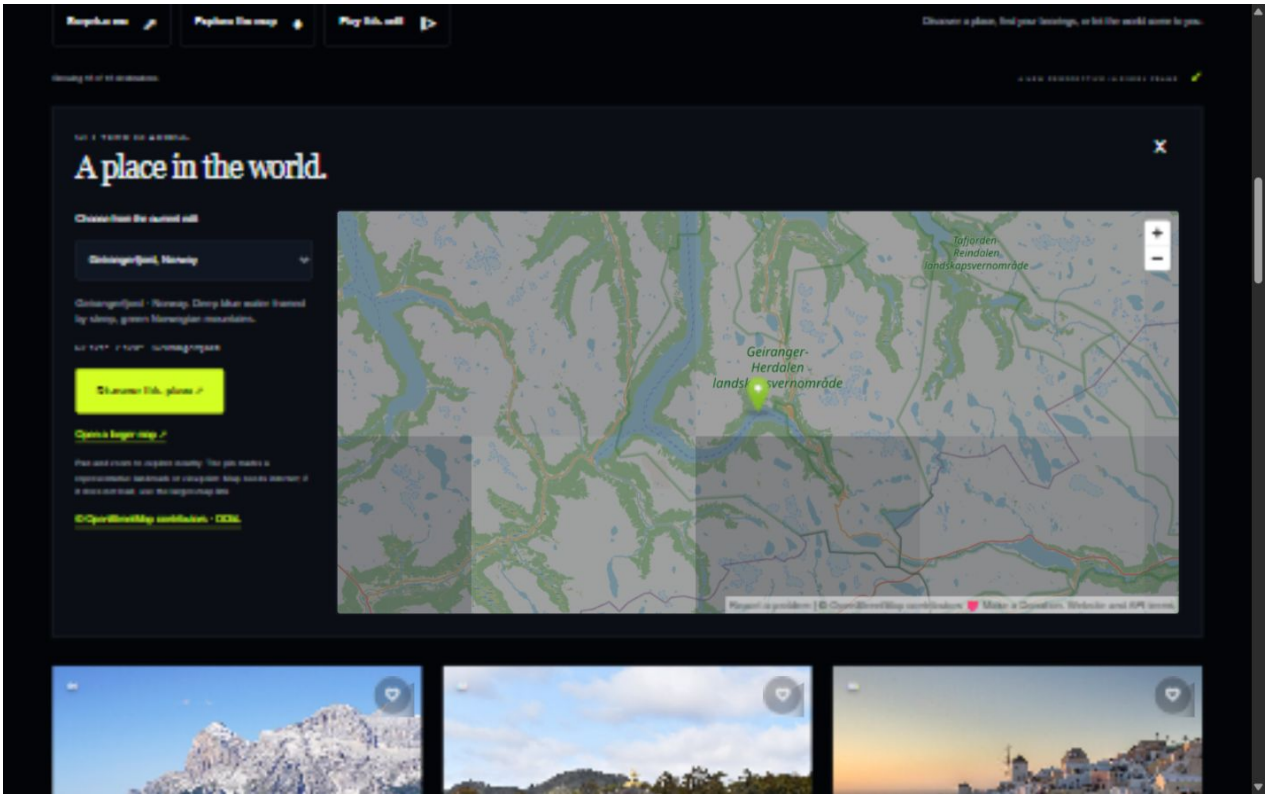


Figure 4.91: Geirangerfjord: representative location on OpenStreetMap [Visitor]

A location reference rather than a route or precise visitor entrance.

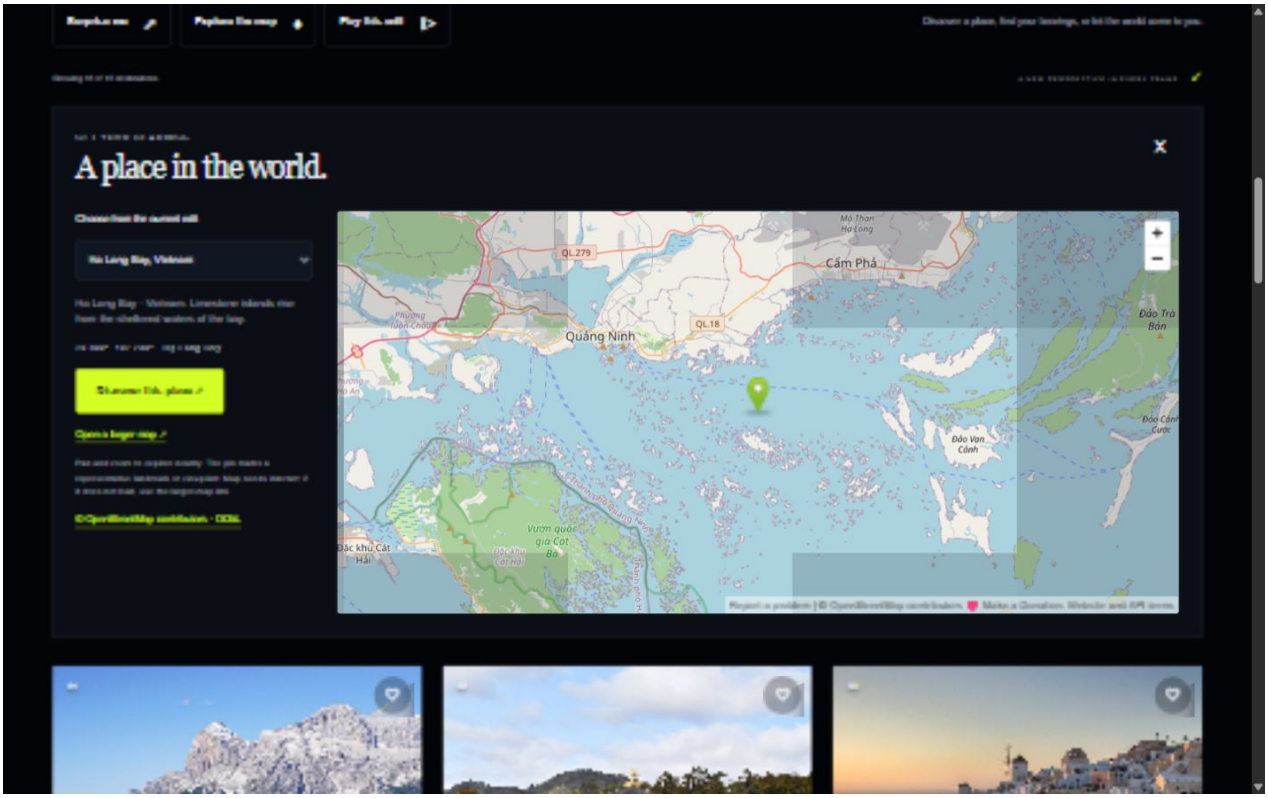


Figure 4.92: Ha Long Bay: representative location on OpenStreetMap [Visitor]

A location reference rather than a route or precise visitor entrance.

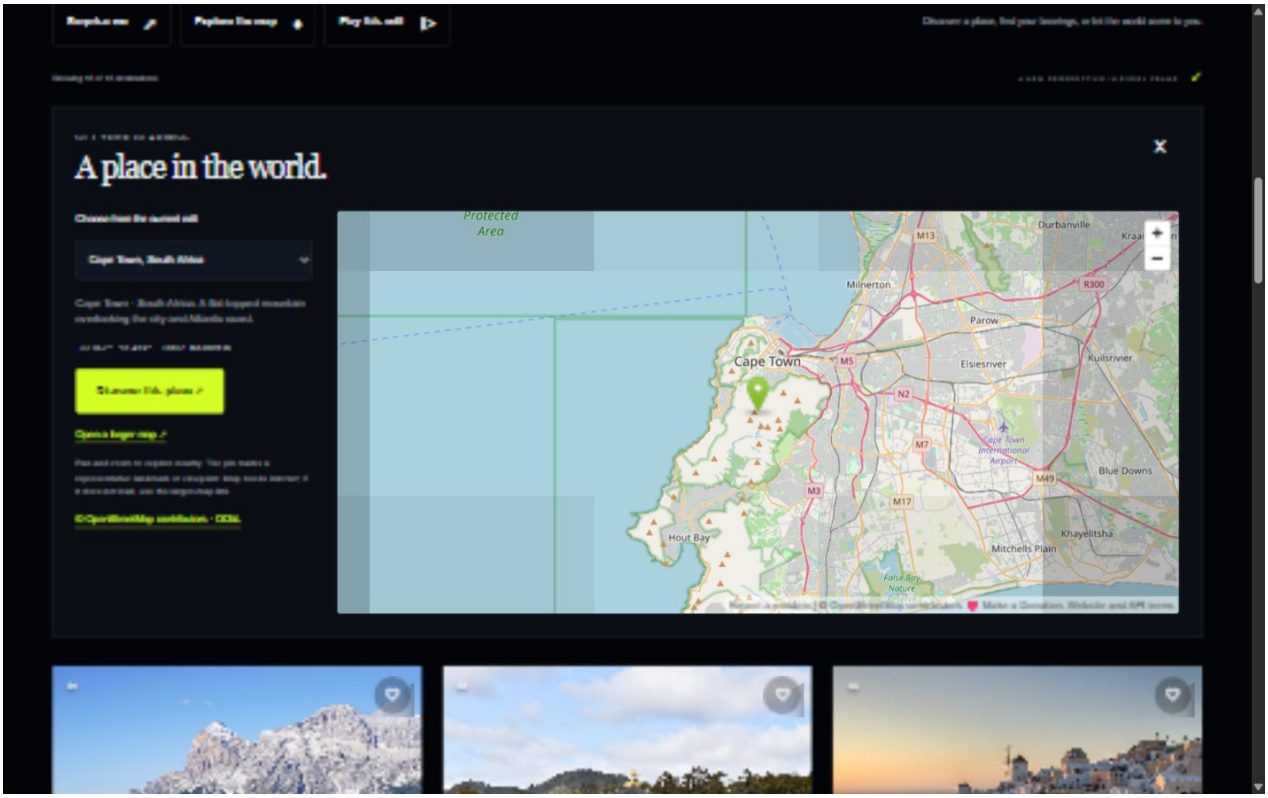


Figure 4.93: Cape Town: representative location on OpenStreetMap [Visitor]

A location reference rather than a route or precise visitor entrance.

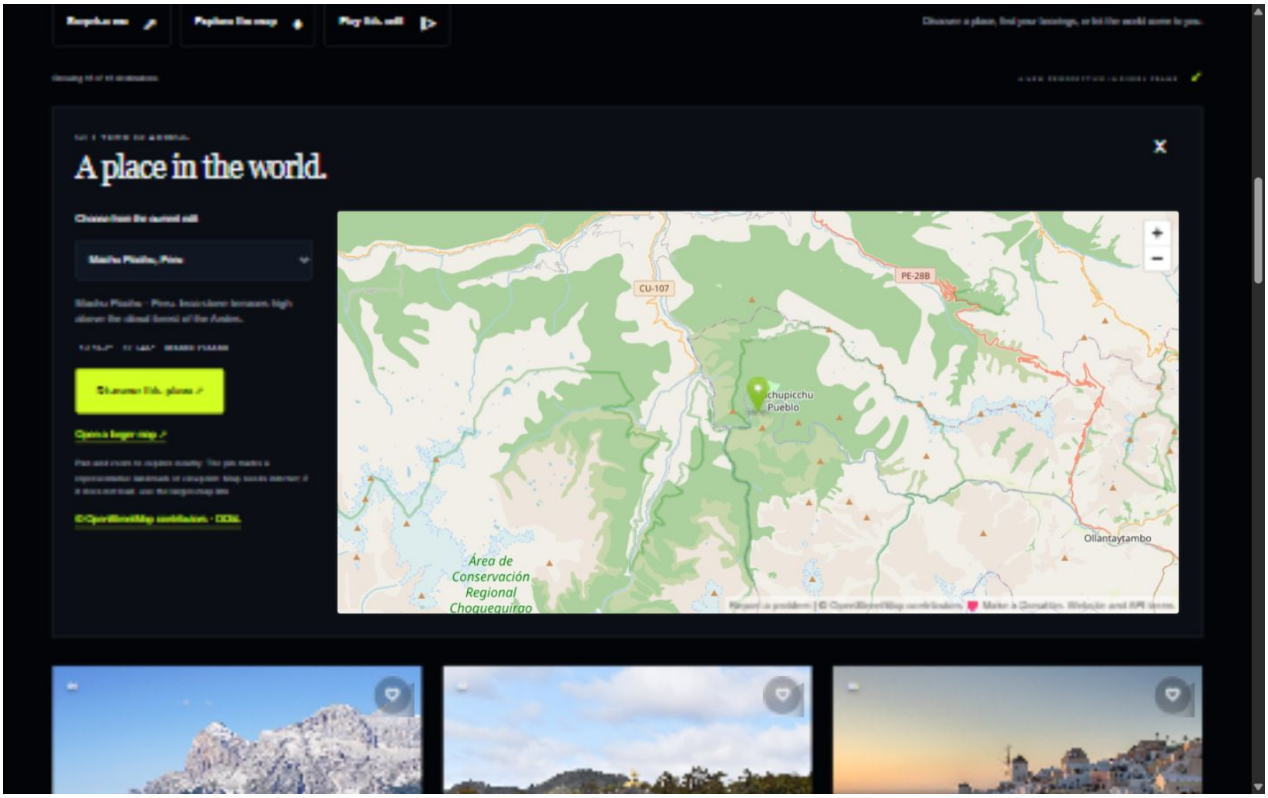


Figure 4.94: Machu Picchu: representative location on OpenStreetMap [Visitor]

A location reference rather than a route or precise visitor entrance.

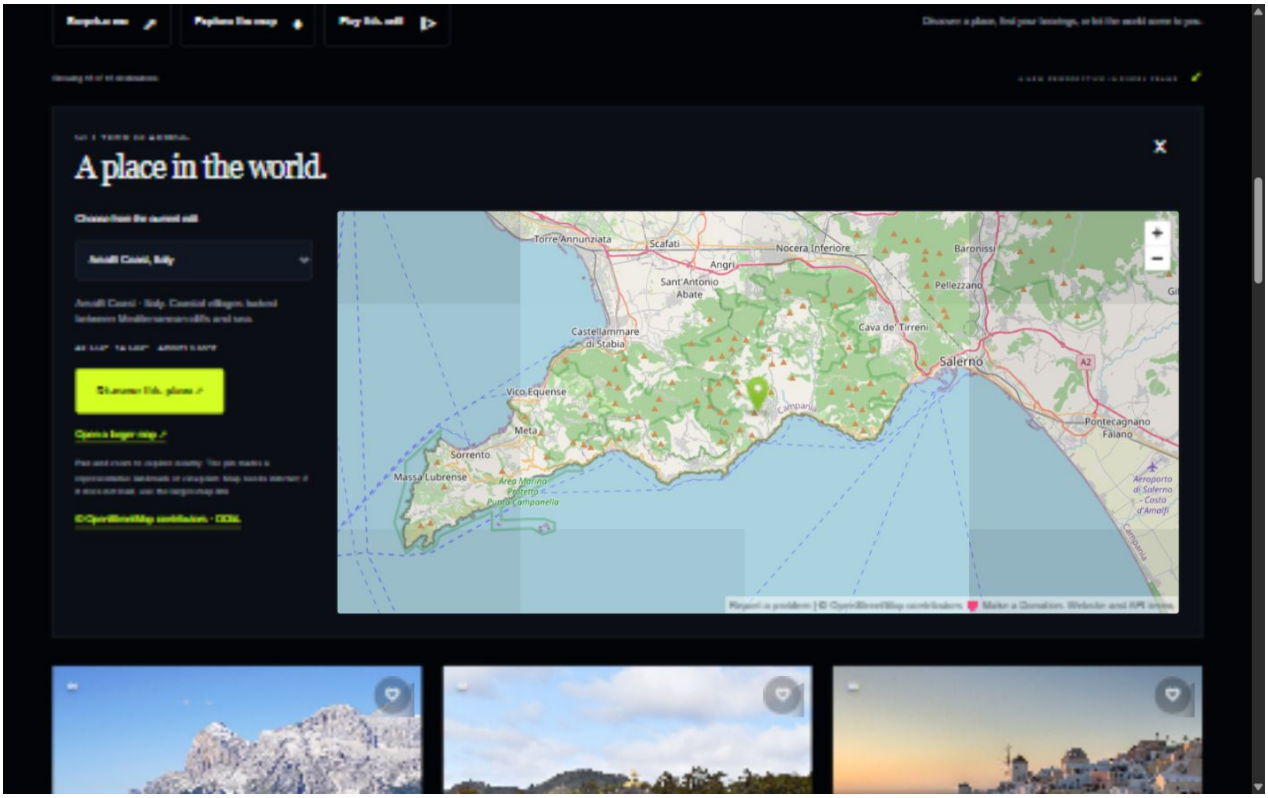


Figure 4.95: Amalfi Coast: representative location on OpenStreetMap [Visitor]

A location reference rather than a route or precise visitor entrance.

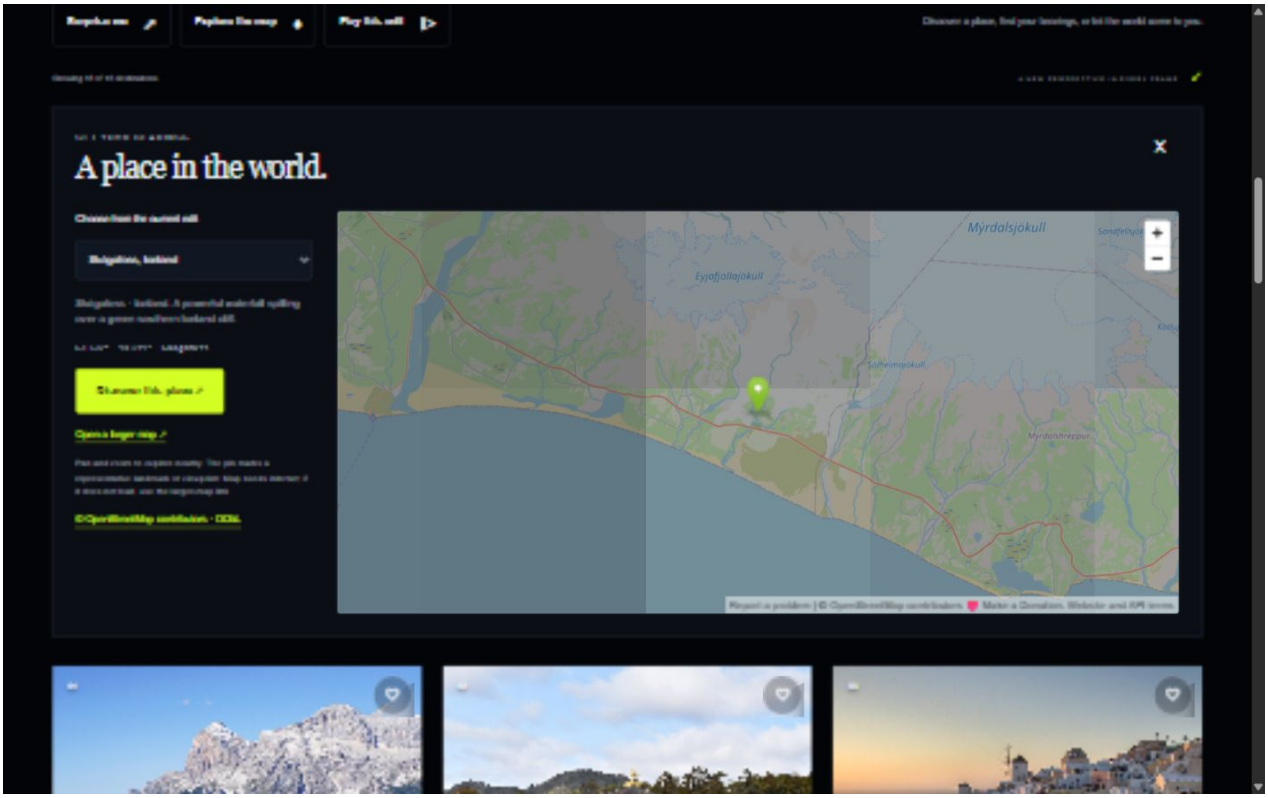


Figure 4.96: Skógafoss: representative location on OpenStreetMap [Visitor]

A location reference rather than a route or precise visitor entrance.

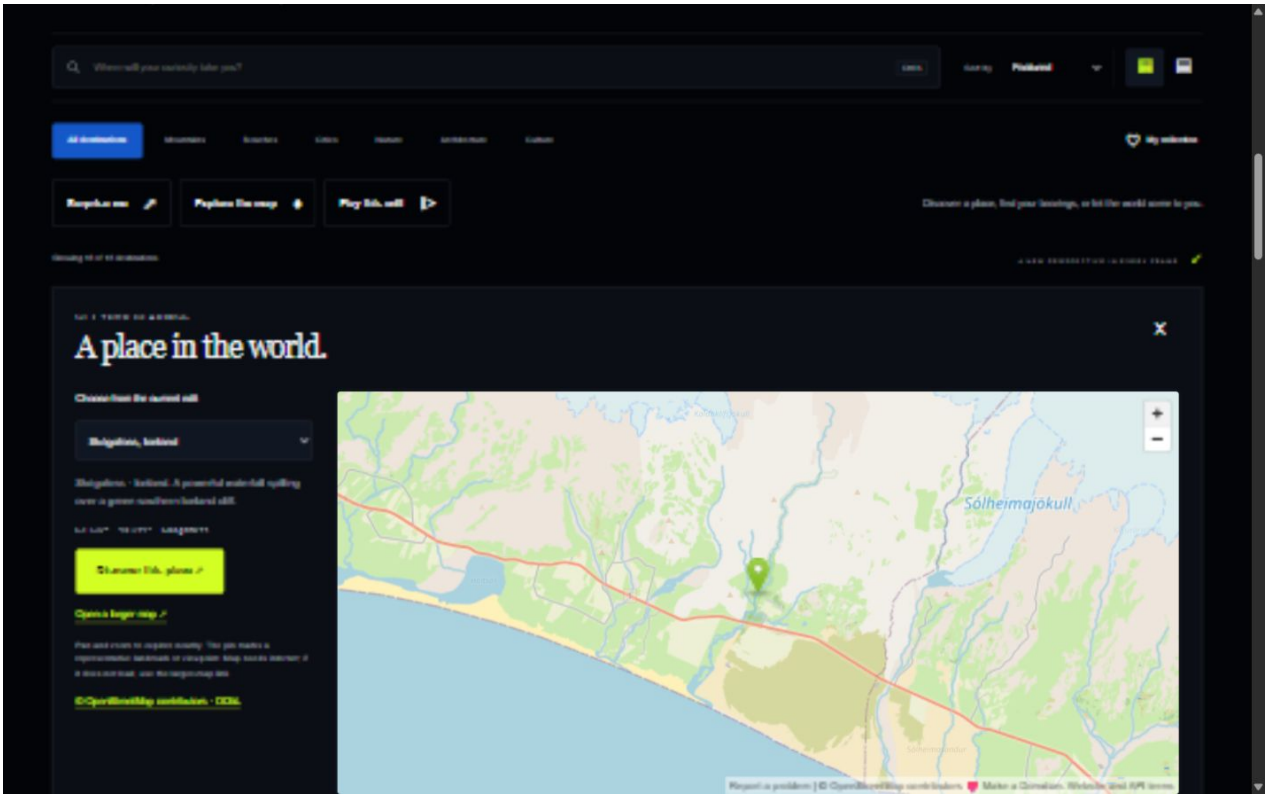


Figure 4.97: Map zoom control in use at Skógafoss [Visitor]

The embedded provider supports nearby exploration using native map controls.

4.12 Filtered Random Discovery

Surprise me chooses within the eligible collection and opens a guide. It avoids an immediate repeat when alternatives exist, permits a single eligible destination and is disabled for an empty edit.

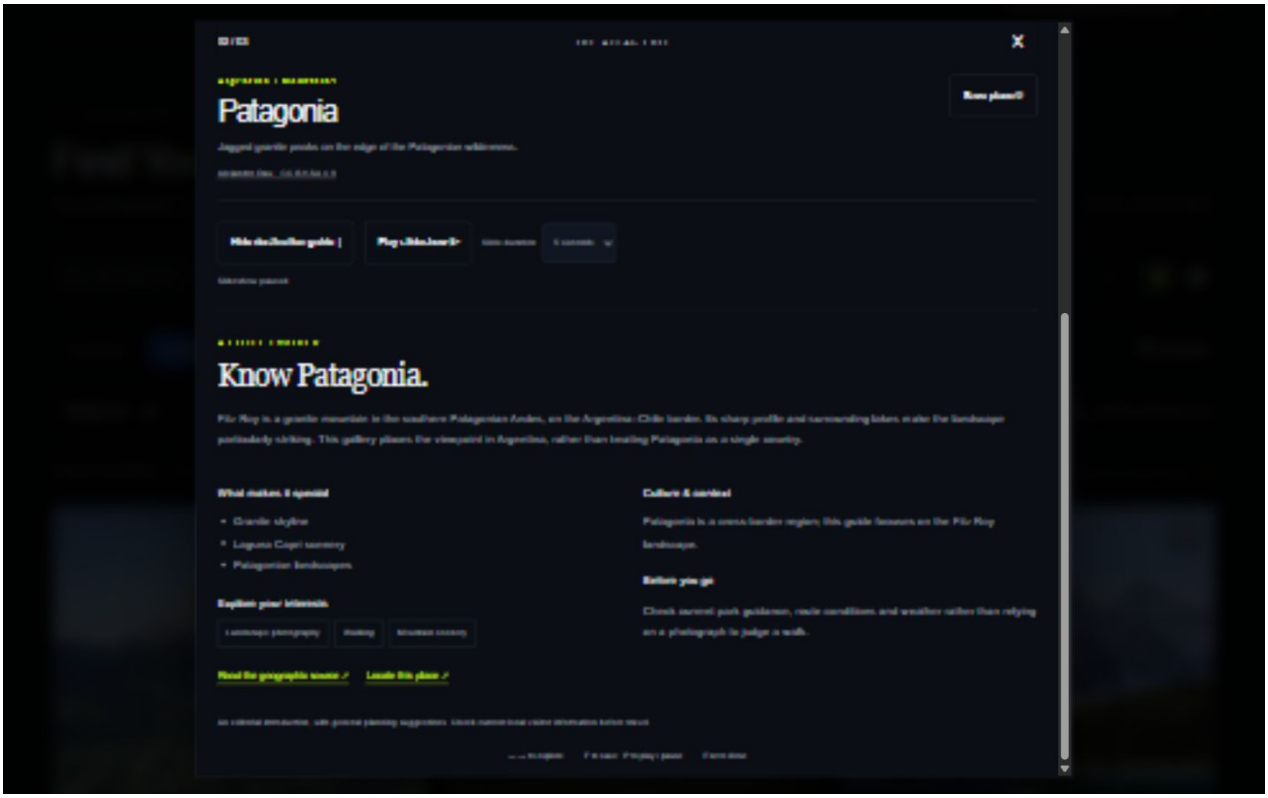


Figure 4.98: Surprise me: Patagonia from the Mountains edit [Visitor]

Random discovery opens an eligible destination guide.

4.13 Photography Slideshow and Playback Controls

Play this edit starts the current eligible sequence. Explicit play/pause, the P shortcut and slide-duration control support user-directed playback. Automatic navigation loops; manual navigation remains bounded. The captures show starting, automatic advance and paused states.

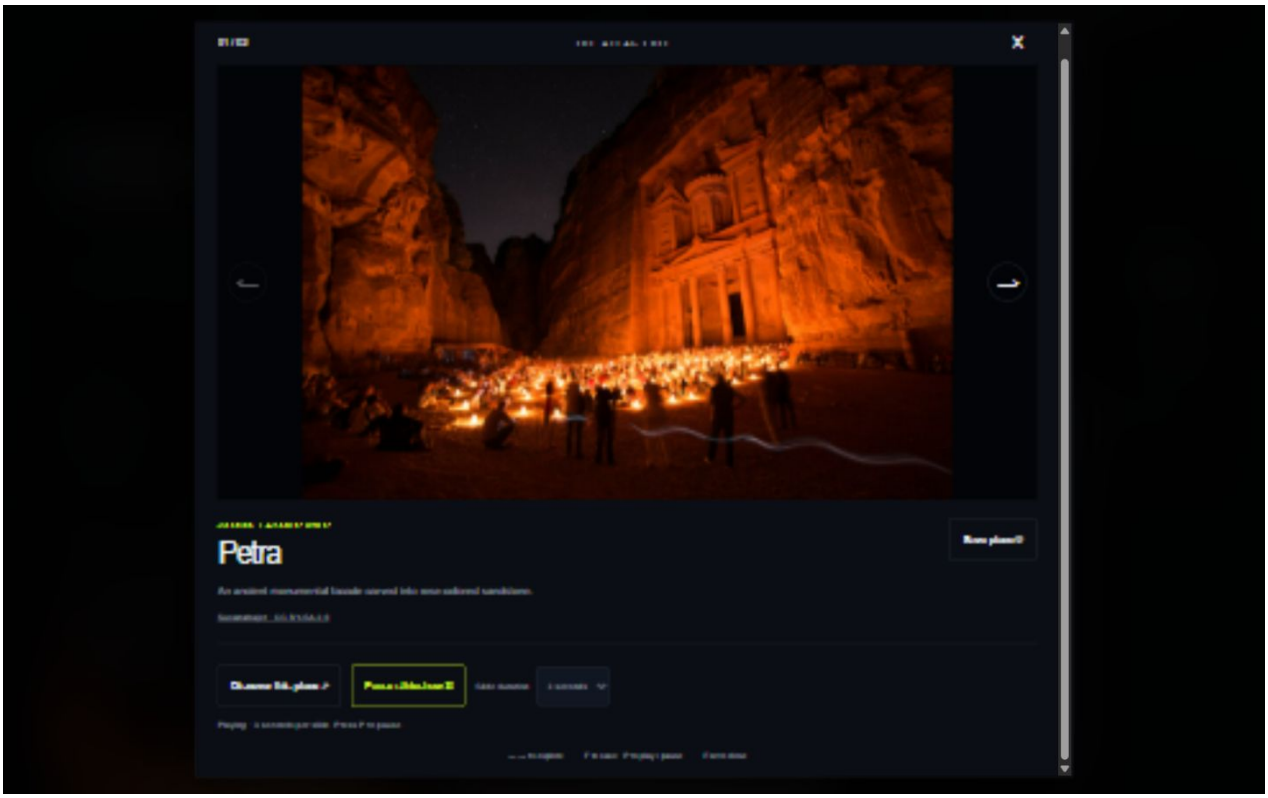


Figure 4.99: Slideshow playing: Architecture edit [Visitor]

Playback is explicitly started and a three-second slide duration is selected.

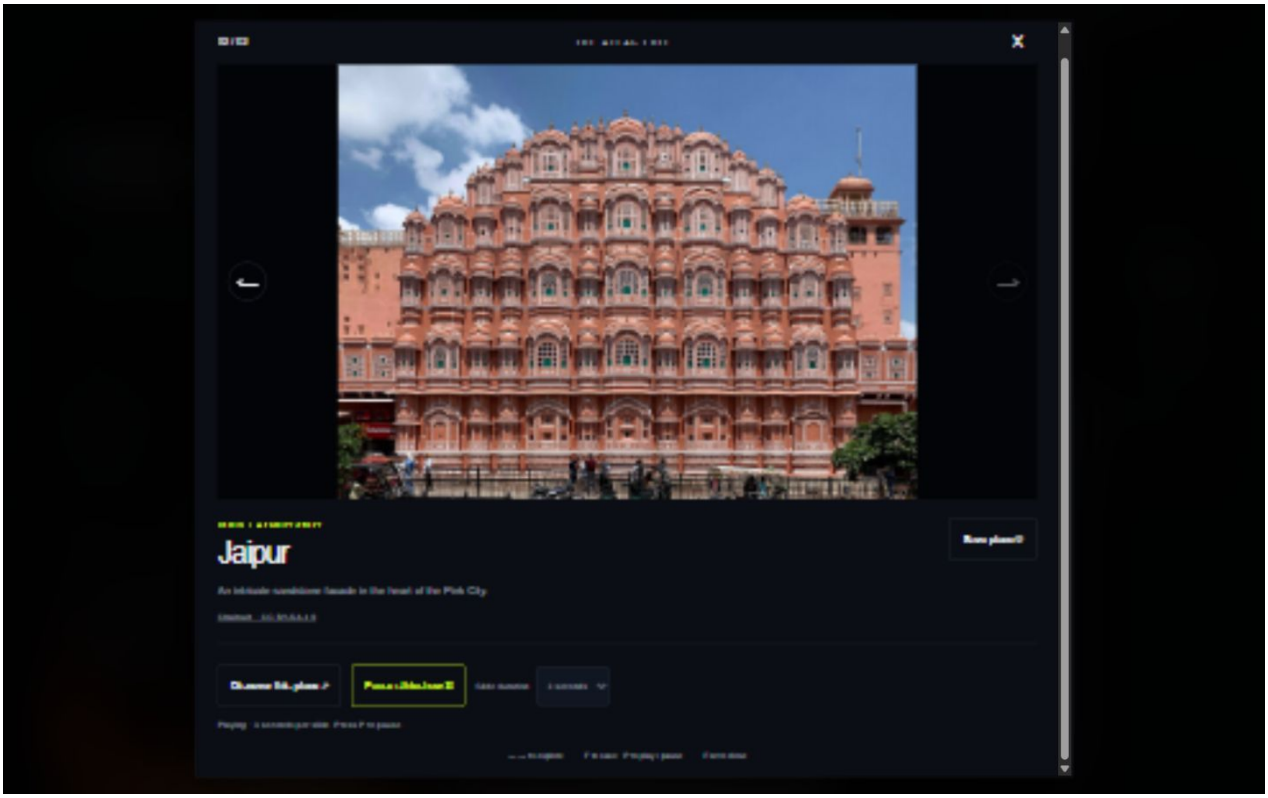


Figure 4.100: Automatic slideshow advance to Jaipur [Visitor]

The next eligible photograph is displayed automatically.

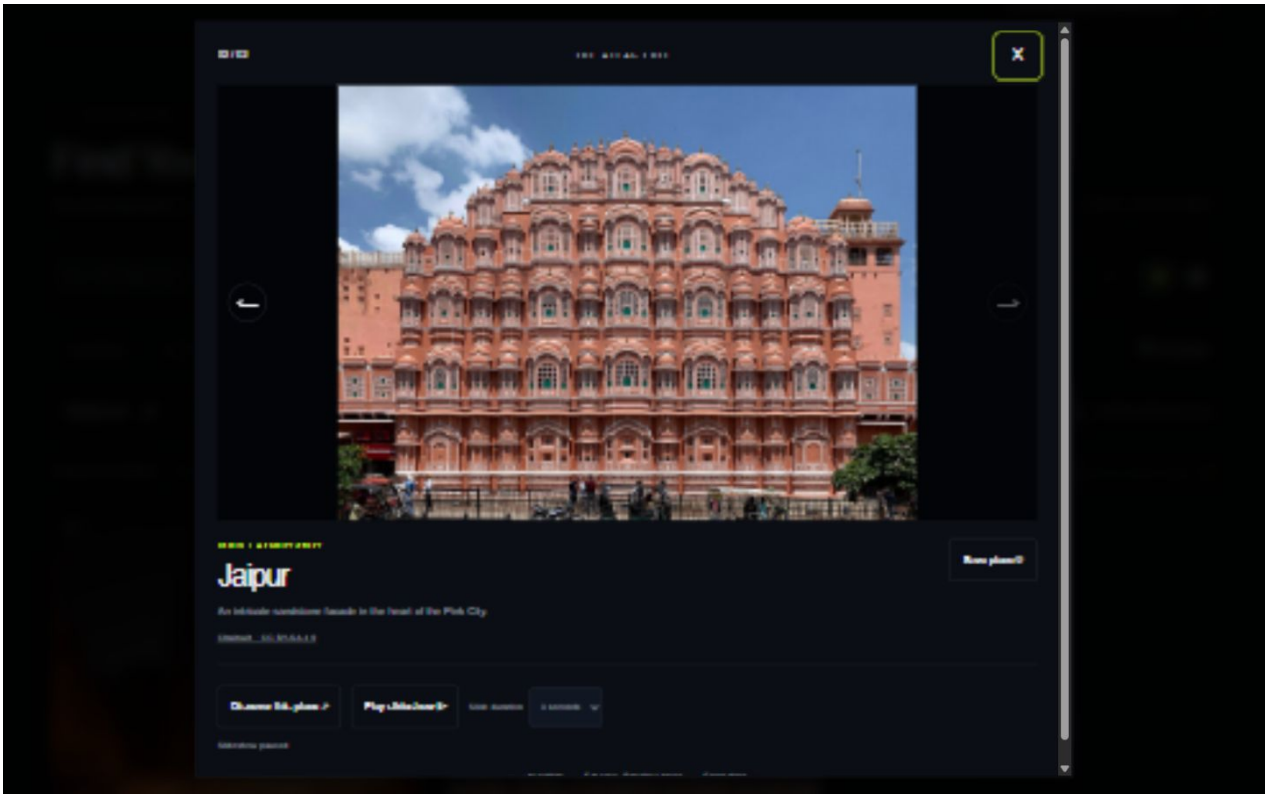


Figure 4.101: Slideshow paused using the P shortcut [Visitor]

Paused status and explicit Play control are visible.

4.14 Responsive Layouts, Phone Navigation and Dialogs

The interface was inspected at 1440, 1024, 768, 390 and 320 CSS pixels. Categories wrap, grid columns adapt and phone navigation becomes a disclosure menu. Guides and comparison details stack on phones. Scrollable dialogs retain visible close controls.



Figure 4.102: Landing page at 1024 px viewport [Visitor]

The same semantic interface adapts to this screen width.

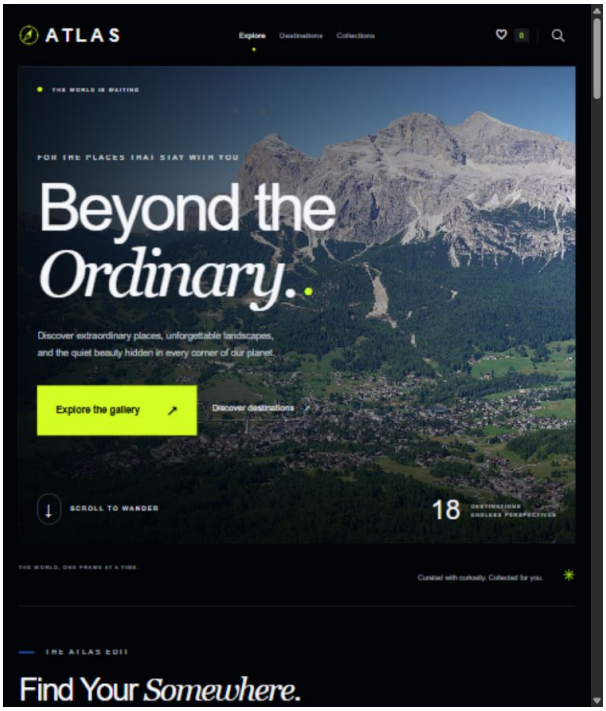


Figure 4.103: Landing page at 768 px viewport [Visitor]

The same semantic interface adapts to this screen width.

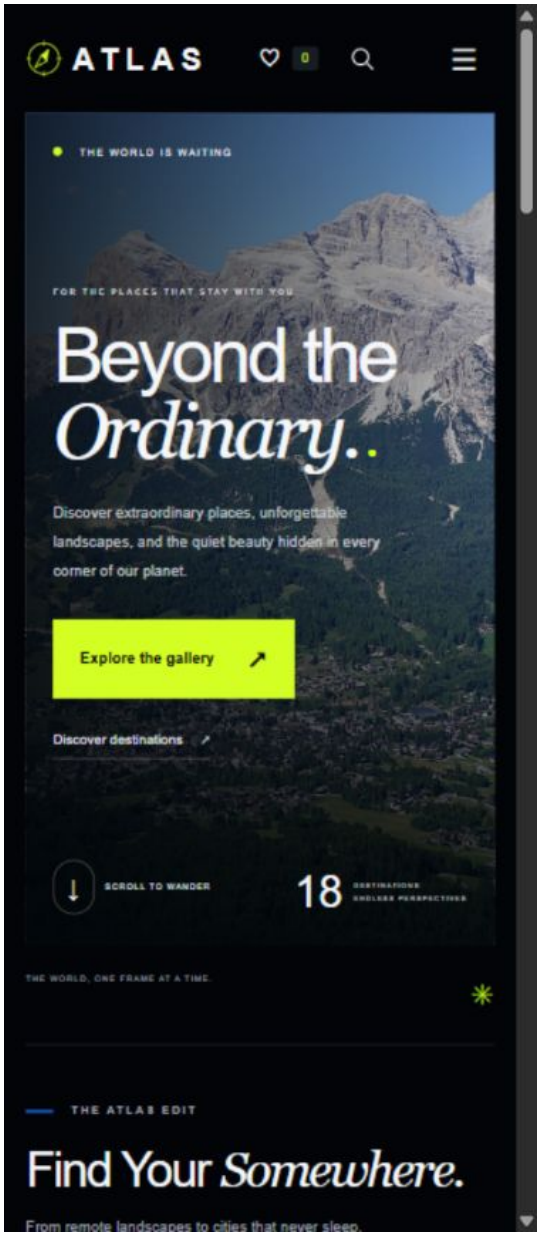


Figure 4.104: Landing page at 390 px viewport [Visitor]

The same semantic interface adapts to this screen width.

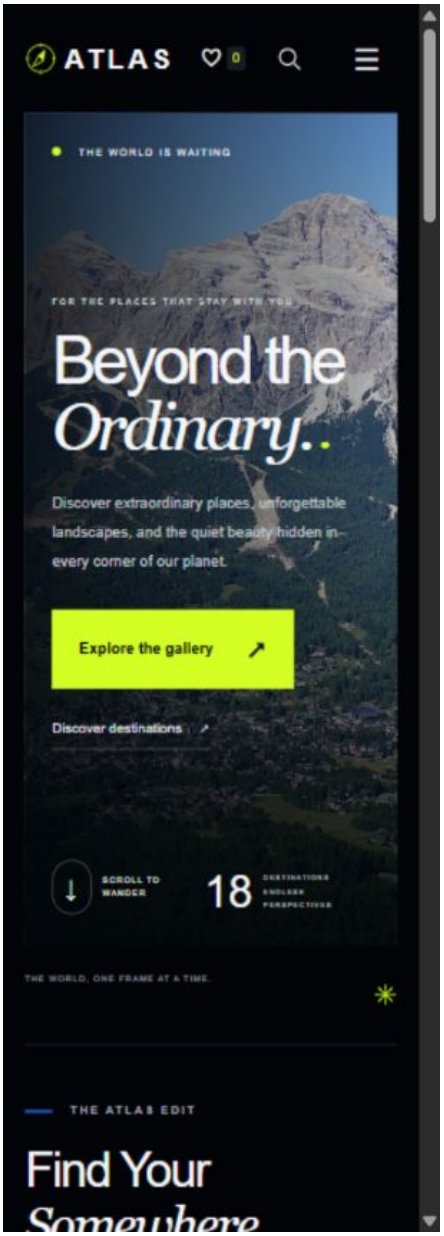


Figure 4.105: Landing page at 320 px viewport [Visitor]
The same semantic interface adapts to this screen width.

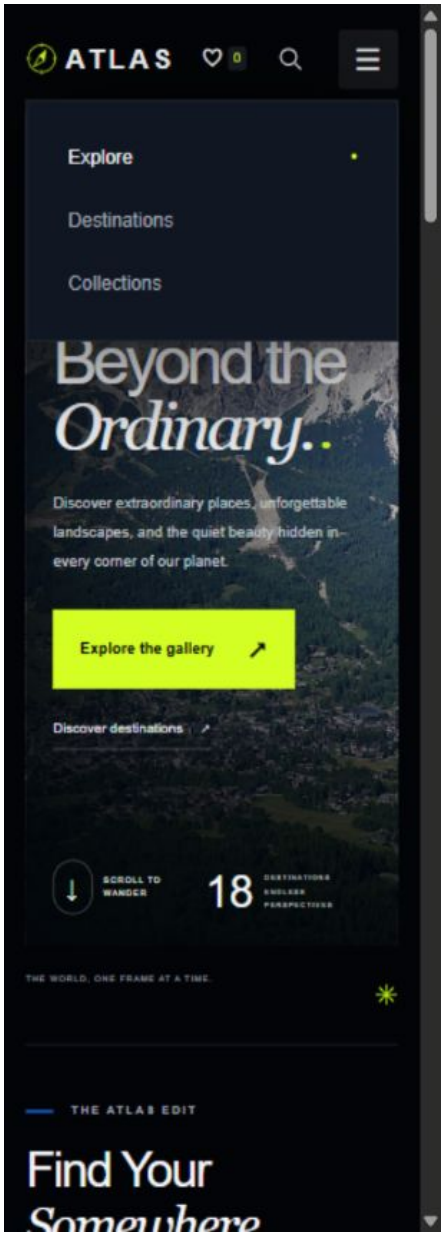


Figure 4.106: Phone navigation expanded at 320 px [Visitor]
Accessible expanded state and links to real page sections.



Figure 4.107: Photograph viewer at 320 px [Visitor]

Phone-scale viewer keeps navigation, saving and close controls available.

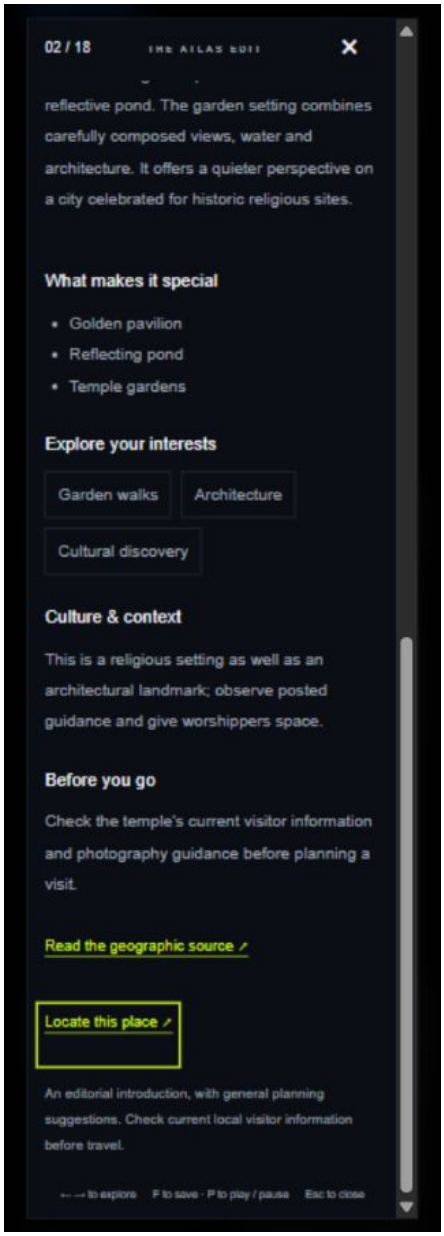


Figure 4.108: Destination guide at 320 px [Visitor]

Guide content stacks into one column with source and map links.



Figure 4.109: Destination comparison at 390 px [Visitor]

Comparison details become stacked cards on a phone viewport.

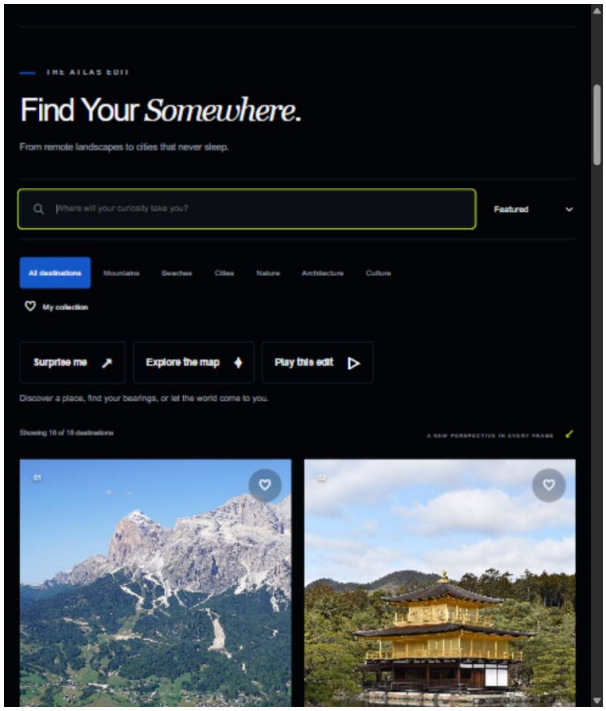


Figure 4.110: Tablet gallery at 768 px [Visitor]

Filters, search and a two-column photo grid remain available.

5 Testing and Verification

The report preparation run executed the existing dependency-free Node suite and fresh browser checks while capturing the application. Node reported 36 executed tests, 36 passed and zero failed. The table below consolidates 40 report cases, TC-01 through TC-40, with observed outcomes. Browser cases use actual click, fill, select and key actions with DOM observations; TC-40 records the executable Node suite. These consolidated cases overlap the earlier project test records and must not be added to them as independent totals.

Test environment and execution record

Environment: Windows PowerShell; Node.js v22.18.0; Python 3.14.7; a static server at http://127.0.0.1:8000; the available Chromium-based in-app browser. Screen widths included 1440, 1024, 768, 390 and 320 px. The source evidence is docs/report-evidence/manifest.json and node-results.tap. No external browser-testing package was installed.

All 40 final consolidated cases passed. The attempt history contains 41 records because an immediate map-zoom observation initially sampled the animation before it settled. TC-35 was observed again after the map updated and passed; that earlier attempt is retained rather than hidden. This is a timing/observation correction, not an invented pass. Eleven broader manual scenarios in the project documentation remain open, including screen readers, other browser engines, forced failures and background-tab playback pause.

ID	Module	Scenario	Result	Type
TC-01	Rendering	Load the page and render all 18 destination cards Observed: 18 destination cards rendered.	Passed	Browser
TC-02	Accessibility	Check meaningful alternative text on gallery photographs Observed: All 18 gallery images have descriptive alternatives.	Passed	Browser
TC-03	Search	Search for destinations and narrow the gallery Observed: Single matching destination shown for the searched place.	Passed	Browser
TC-04	Lightbox	Open photographs and display their matching metadata Observed: Six destination viewers opened correctly and closed with Escape.	Passed	Browser
TC-05	Guides	Expand detailed guides and verify three highlights Observed: Overview and three highlights shown for the selected destination.	Passed	Browser
TC-06	Guides	Provide source and location links in destination guides Observed: Geographic reference and map links are populated.	Passed	Browser
TC-07	Keyboard	Close the native viewer using Escape Observed: Viewer closes and page becomes available again.	Passed	Browser
TC-08	Search	Ignore case and surrounding whitespace when matching country Observed: Padded mixed-case Japan search returns Kyoto.	Passed	Browser
TC-09	Search	Clear only the search query Observed: All 18 eligible cards return.	Passed	Browser
TC-10	Empty states	Display a useful empty state for an unmatched query Observed: No-place message and recovery action shown.	Passed	Browser
TC-11	Filters	Reset to the complete eligible edit Observed: Reset restores all 18 places.	Passed	Browser
TC-12	Categories	Filter the gallery by Mountains Observed: 3 matching Mountains destinations, no other category.	Passed	Browser
TC-13	Categories	Filter the gallery by Beaches Observed: 3 matching Beaches destinations, no other category.	Passed	Browser
TC-14	Categories	Filter the gallery by Cities Observed: 3 matching Cities destinations, no other category.	Passed	Browser
TC-15	Categories	Filter the gallery by Nature Observed: 4 matching Nature destinations, no other category.	Passed	Browser
TC-16	Categories	Filter the gallery by Architecture Observed: 2 matching Architecture destinations, no other category.	Passed	Browser
TC-17	Categories	Filter the gallery by Culture Observed: 3 matching Culture destinations, no other category.	Passed	Browser
TC-18	Sorting	Sort destinations alphabetically A-Z Observed: Alphabetical order starts with Amalfi Coast.	Passed	Browser
TC-19	Sorting	Sort destinations alphabetically Z-A Observed: Reverse alphabetical order starts with Skógafoss.	Passed	Browser
TC-20	Layout	Switch to the compact grid Observed: Compact grid class and selected control updated.	Passed	Browser

ID	Module	Scenario	Result	Type
TC-21	Favorites	Toggle favorites without opening a photograph Observed: Two saved places; photograph dialog remains closed.	Passed	Browser
TC-22	Persistence	Retain favorites after a page refresh Observed: Saved count remains two after reload.	Passed	Browser
TC-23	Favorites	Show only saved places Observed: Only Dolomites and Kyoto remain in the collection.	Passed	Browser
TC-24	Keyboard	Activate a focused photograph using Enter Observed: Native button activation opens the matching photograph.	Passed	Browser
TC-25	Keyboard	Navigate with ArrowRight inside the viewer Observed: The next photograph is Kyoto.	Passed	Browser
TC-26	Keyboard focus	Keep focus within the open viewer Observed: Shift+Tab from Close wraps to the visible duration selector.	Passed	Browser
TC-27	Keyboard focus	Restore the opening card after Escape Observed: Focus returns to the Dolomites photo button.	Passed	Browser
TC-28	Discovery	Choose a random destination from the active category Observed: Patagonia belongs to the selected Mountains edit.	Passed	Browser
TC-29	Comparison	Require at least two destinations to compare Observed: One selection leaves Compare places disabled.	Passed	Browser
TC-30	Comparison	Compare two selected destinations Observed: Two destination columns contain location, highlights, interests and context.	Passed	Browser
TC-31	Comparison limits	Reject a fourth selected destination Observed: Three selections retained and a limit message displayed.	Passed	Browser
TC-32	Comparison focus	Close comparison with Escape and restore focus Observed: Focus returns to Compare places.	Passed	Browser
TC-33	Map	Offer map choices from the eligible edit Observed: All 18 destinations are offered when no filter is active.	Passed	Browser
TC-34	Map	Center the embed on the selected destination Observed: Kyoto map marker uses the sourced Kinkaku-ji coordinate pair.	Passed	Browser
TC-35	Map interaction	Operate the embedded map zoom control Observed: After the zoom animation settles, the provider reports the updated zoom level.	Passed	Browser
TC-36	Slideshow	Start, advance automatically, and pause playback with P Observed: Automatic advance to Jaipur observed; P changes playback to paused.	Passed	Browser
TC-37	Mobile navigation	Open the phone navigation menu Observed: Menu is expanded with actual section links.	Passed	Browser
TC-38	Responsive layout	Check 1024, 768, 390 and 320 px widths for horizontal overflow Observed: All four additional widths fit without horizontal document overflow.	Passed	Browser
TC-39	Browser diagnostics	Check application-script warning and error entries Observed: No application-script warning or error entries observed.	Passed	Browser
TC-40	Automated verification	Run the complete dependency-free Node test suite Observed: 36 tests executed, 36 passed, zero failed; TAP output retained.	Passed	Automated

Verification boundaries

The test run establishes the listed behavior in the available browser, not complete cross-browser, assistive-technology or performance conformance. Invalid stored-data parsing is covered by the Node tests; actual denied-storage and image-host failure injection were not executed in this report session. Map zoom was observed; native map panning is available but was not separately asserted. The screenshot count is 110 distinct image files, verified by content hashes, and is separate from the count of test cases.

6 Deployment

ATLAS is delivered locally as a static website and can be hosted by GitHub Pages or any static server. No public deployment or repository URL has been supplied or verified; this report does not claim a live hosted service. All application paths are relative and there is no framework build, environment secret, backend process or database migration.

Item	Configuration / procedure
Entry point	index.html at the project root
Local server	From the project folder: python -m http.server 8000 --bind 127.0.0.1
Local address	http://127.0.0.1:8000
Runtime prerequisites	A modern JavaScript-enabled browser supporting native dialog; Python is only a local-server convenience.
Static-host files	index.html, css/, js/, assets/, favicon.ico and docs/
GitHub Pages option	Commit the source, select Settings > Pages > Deploy from a branch, and select the repository root. This is a deployment procedure, not evidence of a completed deployment.
External network needs	Wikimedia photography and optional OpenStreetMap embed; guide text and source coordinates are bundled.
Persistence	Favorites use localStorage for the same origin/browser; other view settings and comparison are session-only.

Local demonstration workflow

Start the server, open the local address, browse a card and its guide, combine a country query with a category, save a place and refresh, compare two destinations, open a representative map, and run a two-place Architecture slideshow at three seconds. Demonstrate Tab/Shift+Tab, Enter/Space, arrows, F, P, Escape and Ctrl/Cmd+K. Stop the server with Ctrl+C after evaluation.

7 Limitations and Future Scope

Limitations

- Photographs and interactive map tiles depend on external connectivity and service availability. The local fallback does not cache the whole image collection.
- Favorites are browser/origin-local and can be cleared or restricted. No account or cloud synchronization is present.
- Search is a literal substring over the original destination metadata, without fuzzy ranking or multilingual matching.
- Map pins identify representative landmarks or viewpoints, not precise visitor entrances, routes or the boundary of an entire region.

- Guides contain editorial context and general suggestions; live entry schedules, prices, weather, visa rules and travel advisories are not supplied.
- Comparison is qualitative, session-only and limited to three places. No cost estimator, upload, booking service, personal travel journal or shareable destination link is implemented.
- Testing does not constitute a full WCAG audit, real-device certification, performance benchmark or Firefox/Safari pass. Remaining manual scenarios are retained in the project test documents.

Future scope

- Add a backend and authenticated, cloud-synchronized favorites when a real multi-device requirement arises.
- Extend the map with an all-destination marker layer, optional route planning and richer geographic filters.
- Introduce reviewed multilingual content, advanced search and a small destination-content management workflow.
- Add license-aware image optimization/caching, optional PWA support, visual regression checks and systematic accessibility/cross-browser testing.

8 Conclusion

ATLAS demonstrates how a small static frontend can support a coherent discovery experience without a framework or backend. The completed interface combines photography, contextual guides, maps, favorites, comparison, random exploration and controlled playback through clear browser state. Native controls, semantic markup, restrained styling and responsive layouts support its academic frontend-design objectives. Mouse and keyboard events perform real tasks, while explicit dialog and timer lifecycles reduce inconsistent behavior. This report follows the supplied academic structure, documents the implemented states with 110 actual screenshots, and records TC-01 through TC-40 with final observed outcomes. The source, evidence and remaining verification gaps are retained for demonstration and review.

9 References

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Project evidence and technical references were consulted during implementation and report preparation on 29 October 2026. Source-verified metadata is retained in js/data.js, docs/IMAGE_CREDITS.md and docs/DESTINATION_SOURCES.md. Screenshots and current report test outcomes are retained in docs/report-evidence/.